

Chromatin states

Chapter 9



Main content



NUCLEOSOME



HETEROCHROMATIN
VS EUCHROMATIN



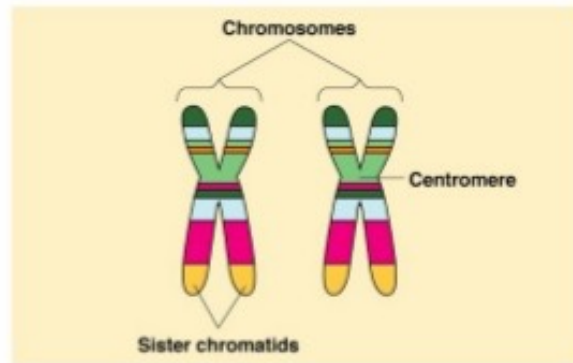
CHROMATIN STATE

Nucleosome

Chromosomes vs. Chromatin

Chromosomes

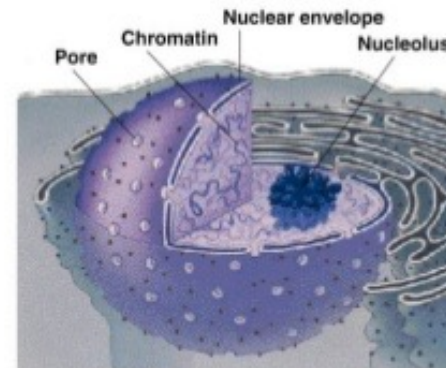
- Tightly packaged DNA
- Found only during cell division
- DNA is not being used for macromolecule synthesis



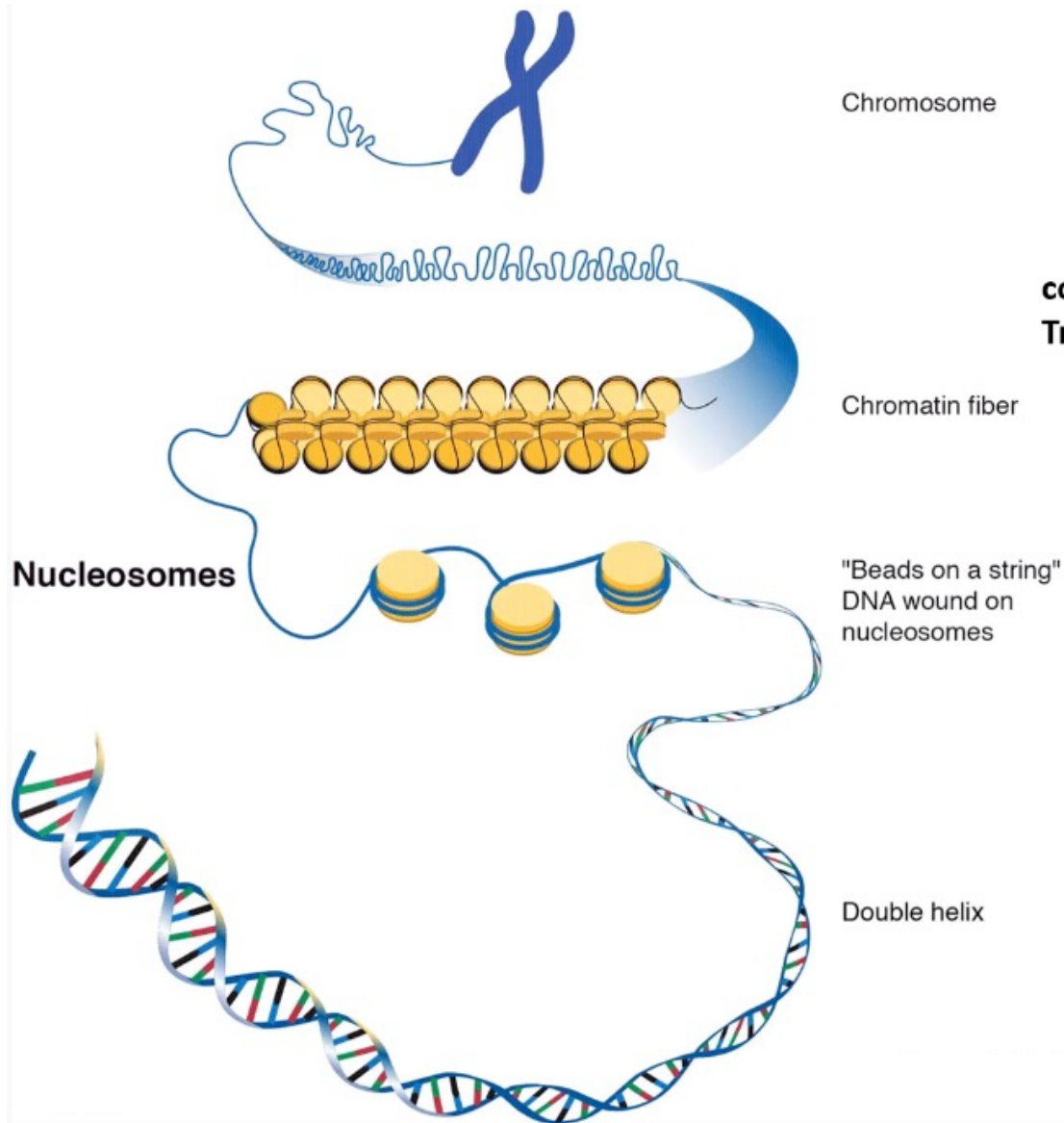
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Chromatin

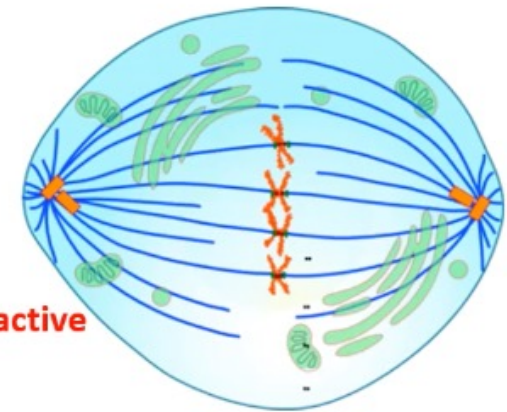
- Unwound DNA
- Found throughout Interphase
- DNA *is* being used for macromolecule synthesis



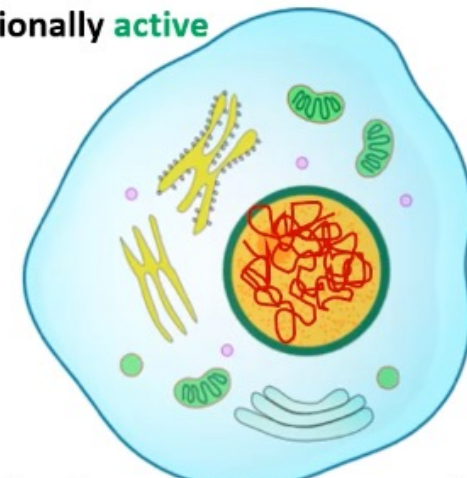
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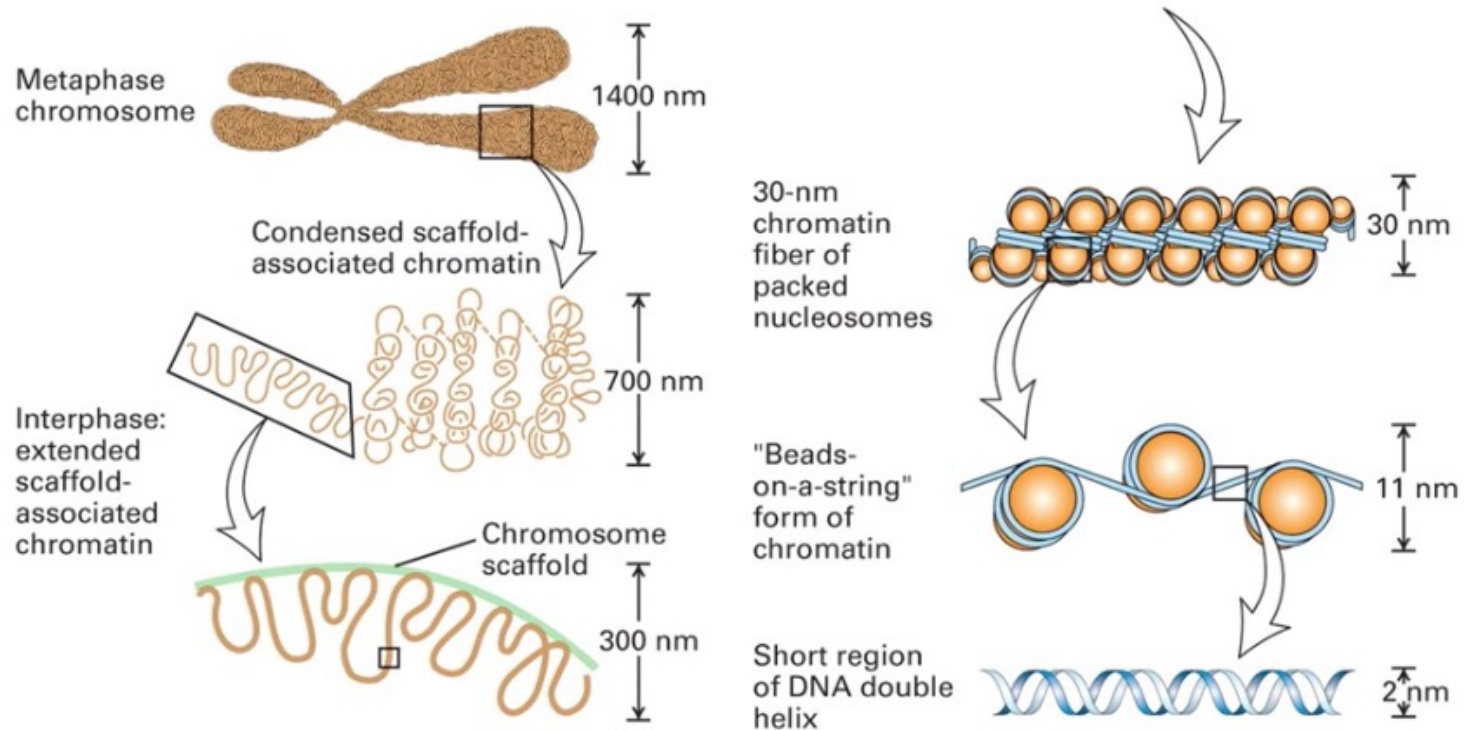
condensed
Transcriptionally inactive



Relatively relaxed
Transcriptionally active



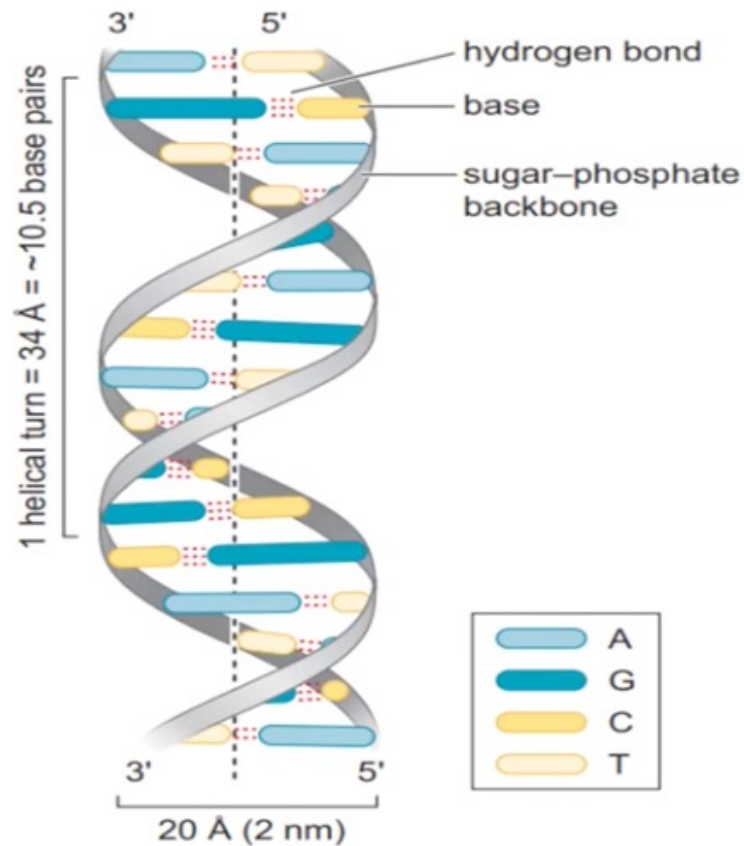
**The DNA in one human cell is about 1.5m long
How is it folded and organised inside the nucleus?**



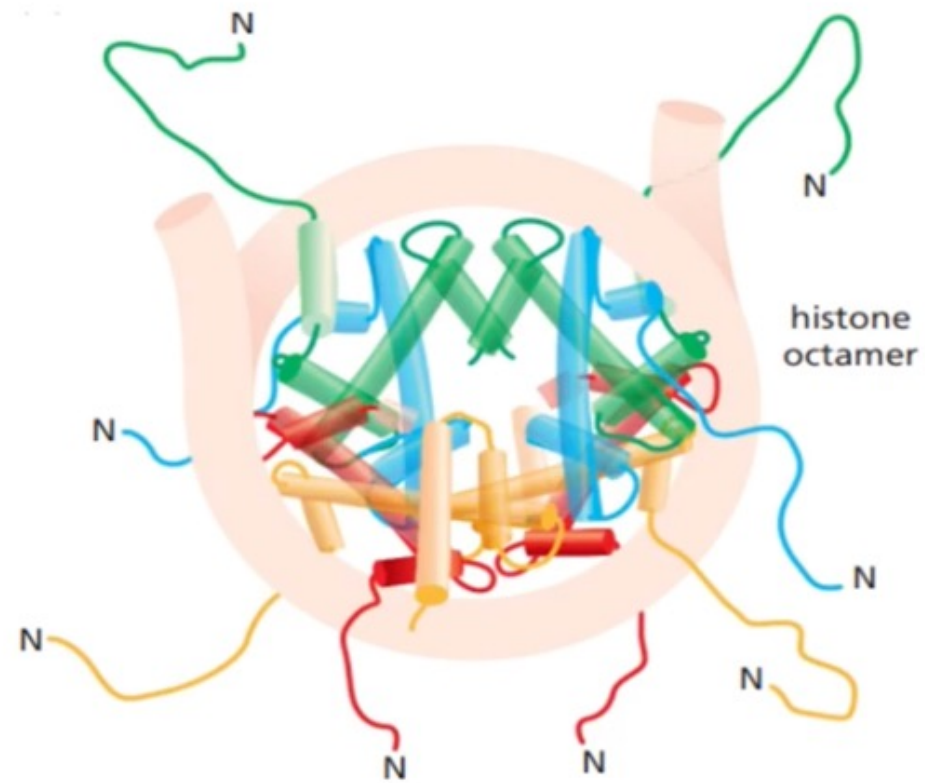
Nucleosome



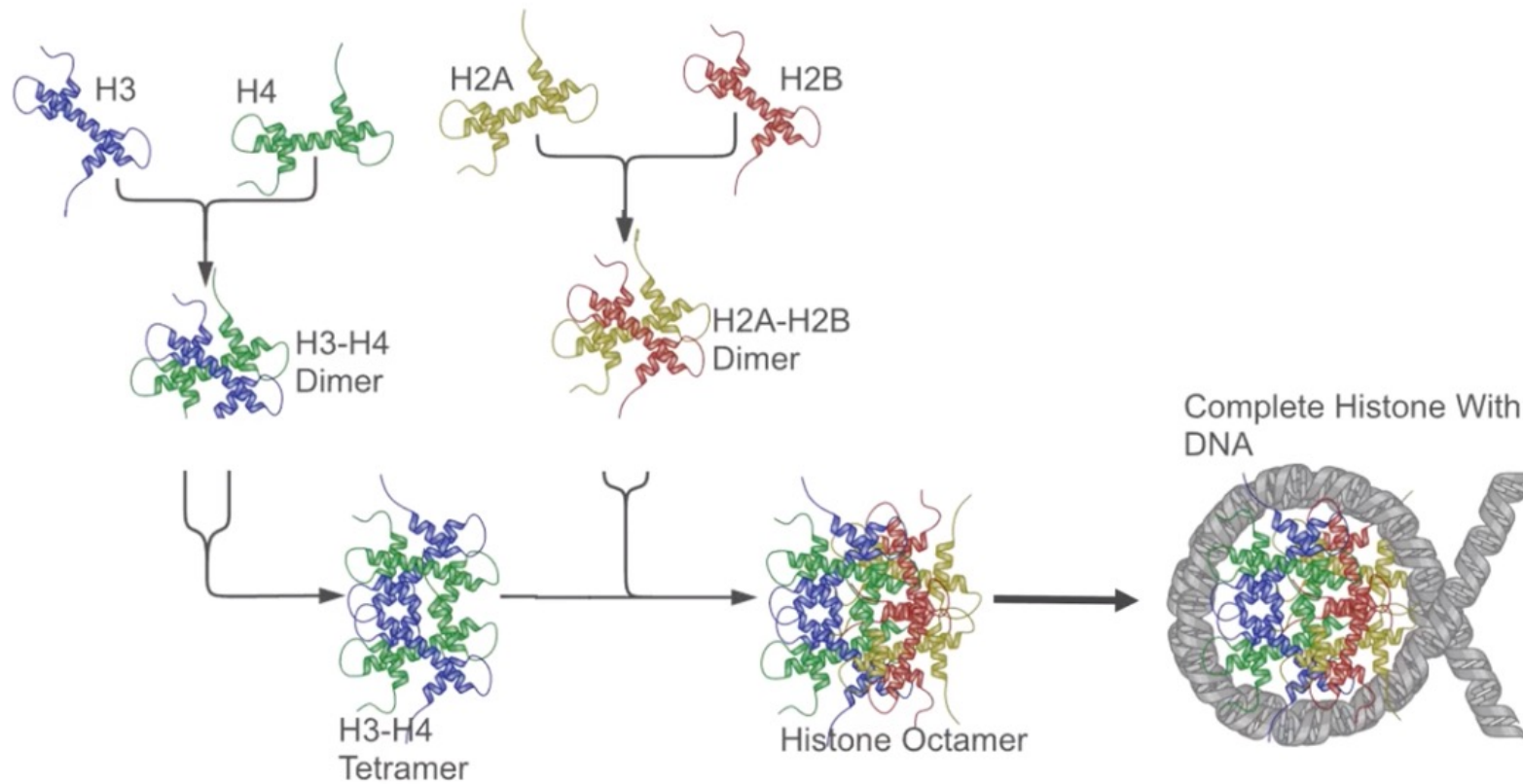
DEOXYRIBONUCLEIC ACID (DNA)



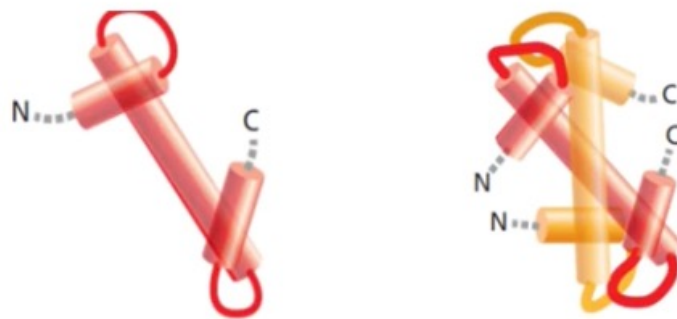
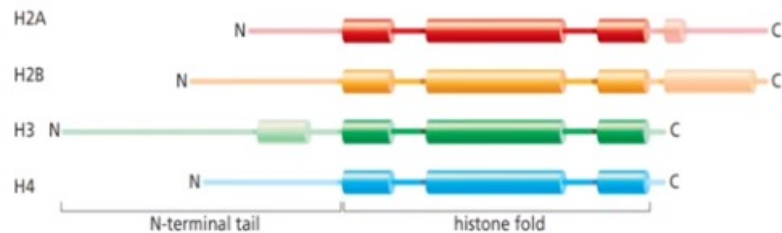
- Rise per base pair- 3.32 Å
- Helix packing-diameter- 23.7 Å
- Helix rotation sense- Right handed
- Base pairs per helix turn- ~10
- Rotation per base pair- 35.9 °
- Pitch per helix turn- 33.2 Å



Histone octamer on DNA

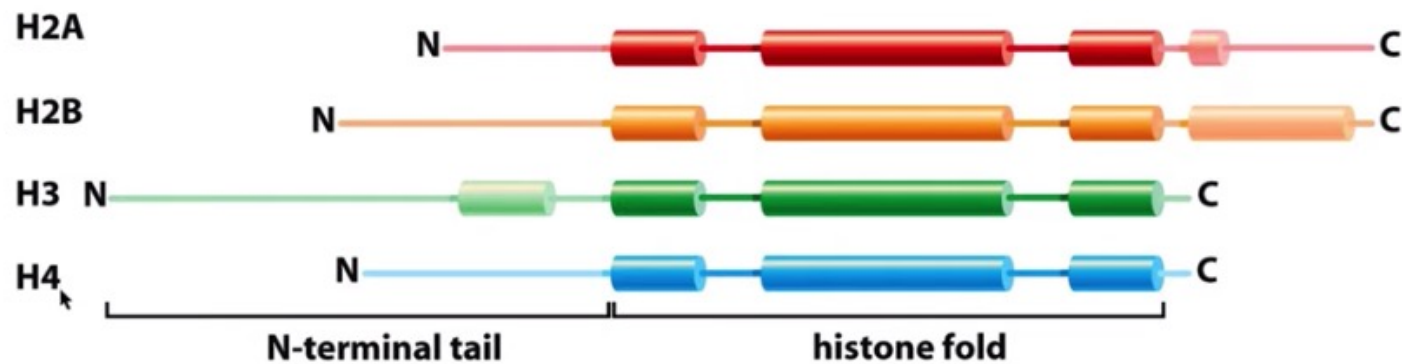


STRUCTURAL ORGANISATIONS OF THE CORE HISTONES



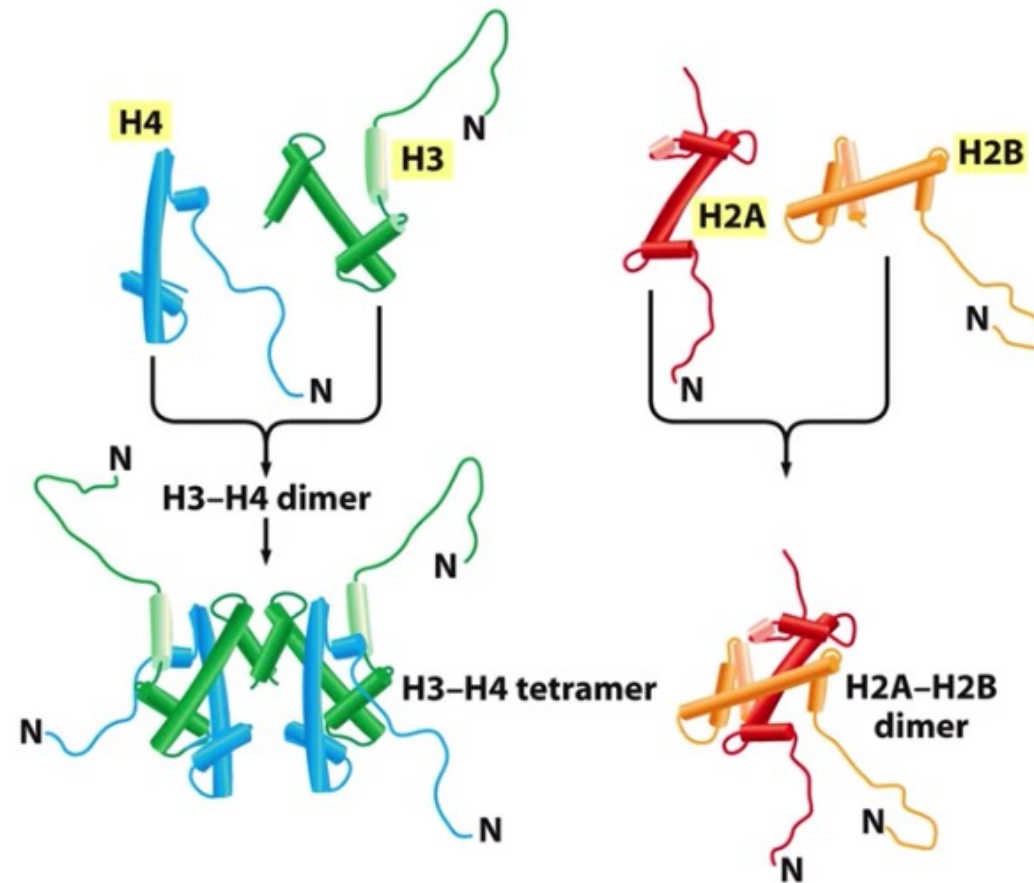
- Each of the core histones contains an N-terminal tail, which is subject to several forms of covalent modification, and a histone fold region
- The structure of the histone fold, which is formed by all four of the core histones
- Histones 2A and 2B form a dimer through an interaction known as the "handshake."
- Histones H3 and H4 form a dimer through the same type of interaction

The core histone proteins

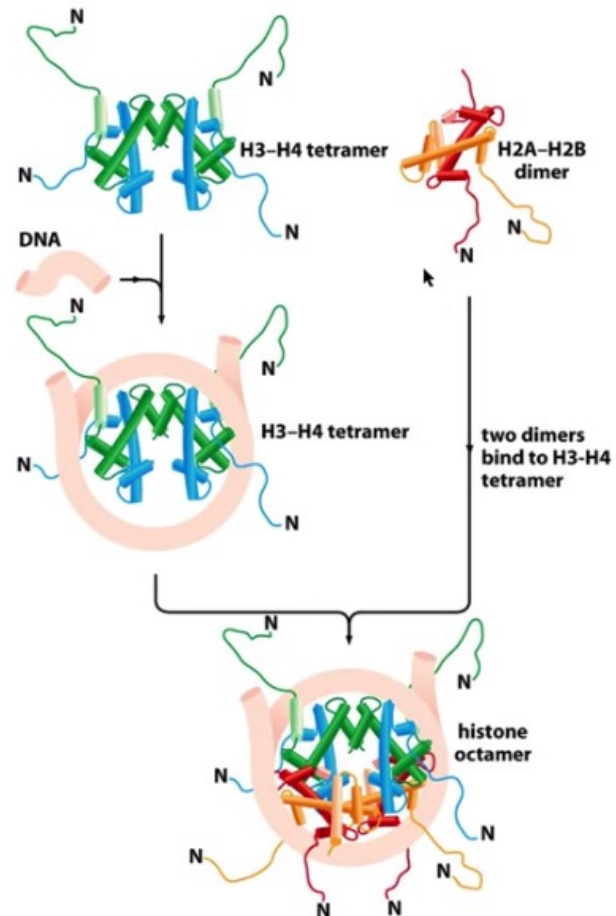


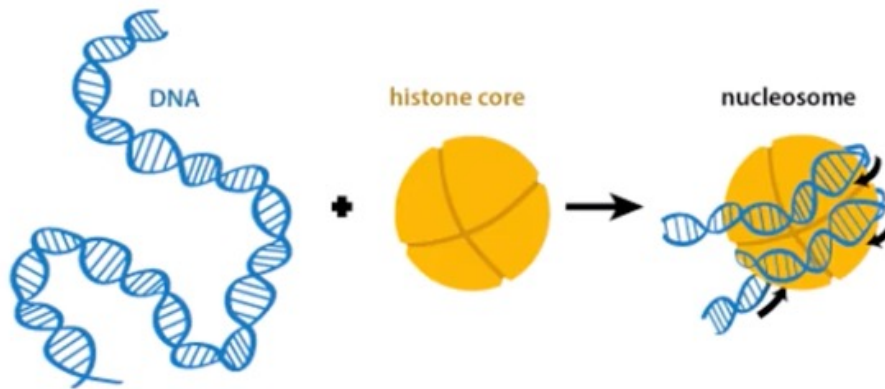
Each core histone has a histone fold
and long unstructured N-terminal tails

Histone folding



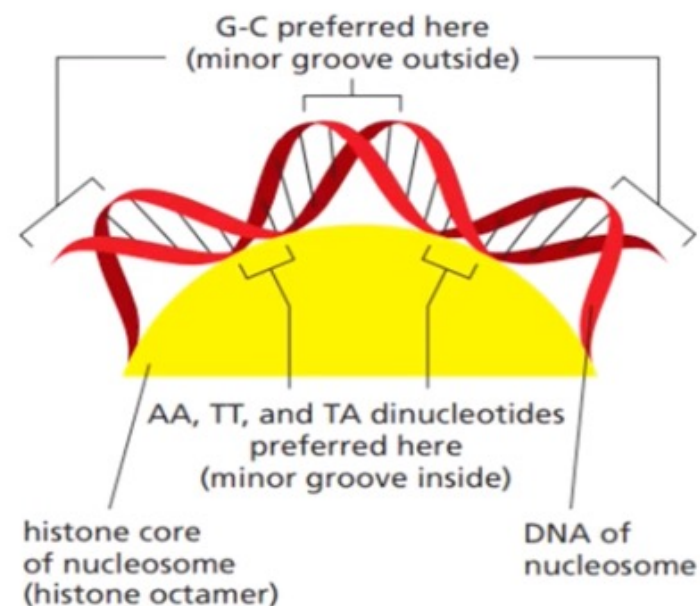
Histone folding

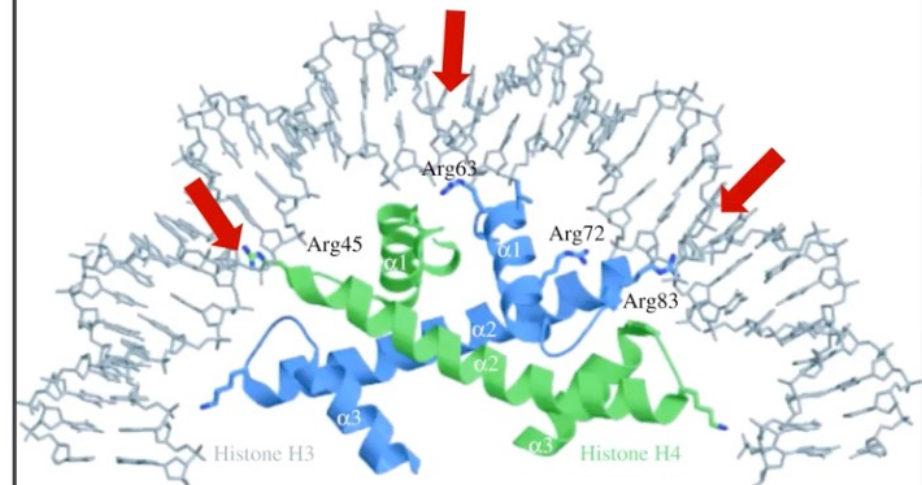
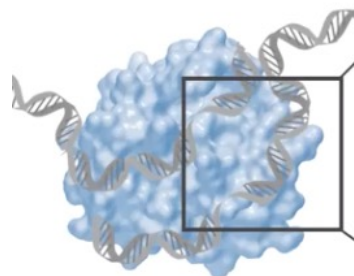
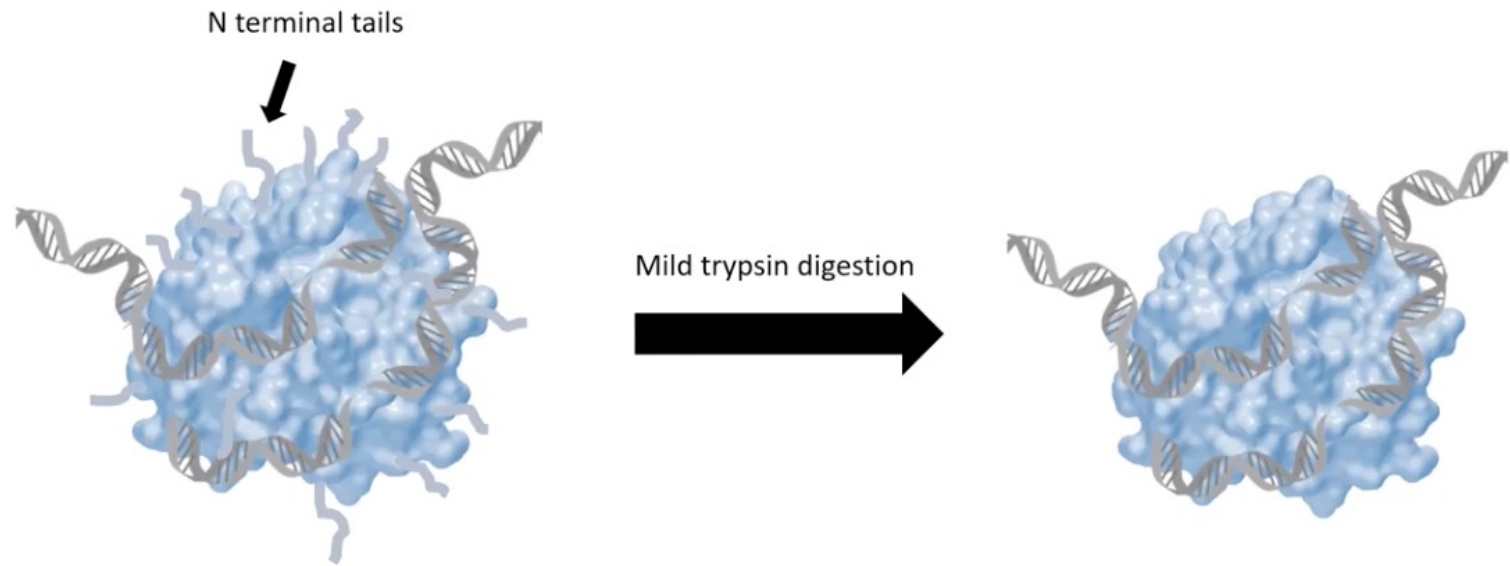




Wrapping of DNA on Histone core

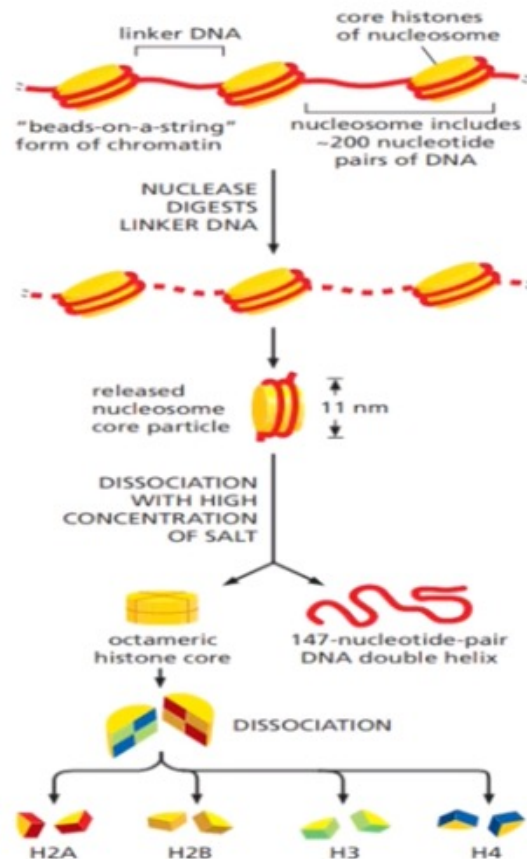
- The path of the DNA around the histone core is not smooth.
- several kinks are seen in the DNA
- The bending requires a substantial compression of the minor groove of the DNA helix
- the indicated dinucleotides are preferentially accommodated in such a narrow minor groove, which helps to explain why certain DNA sequences will bind more tightly than others to the nucleosome core.





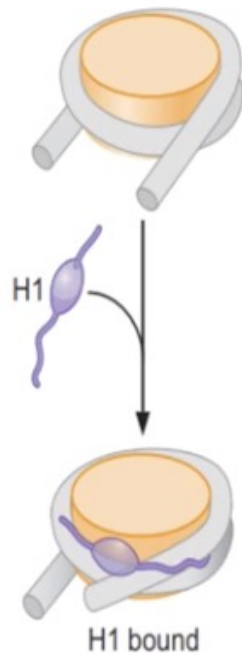
Histone interaction is with minor groove or phosphate backbone
Interaction type : hydrogen bond and ionic interaction

NUCLEOSOME (DNA + HISTONE PROTEINS)



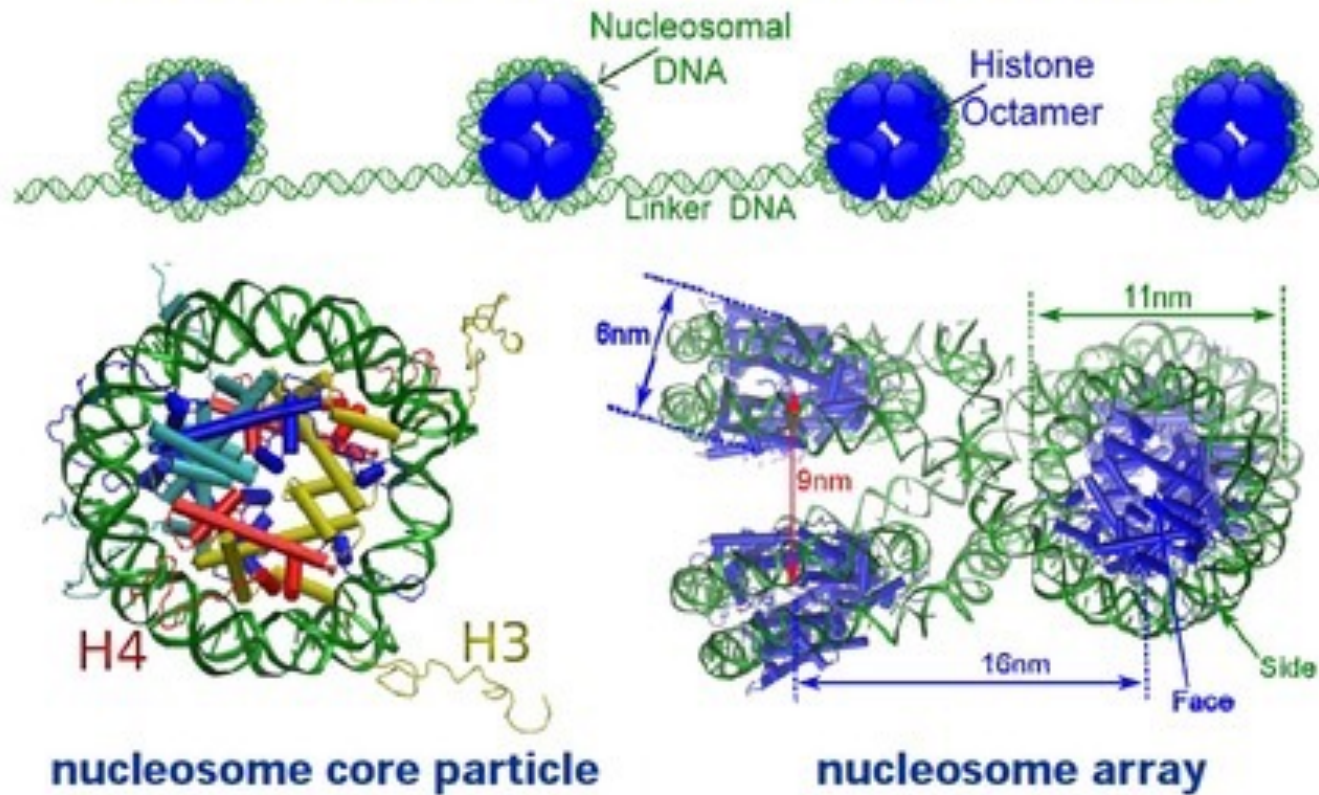
- Histone proteins form the octamer core around which DNA wraps itself and takes 1.67 turn.
- Histone proteins have positive charge as they are rich in basic amino acids (lysine & arginine)
- Each individual nucleosome core particle consists of a complex of eight histone proteins—two molecules each of histones H2A, H2B, H3, and H4—and double-stranded DNA that is 147 nucleotide pairs long.
- The DNA which flanks between the two nucleosomes is called as linker DNA that vary in length from a few nucleotide pairs up to about 80.

HISTONE H1 BINDS TO LINKER DNA BETWEEN NUCLEOSOMES

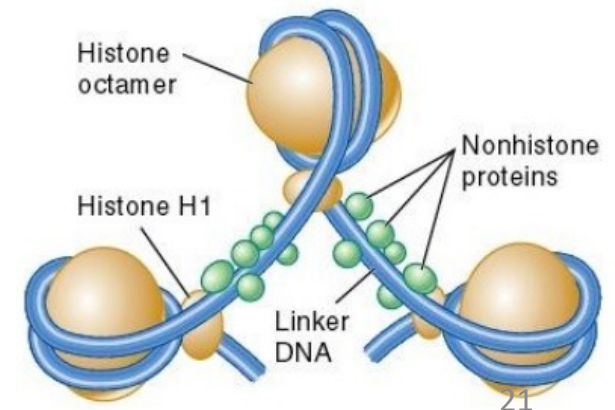


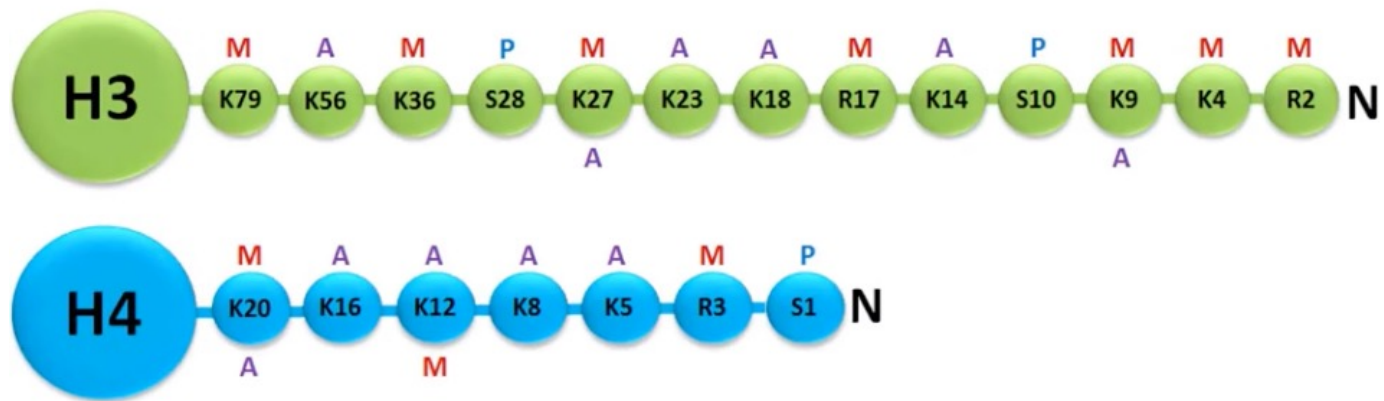
- Like the core histones, H1 is a small, positively charged protein and interacts with the linker DNA between nucleosomes.
- Addition of histone H1 to a nucleosome protects an additional 20 bp of DNA from micrococcal nuclease digestion
- H1 has the unusual property of binding two distinct regions of the DNA duplex
 1. Binds to the linker DNA at one end of the nucleosome
 2. Binds in the middle of the associated 147 bp.
- By bringing these two regions of DNA into close proximity, H1 binding increases the length of the DNA wrapped tightly around the histone octamer.

Chromatin Conformation and Dynamics



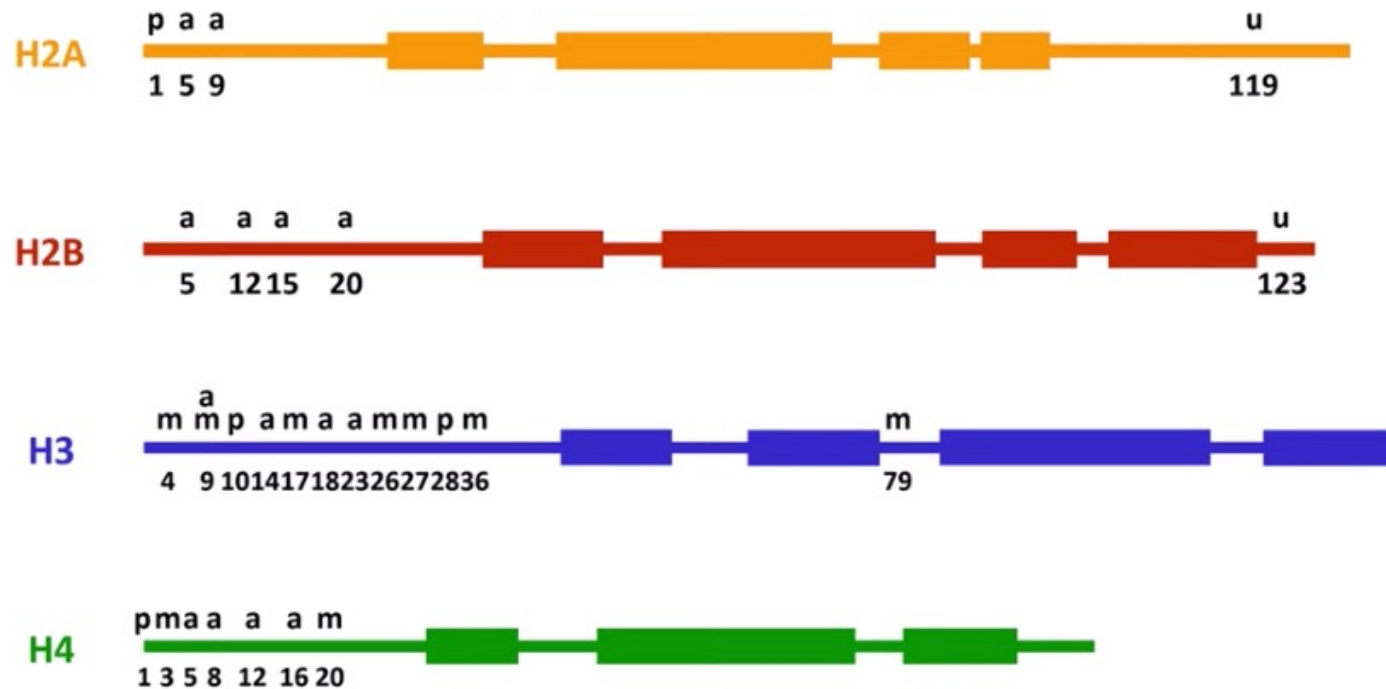
Nucleosomes

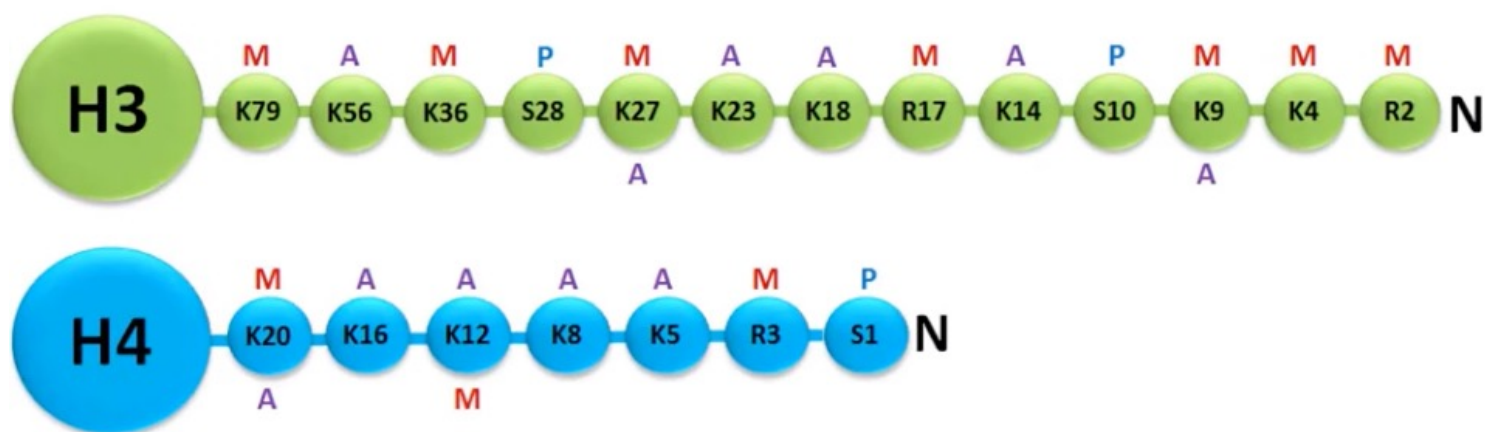




Post-translational modification of core histone 'tails'

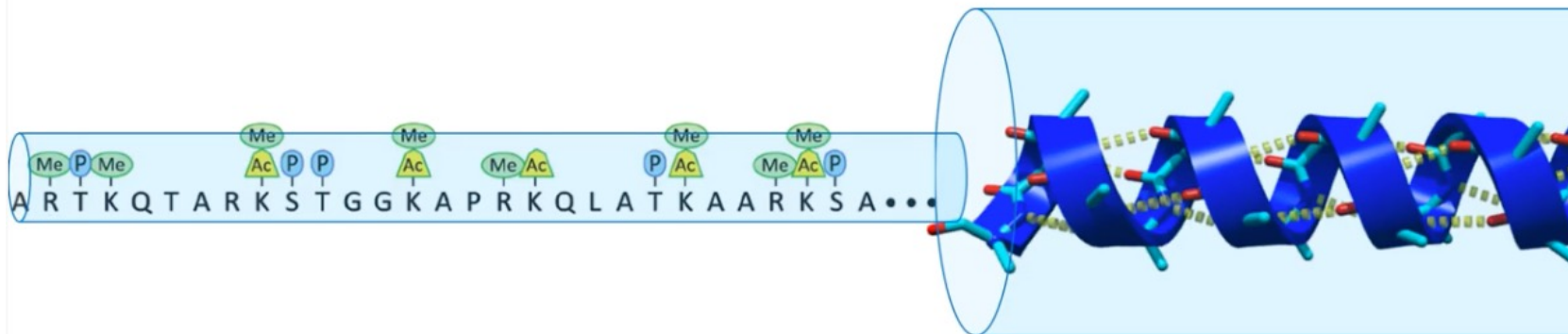
Acetylation, phosphorylation, methylation





中文名称	英文名称	三字母缩写	单字母符号
甘氨酸	Glycine	Gly	G
丙氨酸	Alanine	Ala	A
缬氨酸	Valine	Val	V
亮氨酸	Leucine	Leu	L
异亮氨酸	Isoleucine	Ile	I
脯氨酸	Proline	Pro	P
苯丙氨酸	Phenylalanine	Phe	F
酪氨酸	Tyrosine	Tyr	Y
色氨酸	Tryptophan	Trp	W
丝氨酸	Serine	Ser	S
苏氨酸	Threonine	Thr	T
半胱氨酸	Cystine	Cys	C
蛋氨酸	Methionine	Met	M
天冬酰胺	Asparagine	Asn	N
谷氨酰胺	Glutamine	Gln	Q
天冬氨酸	Asparticacid	Asp	D
谷氨酸	Glutamicacid	Glu	E
赖氨酸	Lysine	Lys	K
精氨酸	Arginine	Arg	R
组氨酸	Histidine	His	H

How these modifications takes place ?
What is the consequences of these modification?



Histone Tail

= Acetylation	= Phosphorylation
= Methylation	= Ubiquitination

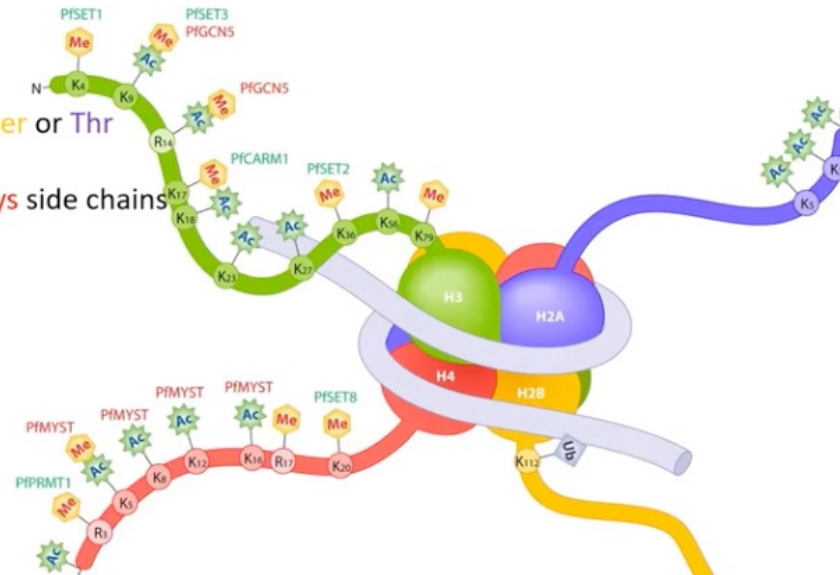
What types of Modification can take place in a histone?

Acetylation : Addition of **acetyl group** on **Lys** or **Arg**

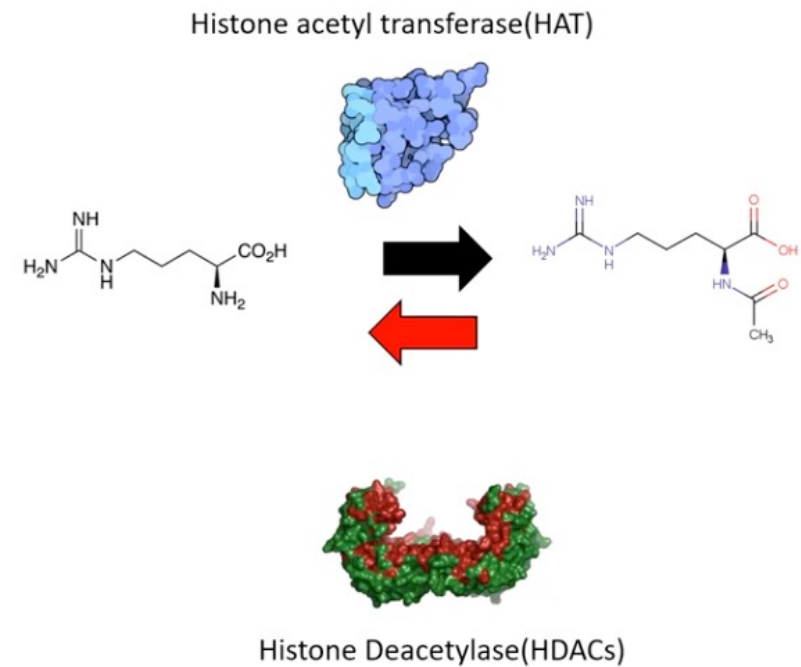
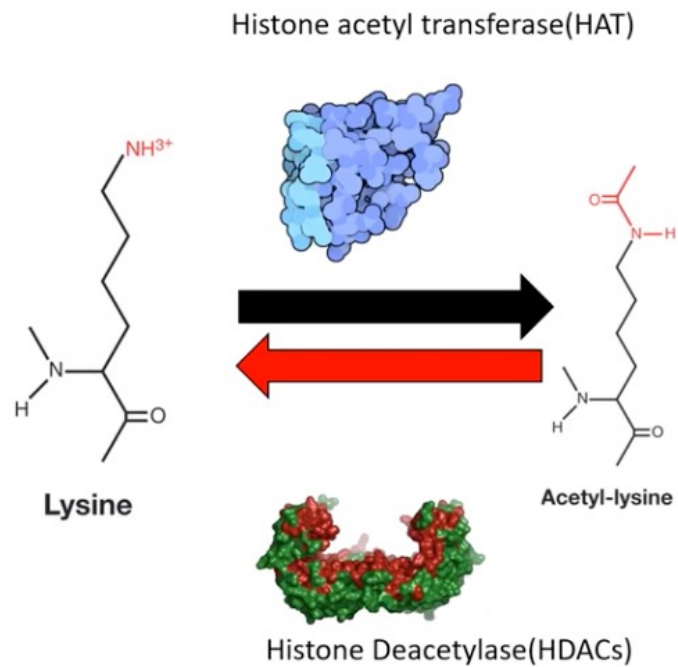
Methylation: Addition of **methyl group** on **Lys** or **Arg**

Phosphorylation: Addition of **phosphate group** on **Ser** or **Thr**

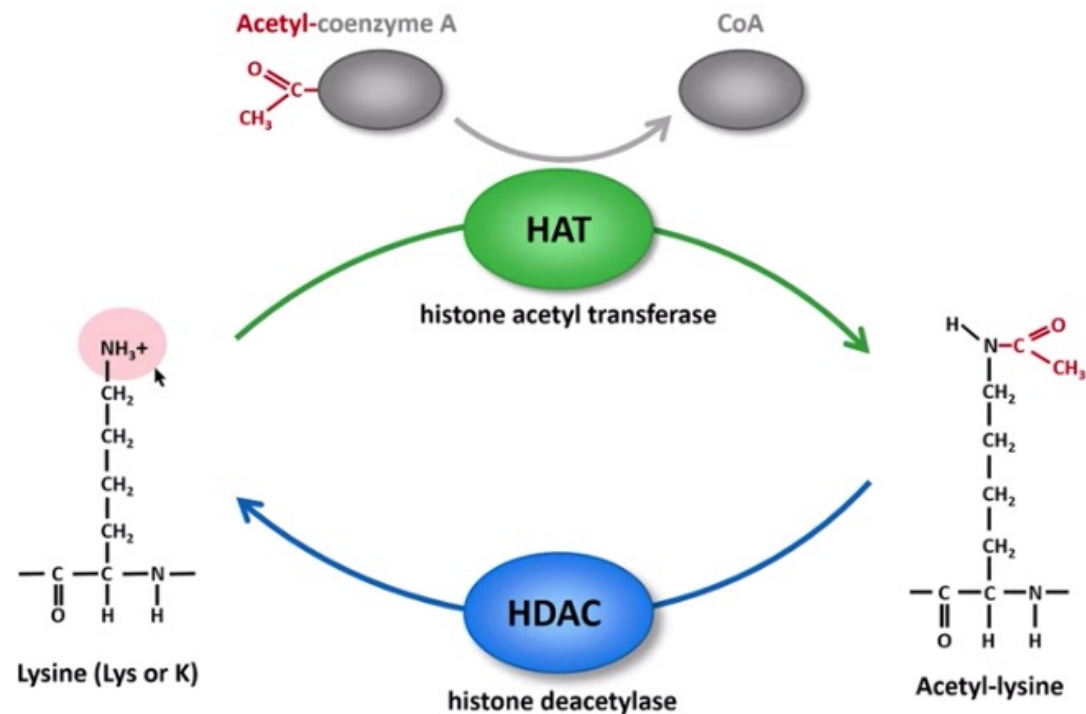
Ubiquitination: Addition of **ubiquitin group** on **Lys** side chains



Acetylation : Addition of acetyl group on **Lys** or **Arg**



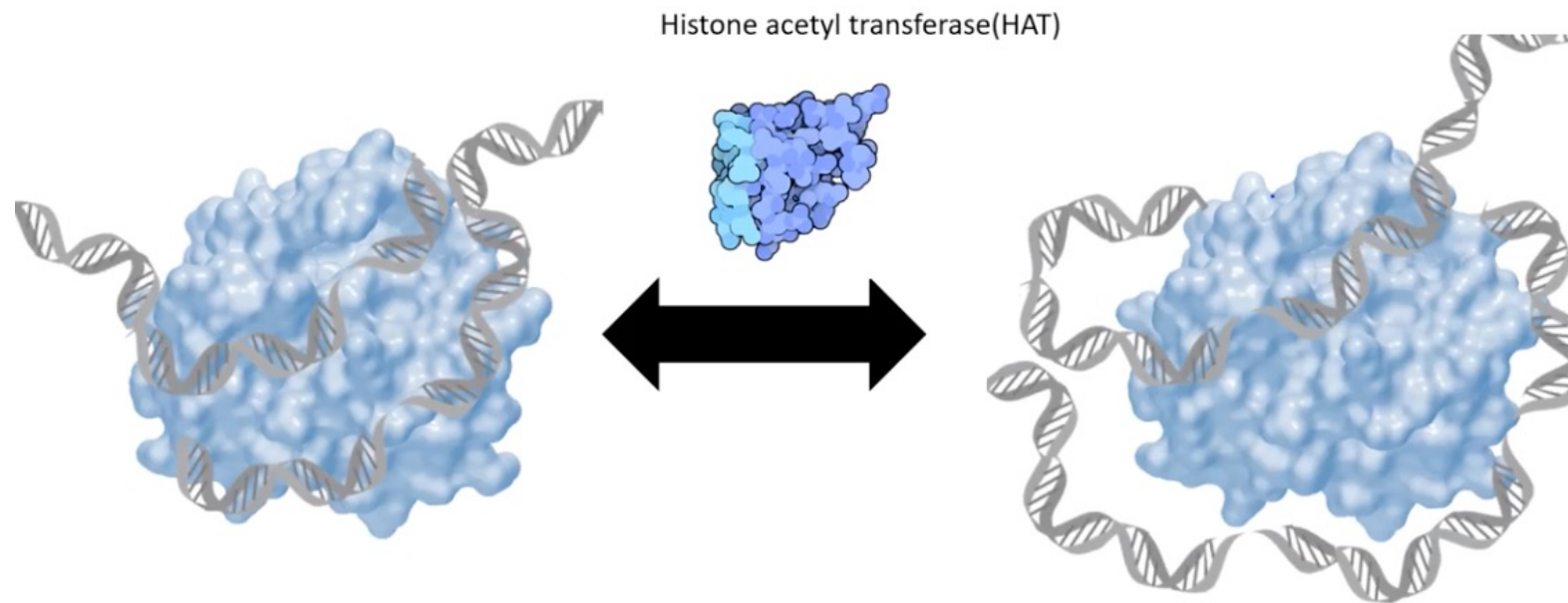
Histone acetylation occurs at lysine residues



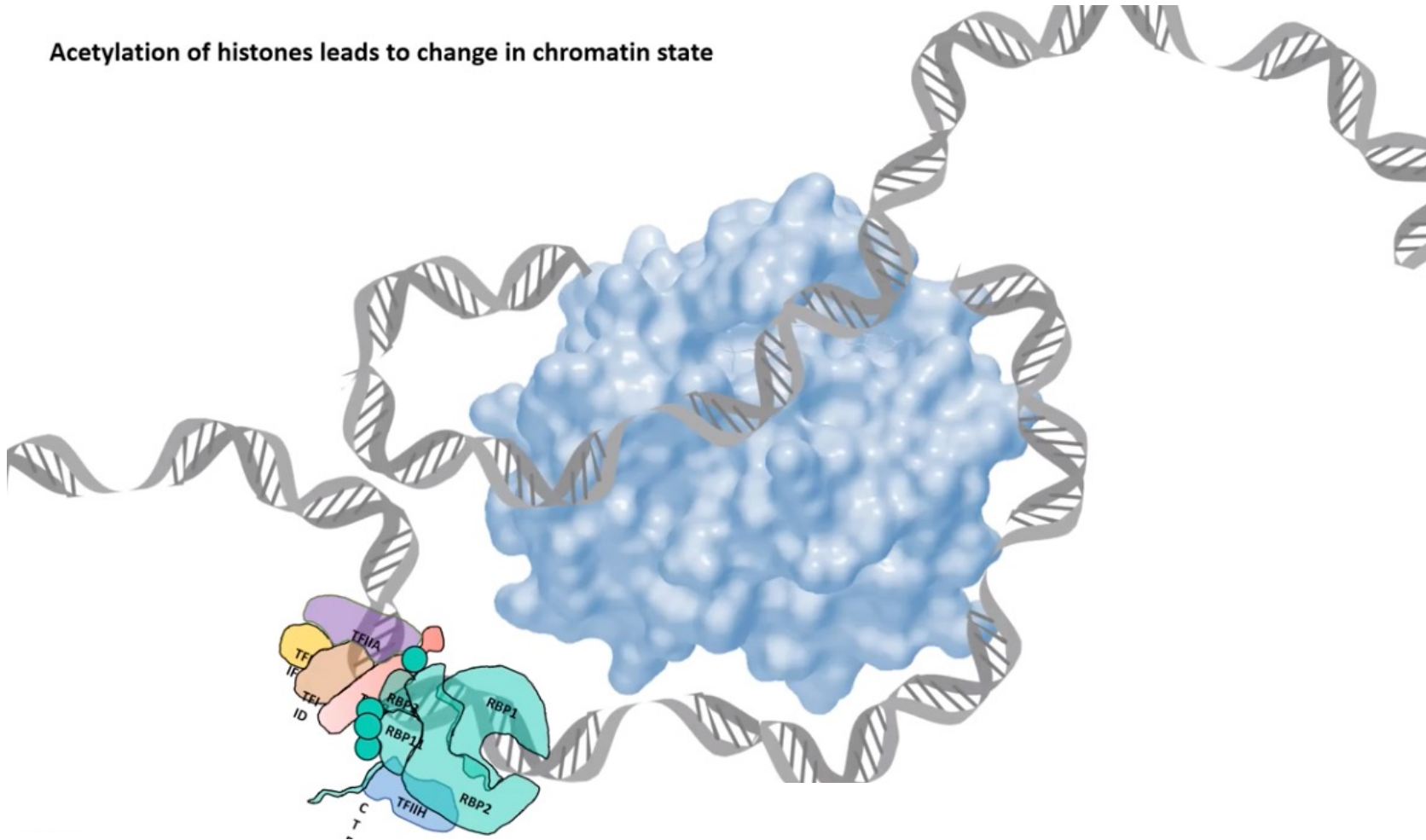
Acetylation neutralises the positive lysine residue

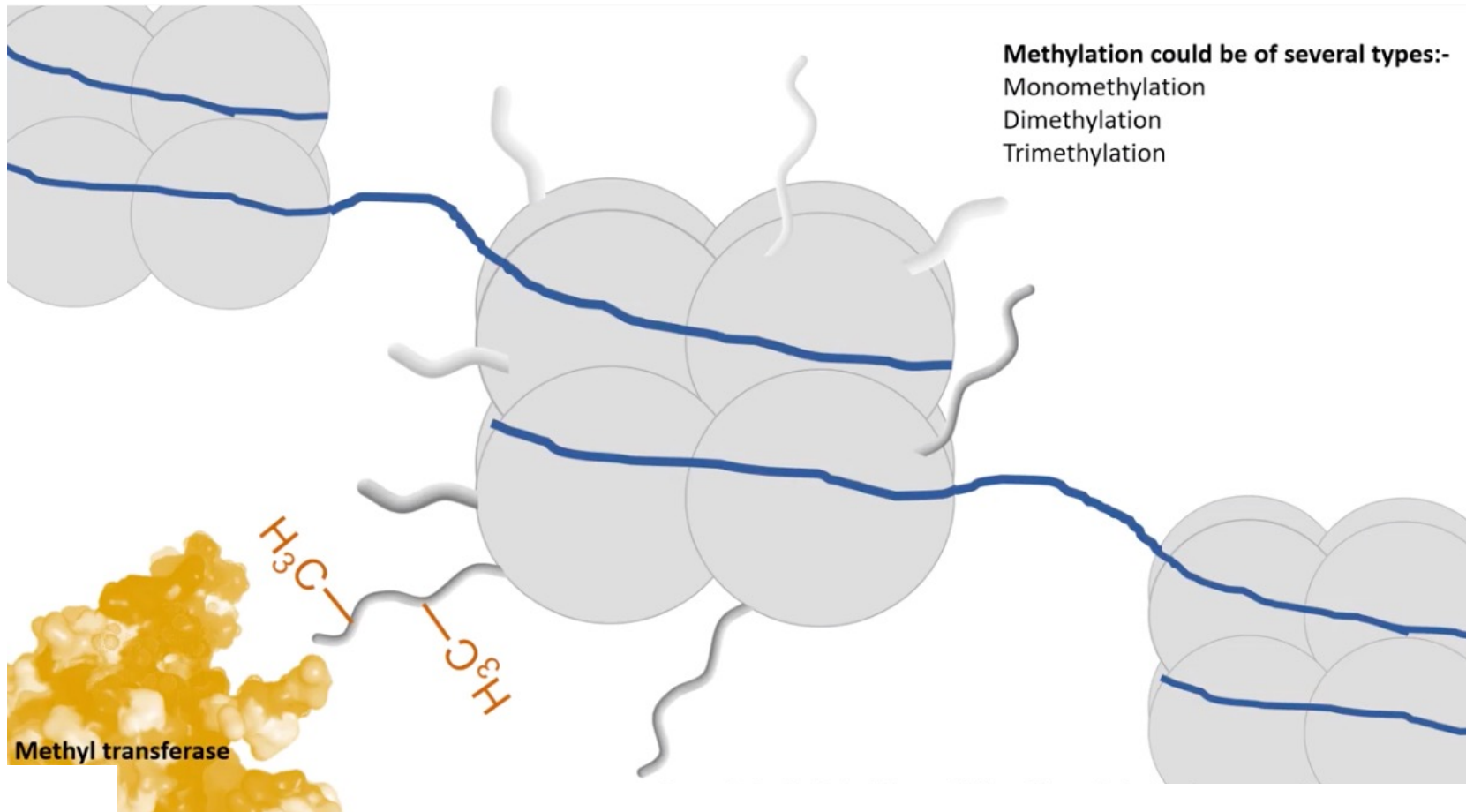
This reduces histone tail : DNA interaction, promotes chromatin unfolding

How does close chromatin to open chromatin switch happen?

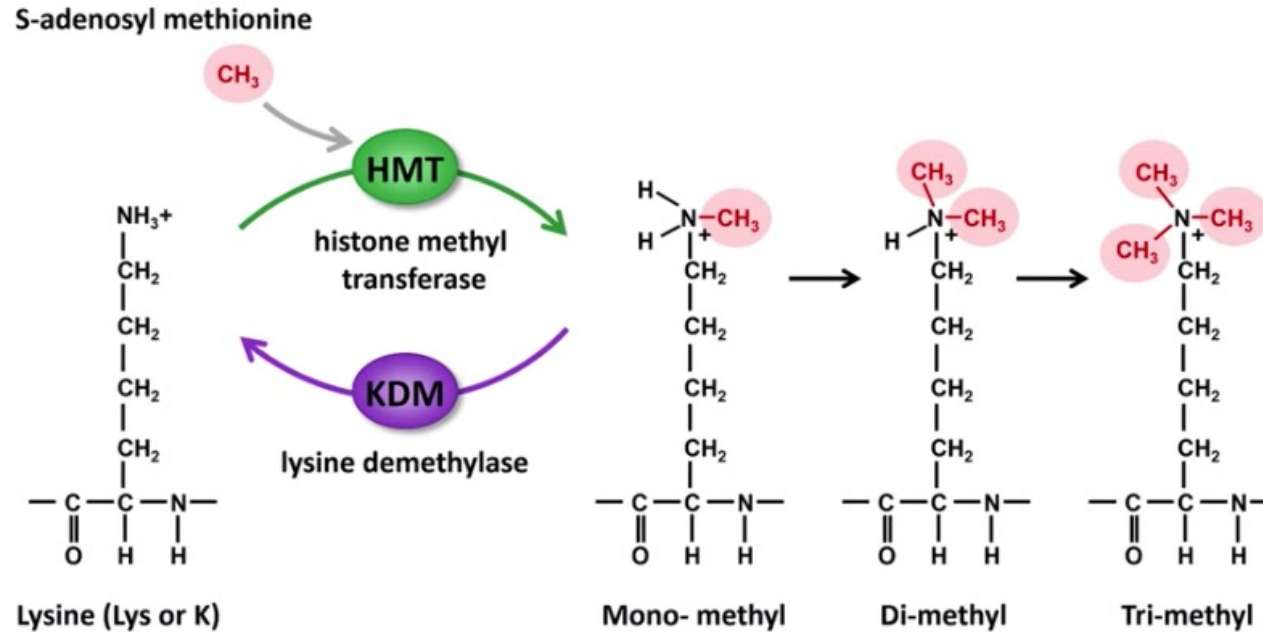


Acetylation of histones leads to change in chromatin state





Histone methylation

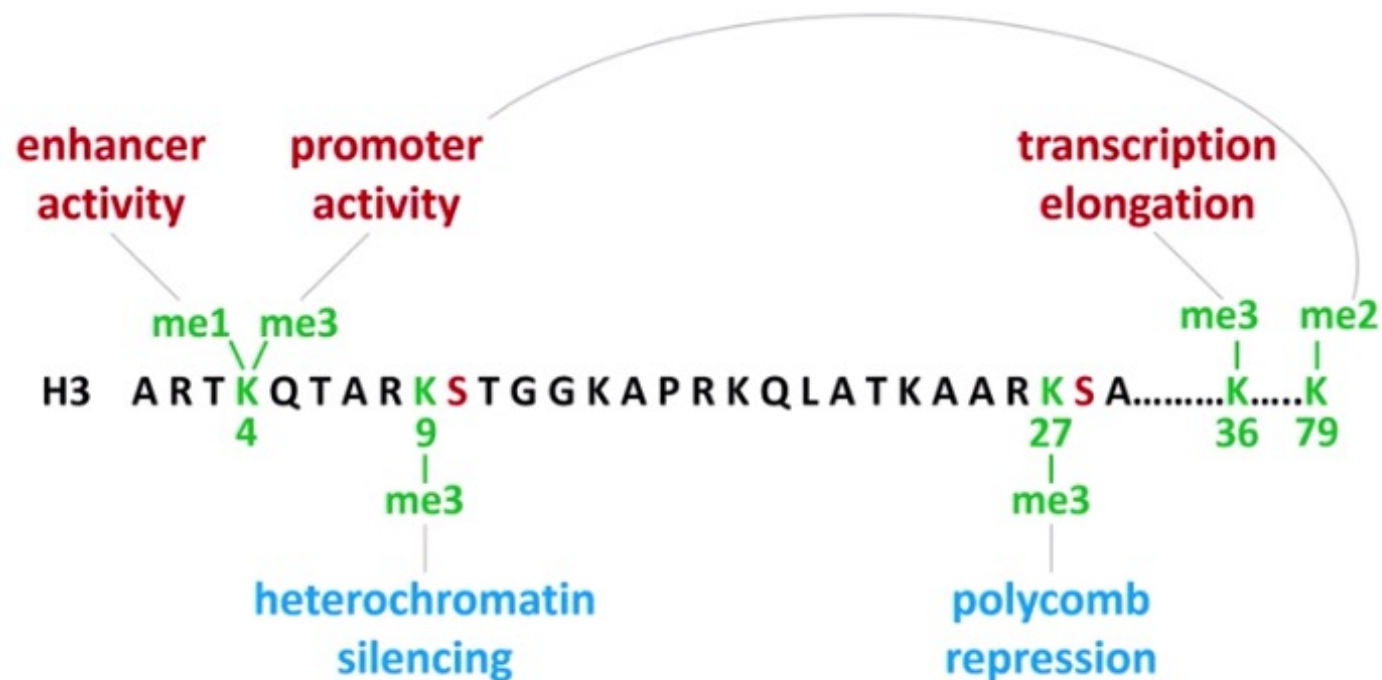


Only deacetylated lysines can be methylated

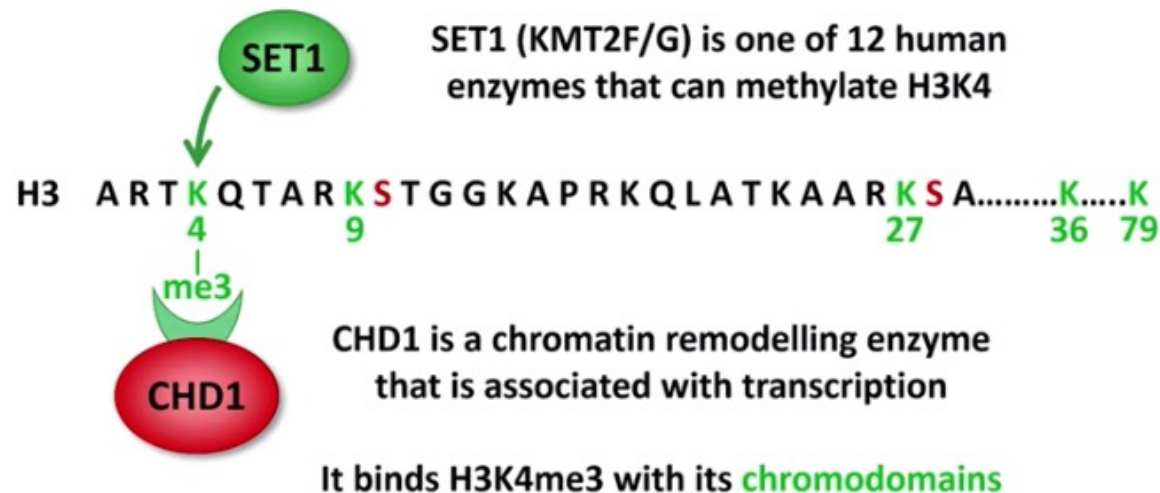
Lysine residues can be methylated multiple times

Different methyl states often have different functional consequences

Lysine methylation is associated with transcription **activation** or **repression**



Lysine methylation creates binding sites for **co-activators**

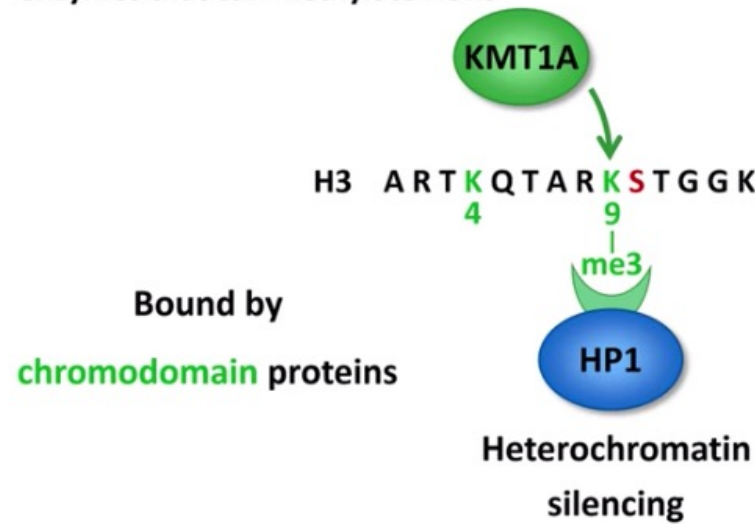


Co-activators with PHD fingers or Tudor domains also bind methylated H3K4

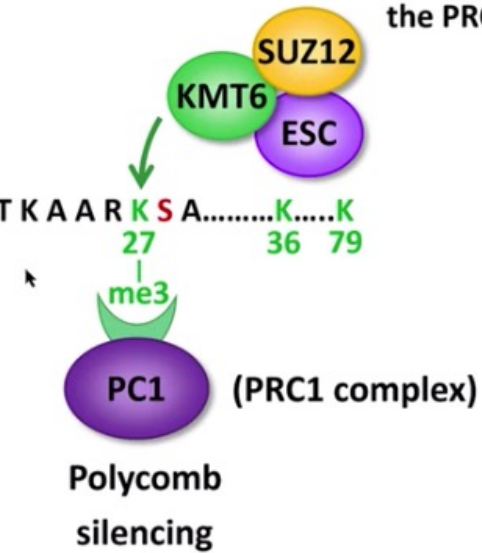
Some **co-repressors** like CHD3/4 in NuRD complex bind unmethylated H3K4

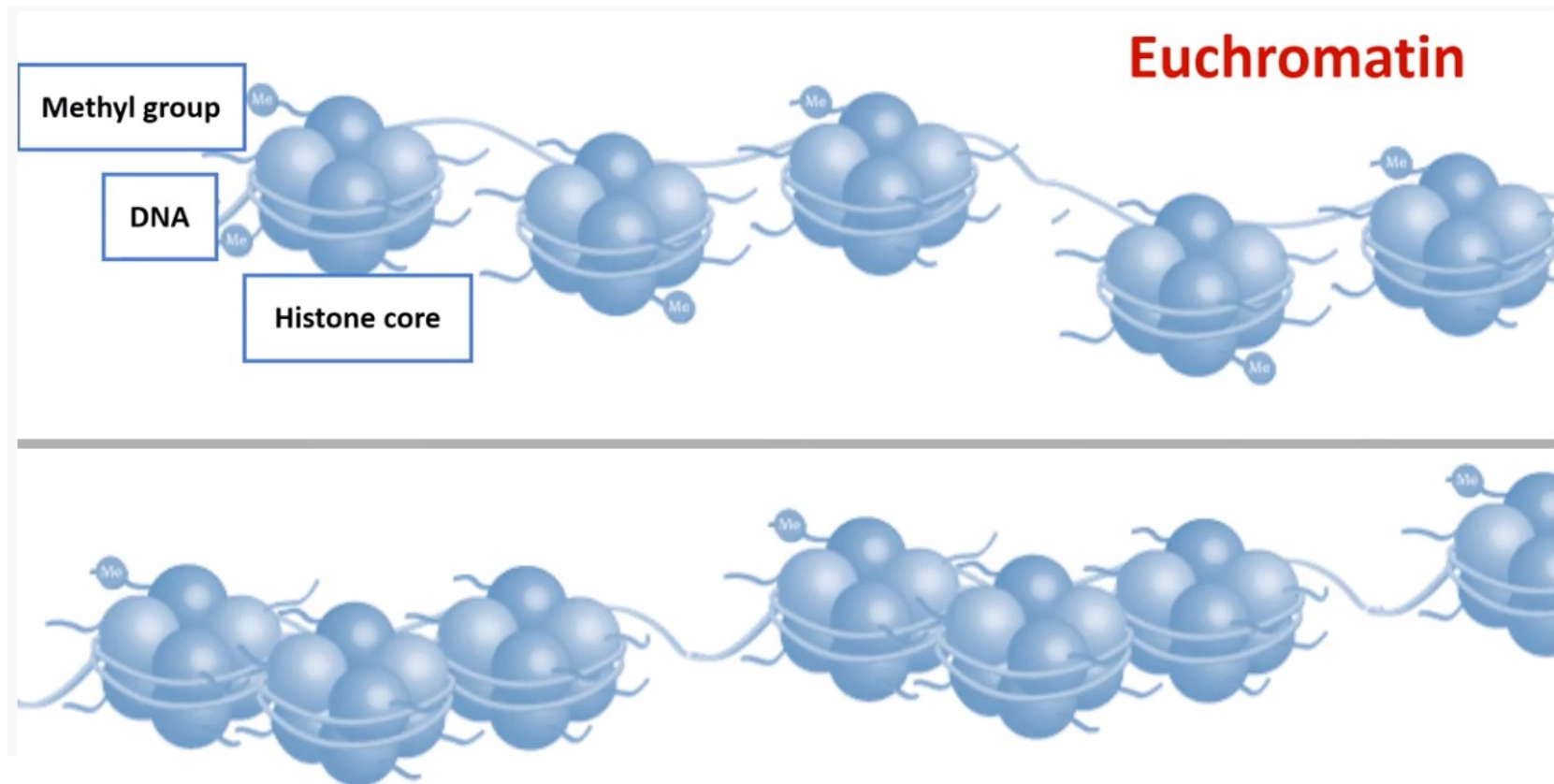
Lysine methylation creates binding sites for co-repressors

KMT1A (Suv39h) is one of 7 human enzymes that can methylate H3K9

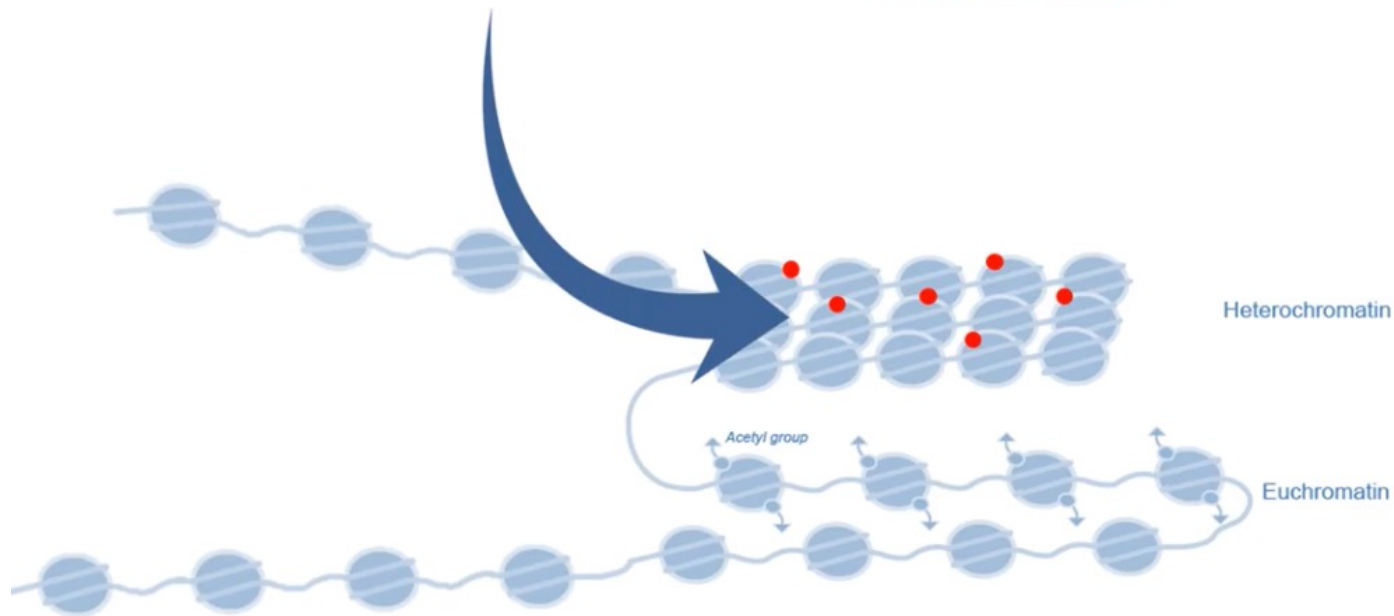


KMT6 (EZH2) is in the PRC2 complex

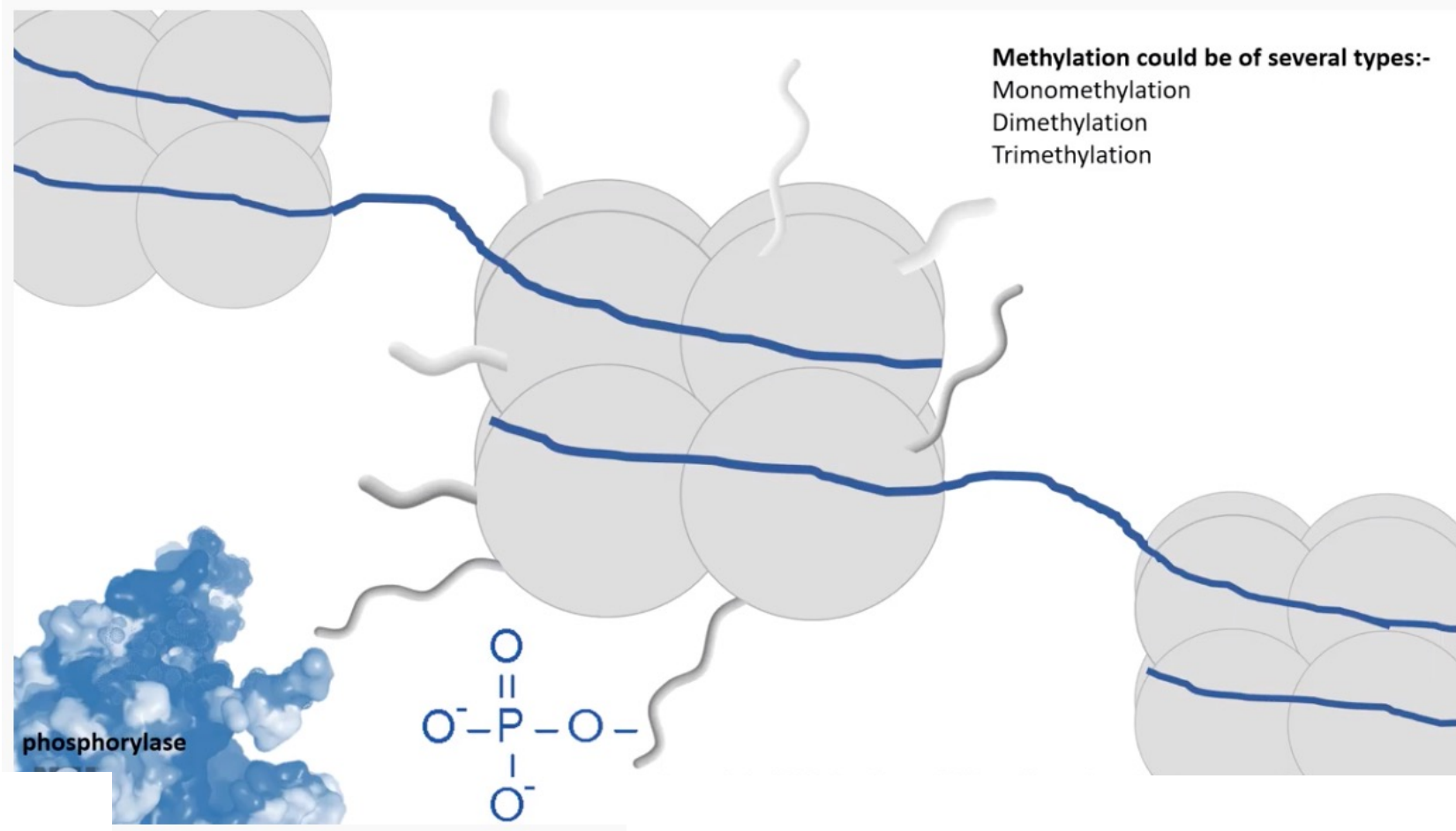


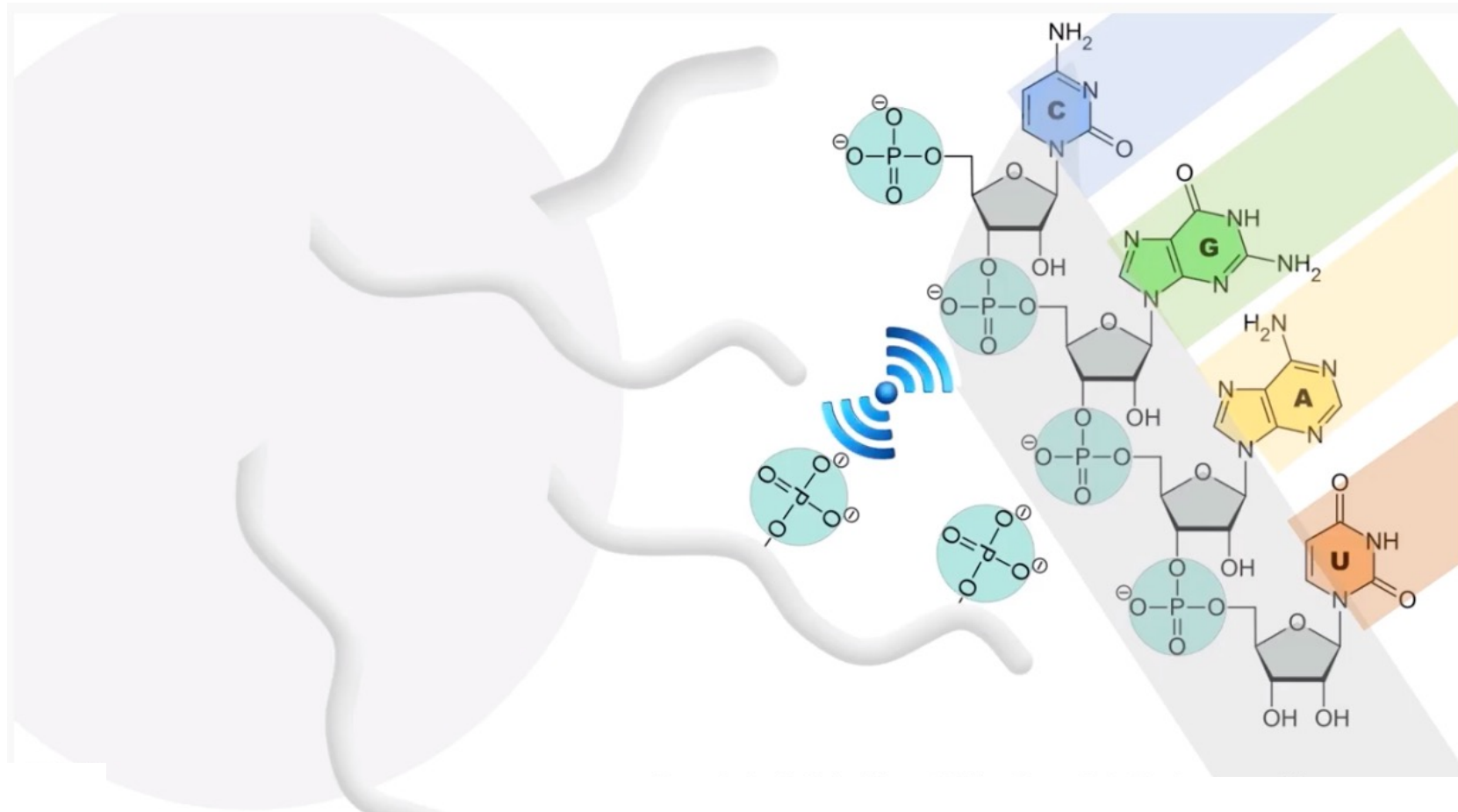


H3K9me3 and H3K27me3 are associated with **heterochromatin** formation

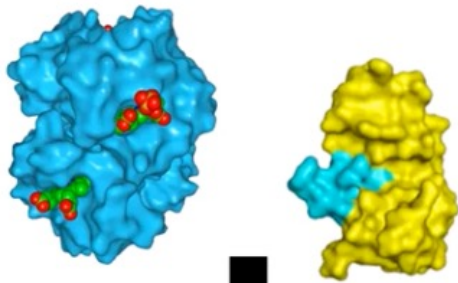


H3K4me2 and H3K4me3 are associated with euchromatin

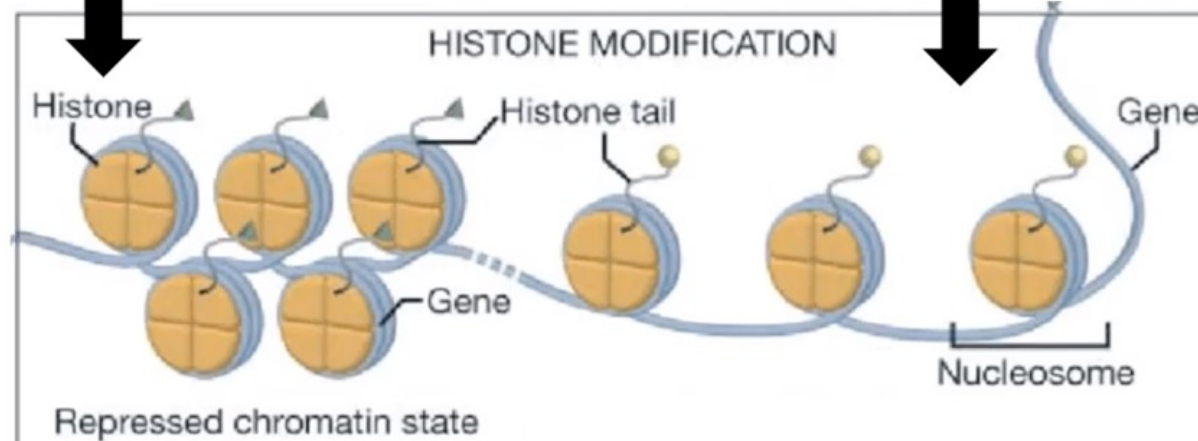
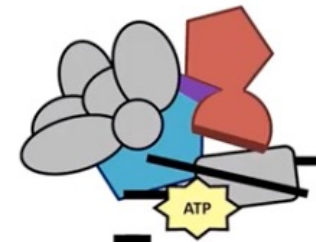


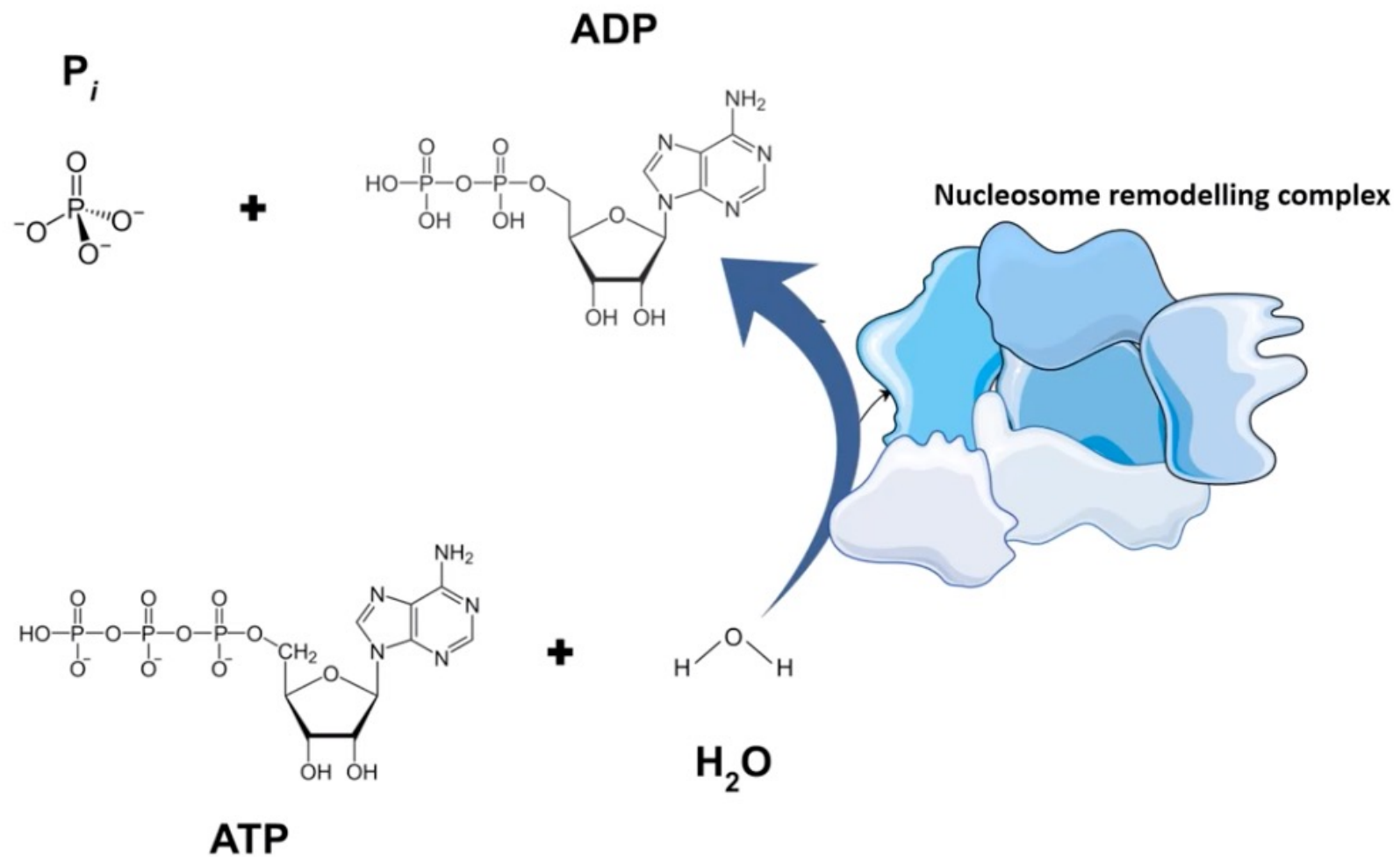


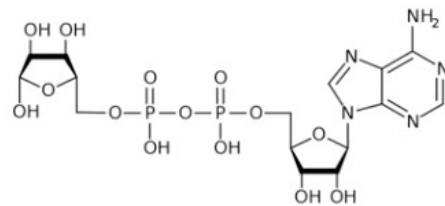
Histone modifiers



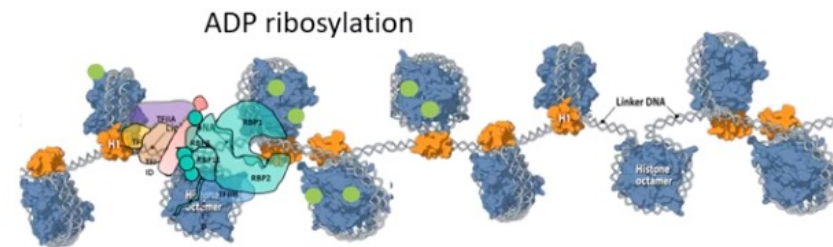
Nucleosome remodelling complex





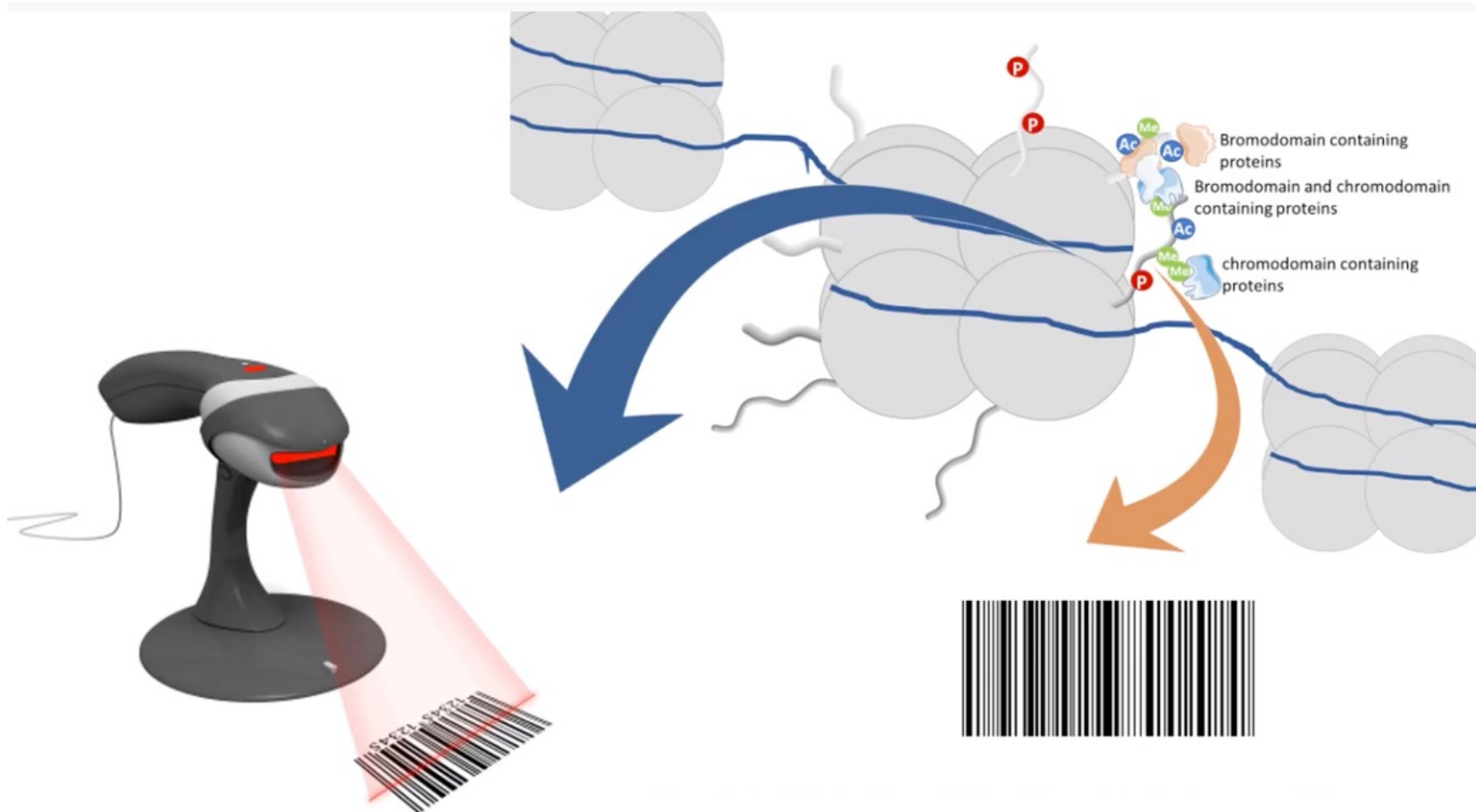


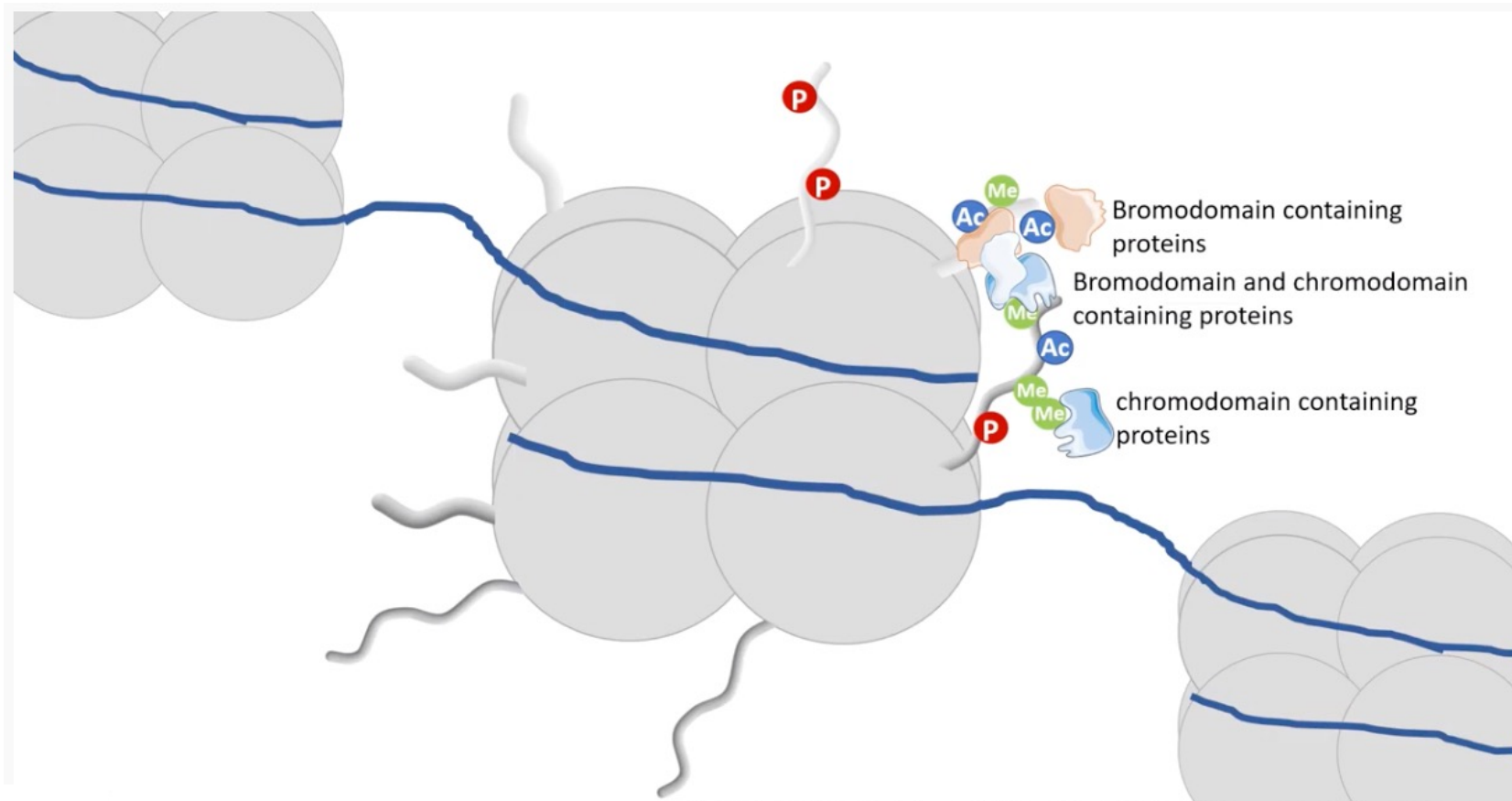
ADP ribosylation imparts a negative charge to its modified residues, suggesting that this modification creates charge repulsion between histones and DNA



Hsp gene associated nucleosomes

During heat shock, poly-ADP ribosylation over heat-shock gene bodies is greatly increased and is linked to nucleosome eviction



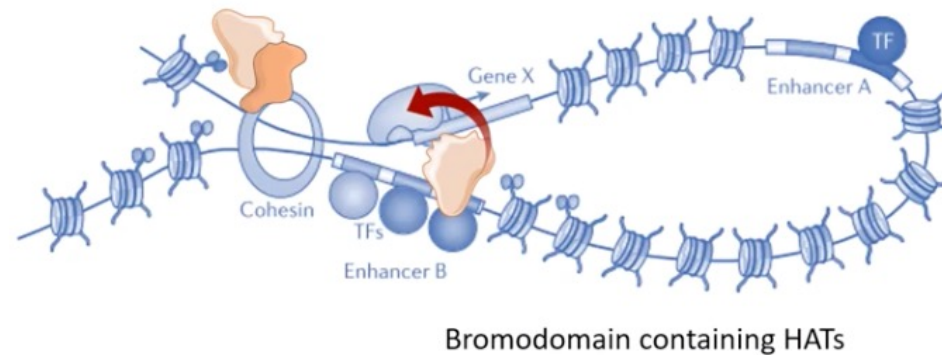


Chromodomain and bromodomain containing proteins could be

1) Histone modifiers



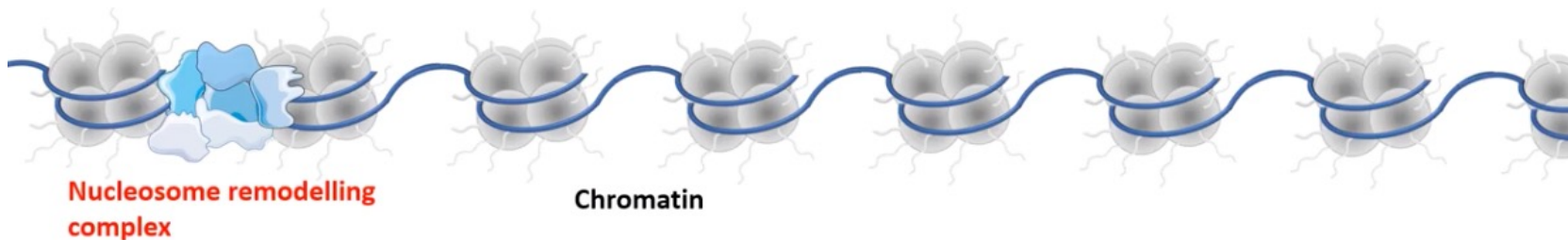
2) Transcriptional regulators



3) Transcription factors

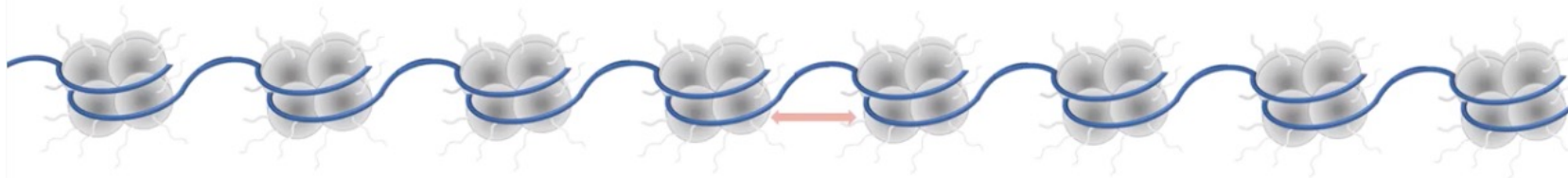
4) Chromatin compaction regulatory proteins

What kind of changes can be evoked by nucleosome remodelling complex at the chromatin level?

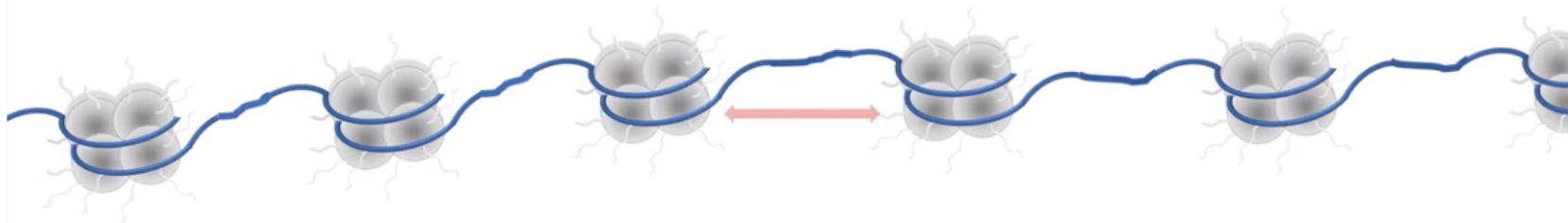


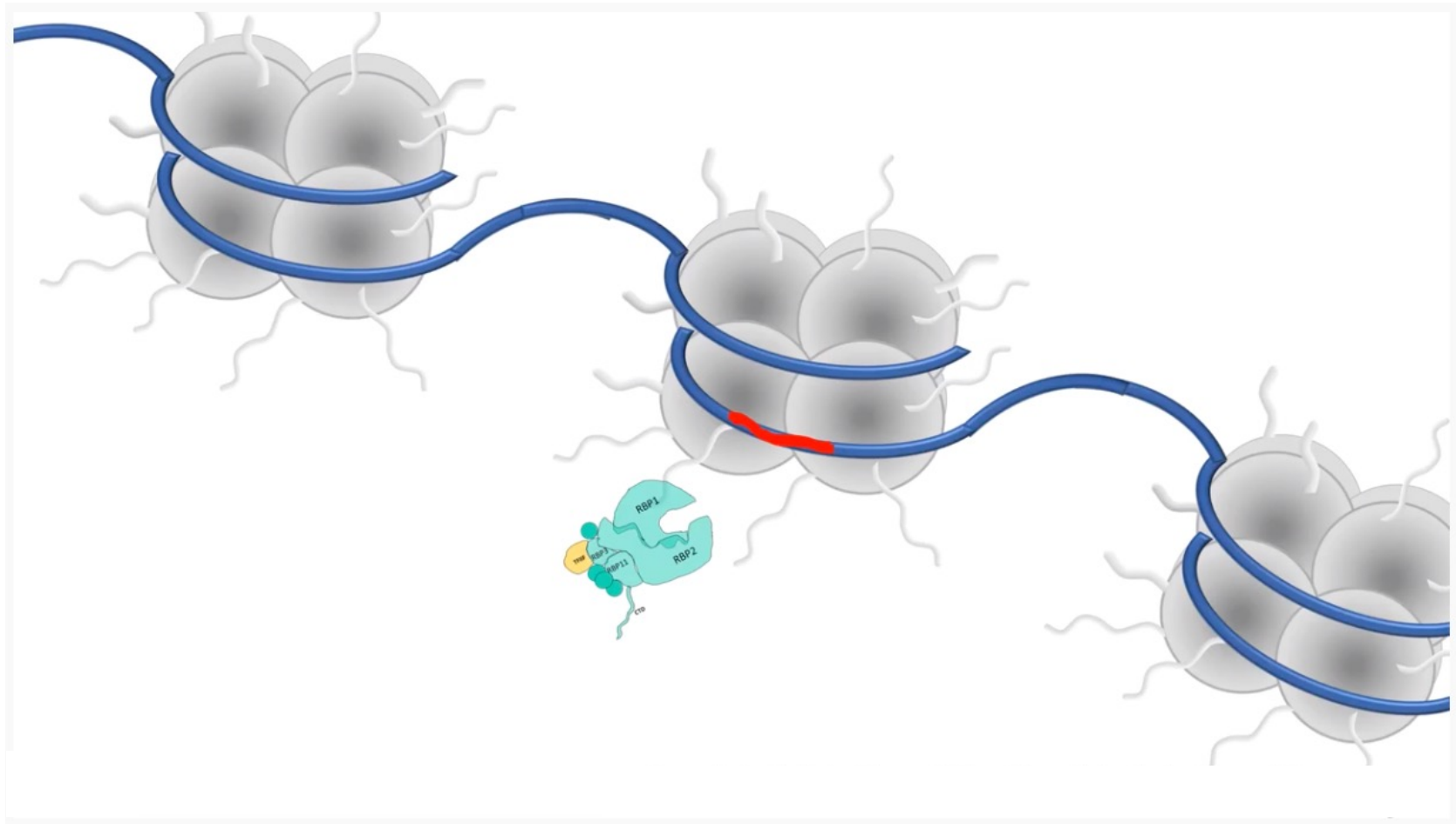
ATP-dependent chromatin-remodeling complexes regulate gene expression by either **moving, ejecting or restructuring nucleosomes**.

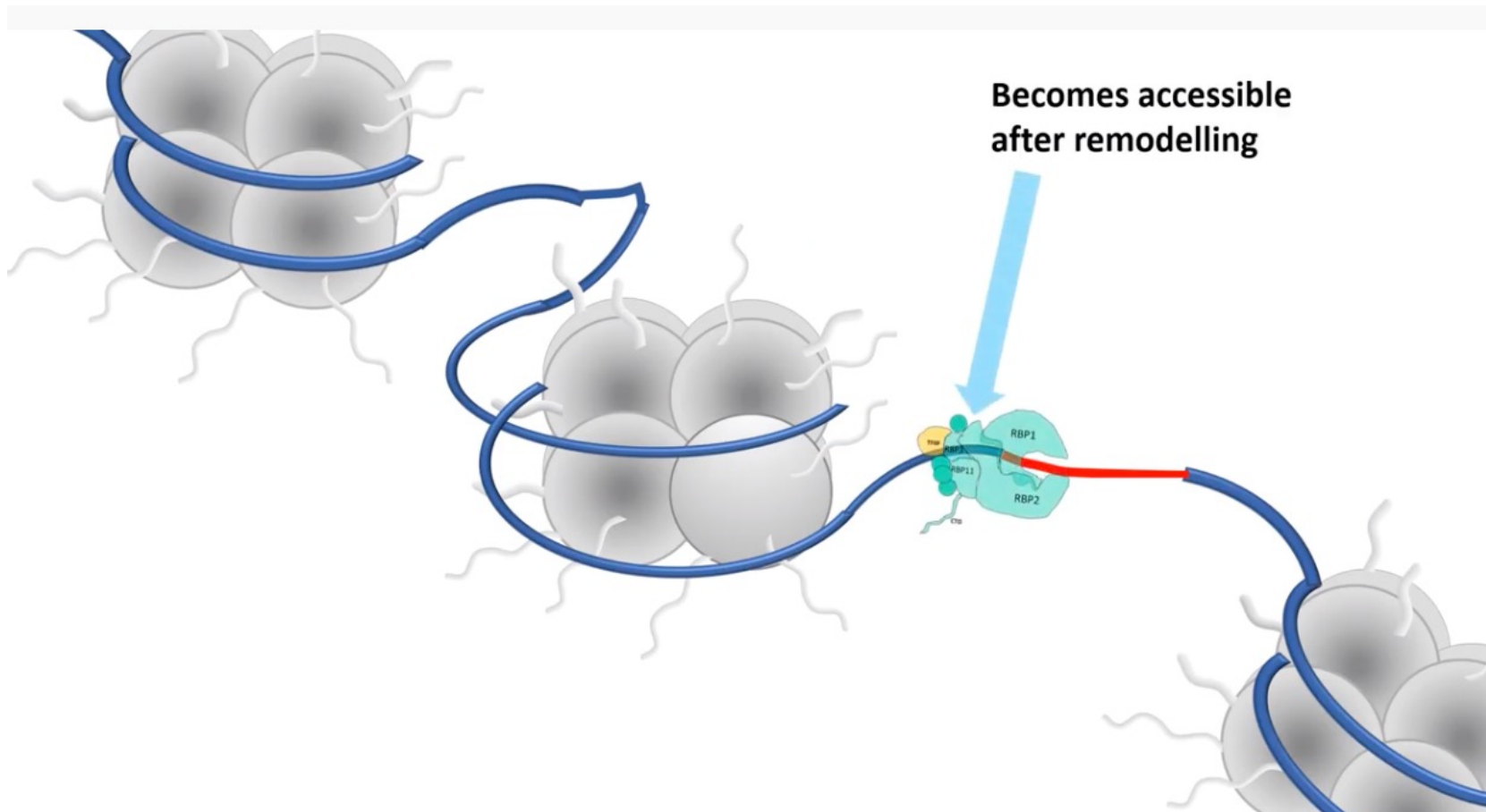
Before remodelling by nucleosome remodelling complex

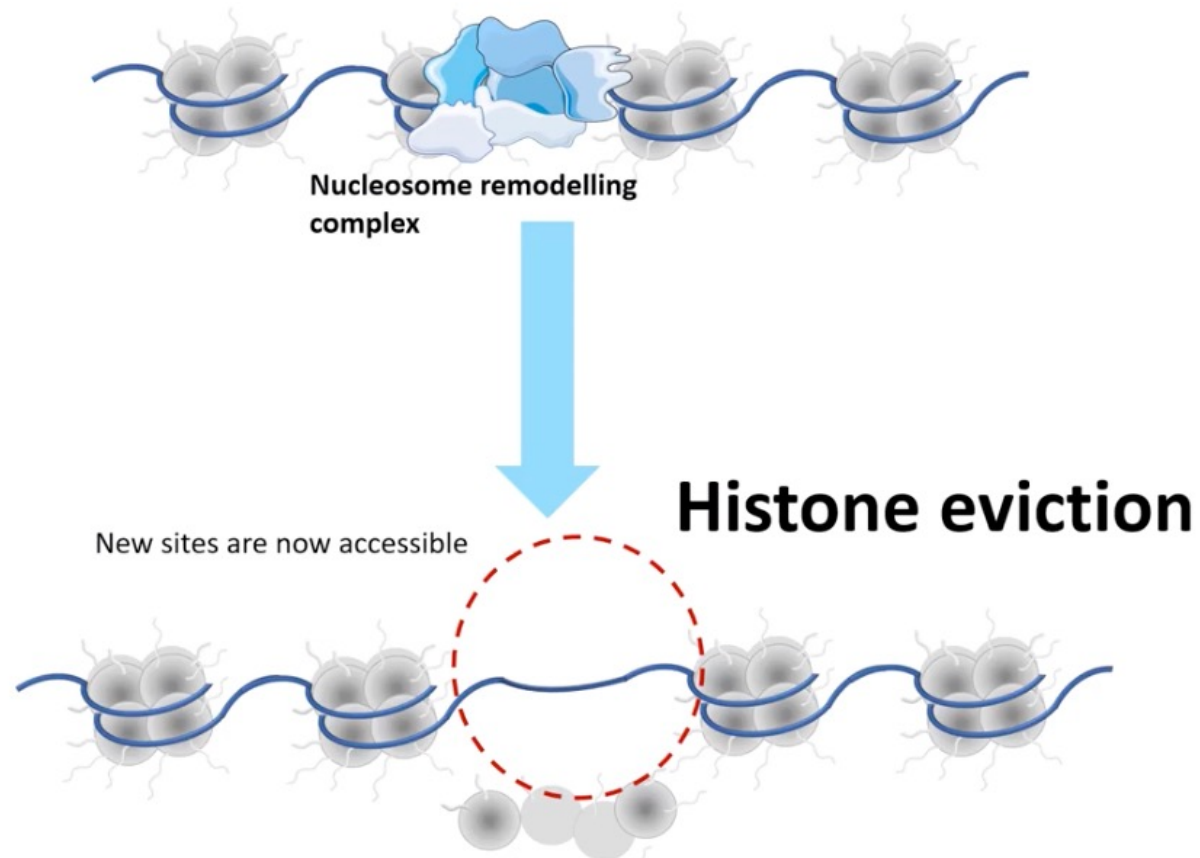


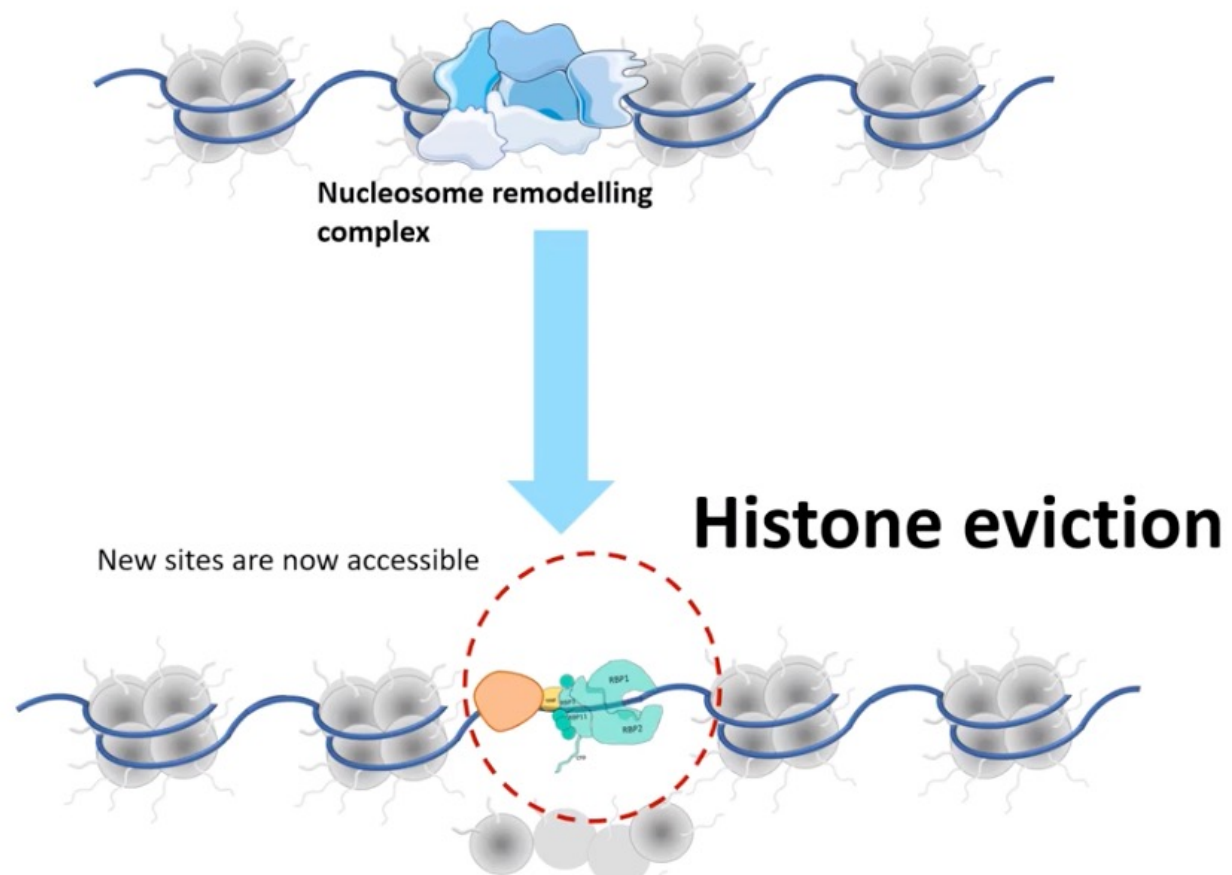
After remodelling by nucleosome remodelling complex

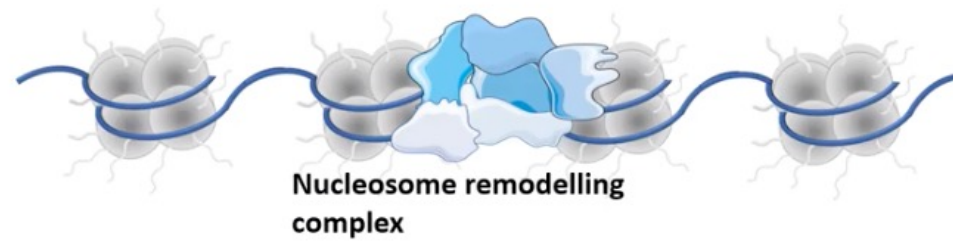




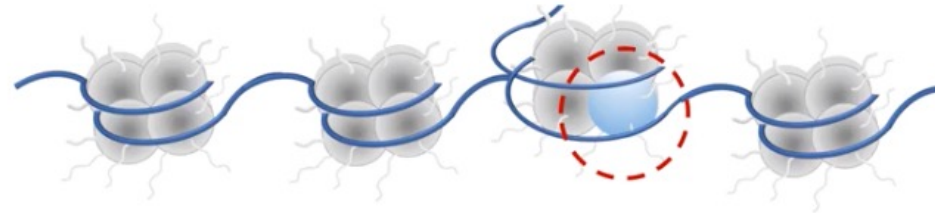


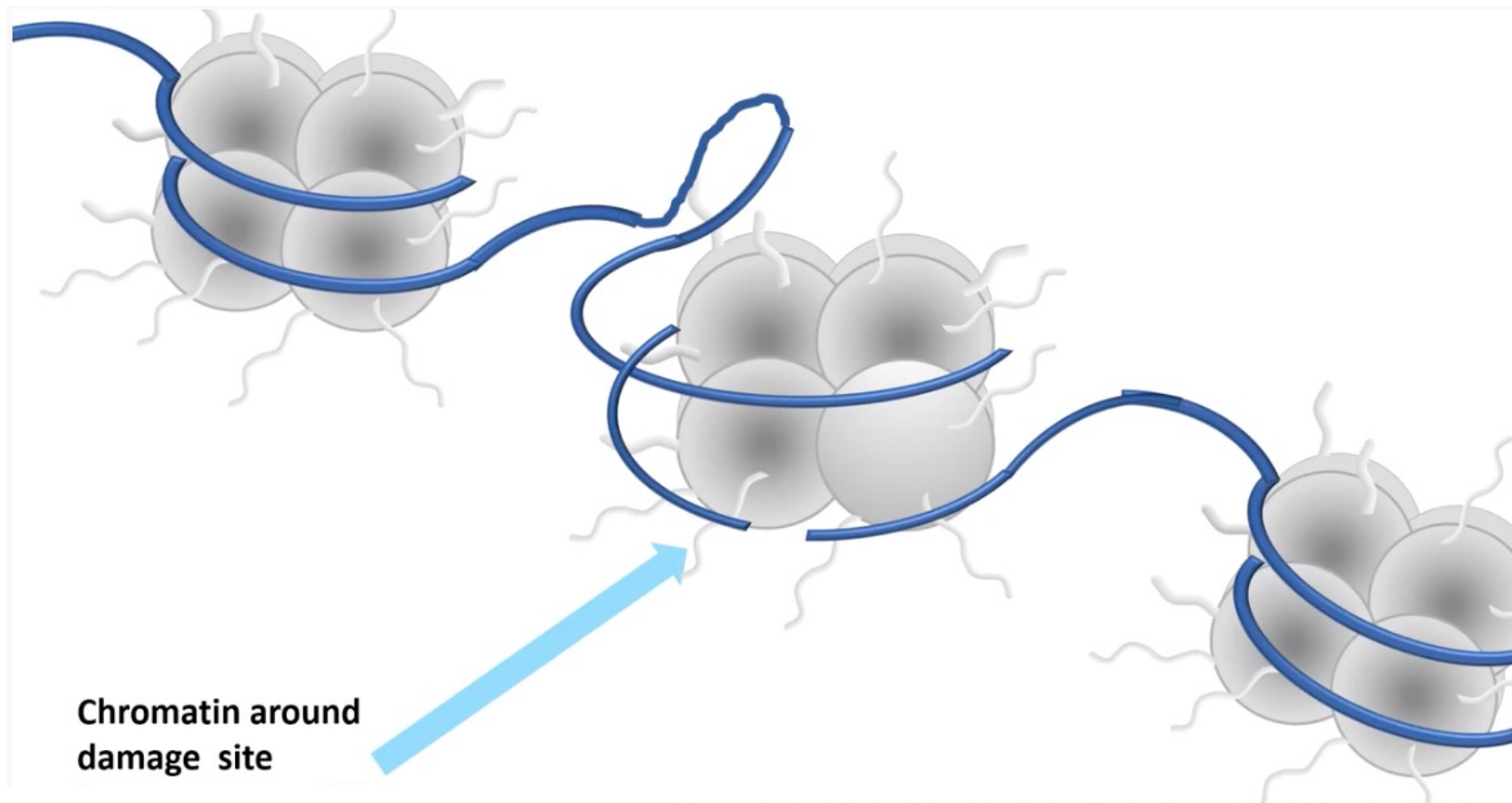


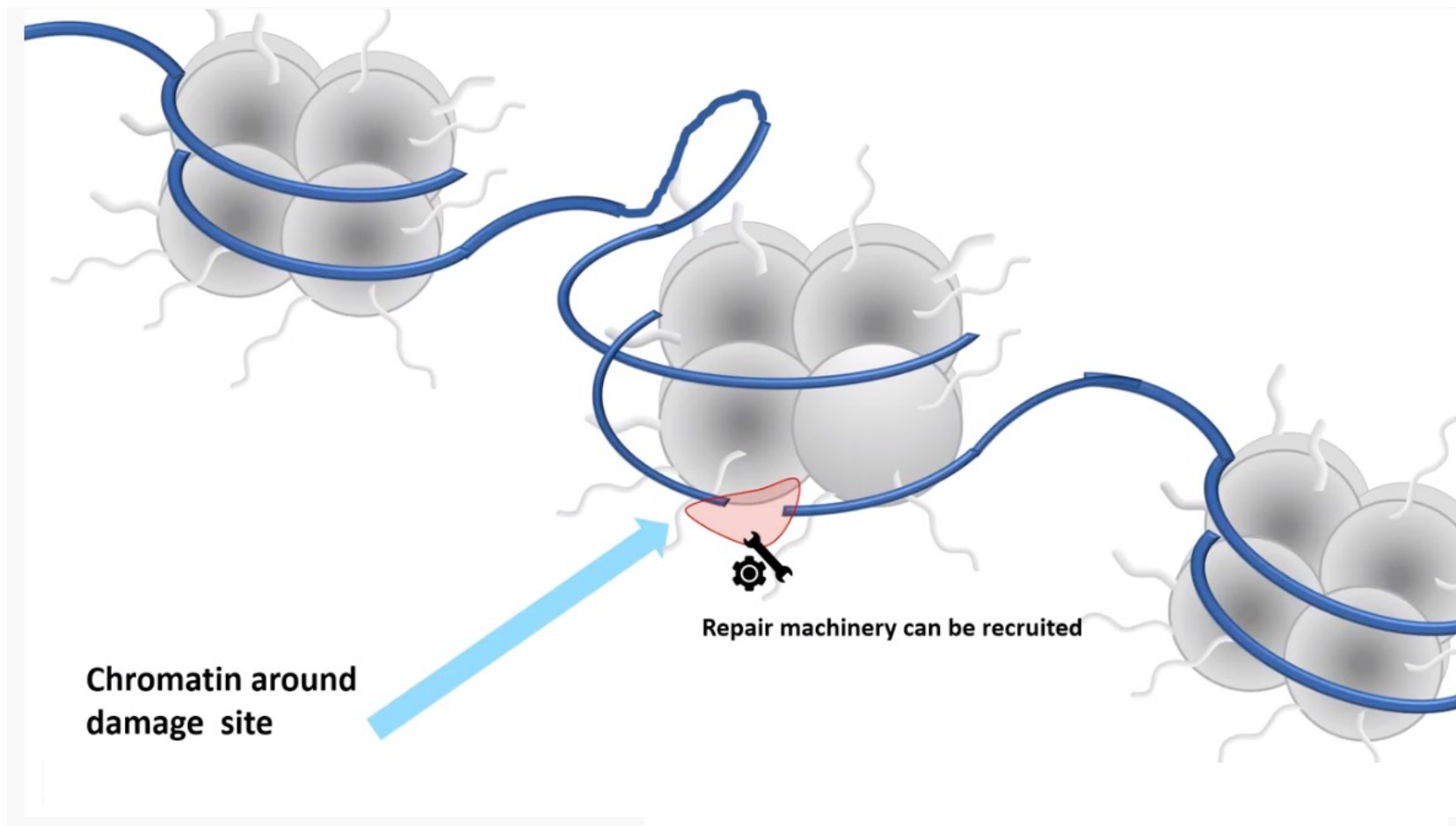


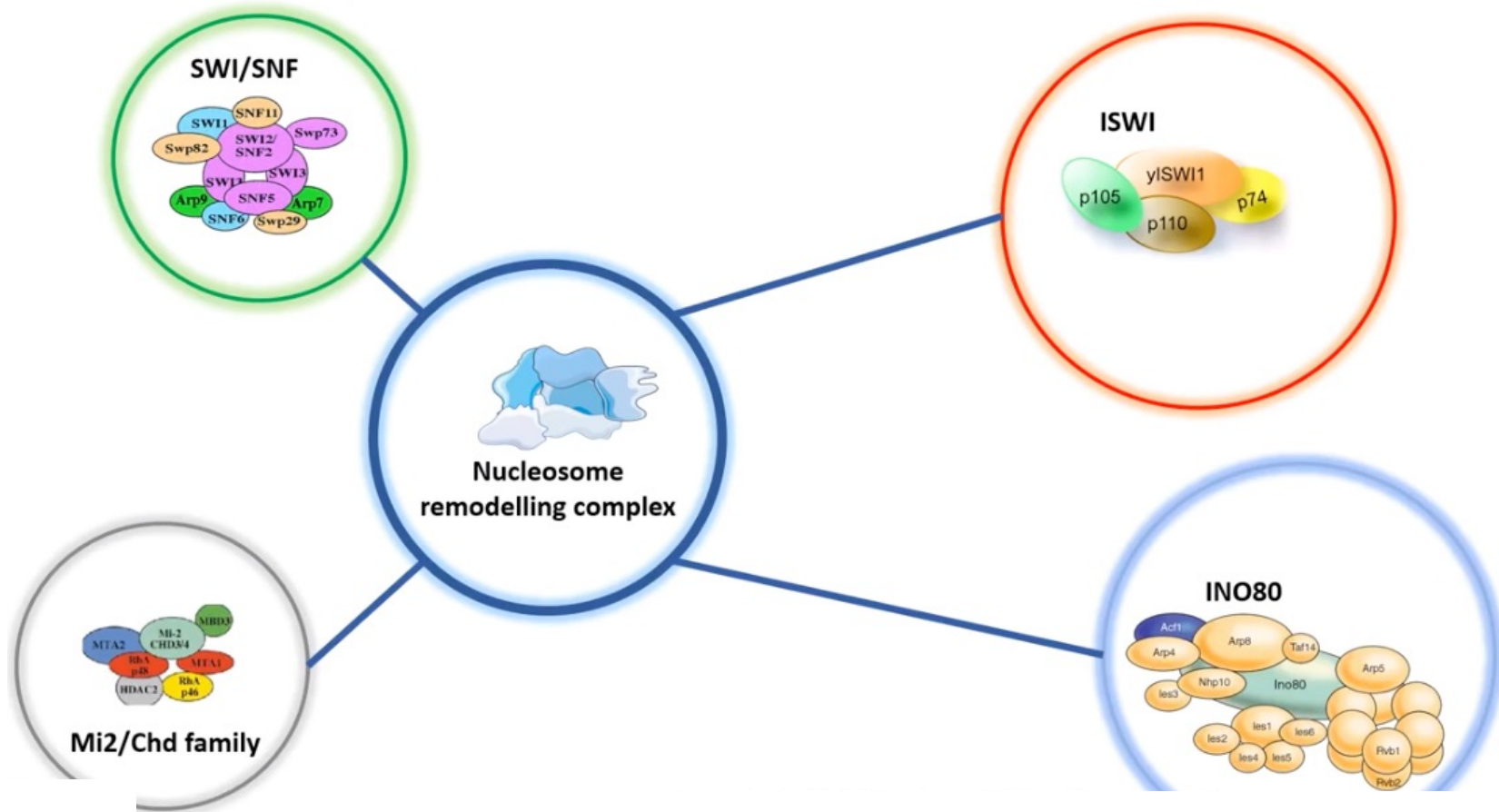


Histone replacement

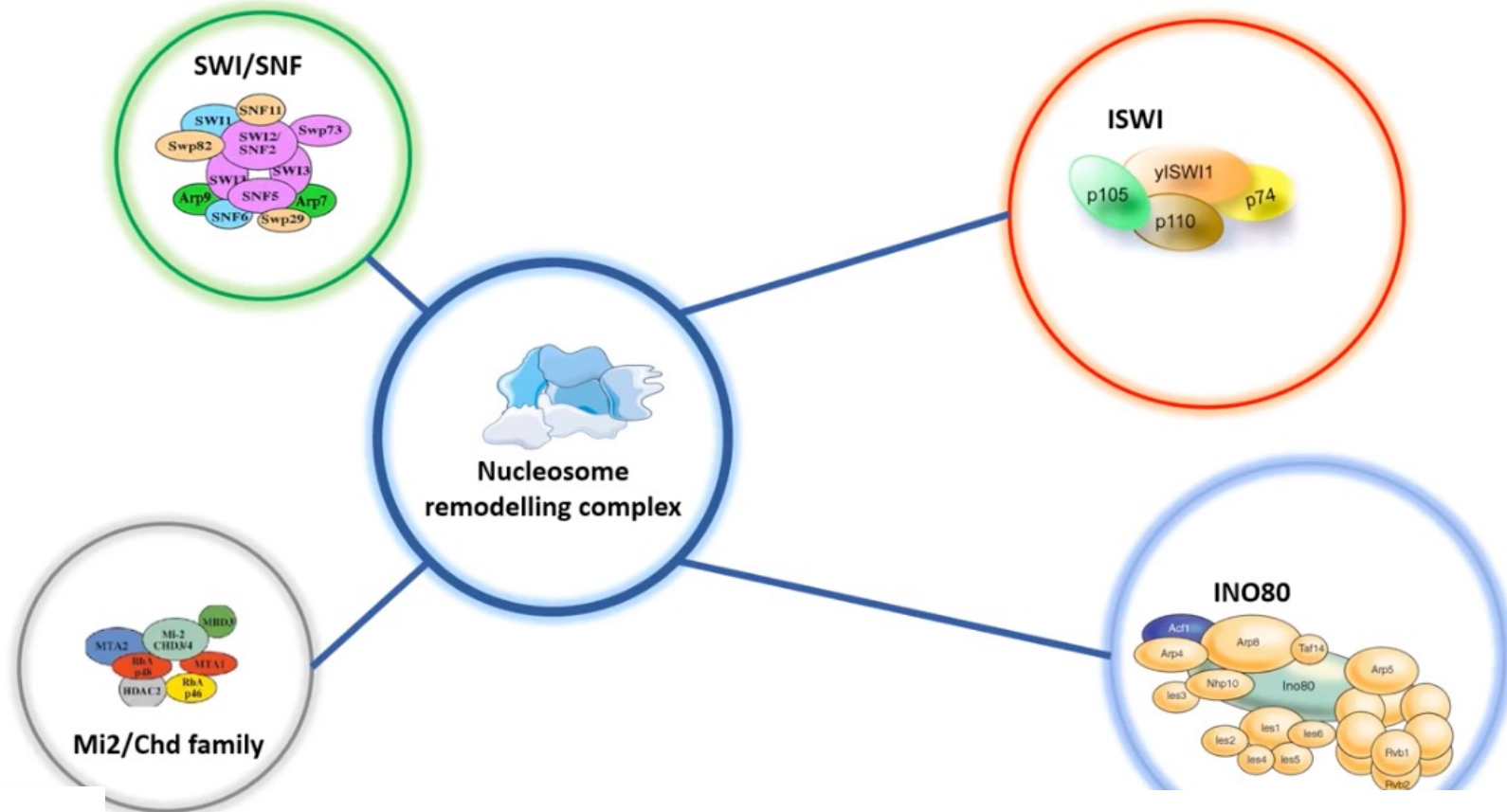


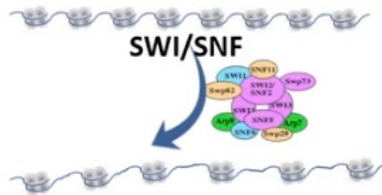






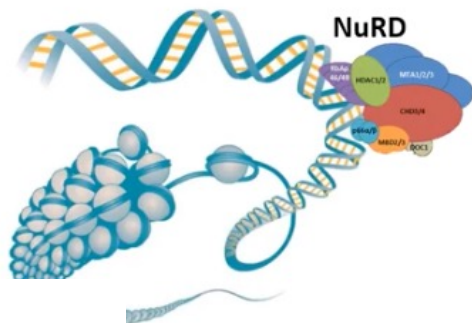
These protein complexes have a common **ATPase domain** and energy from the hydrolysis of ATP allows these remodeling complexes





Several in-vitro experiments suggest that **ISWI remodelers organize nucleosome into proper bundle form** and create equal spacing between nucleosomes, whereas **SWI/SNF remodelers disorder nucleosomes**.

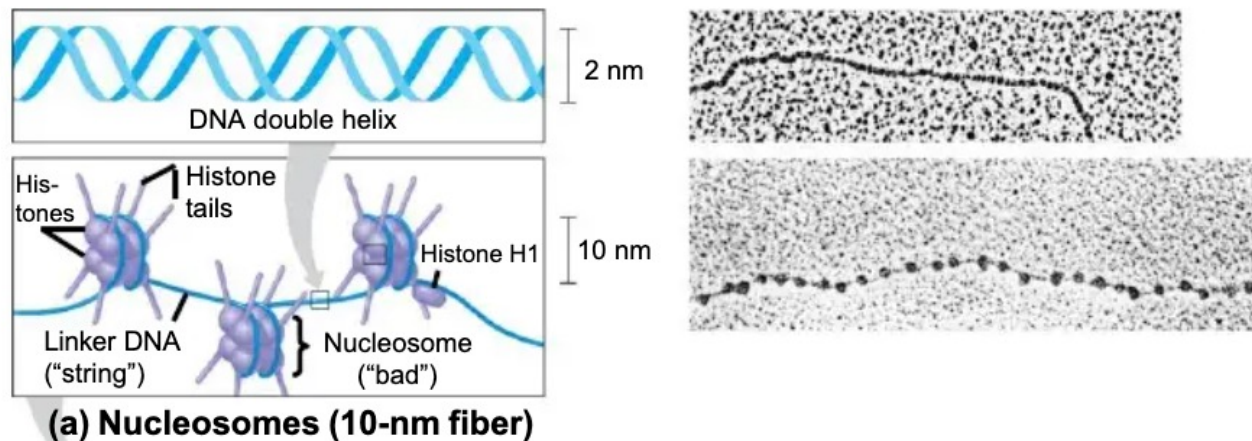
The **ISWI-family** remodelers have been shown to play central roles in chromatin assembly after DNA replication and **maintenance of higher-order chromatin structures**.



NuRD/Mi-2/CHD remodeling complexes primarily **mediate transcriptional repression** in the nucleus and are required for the maintenance of pluripotency of embryonic stem cells.

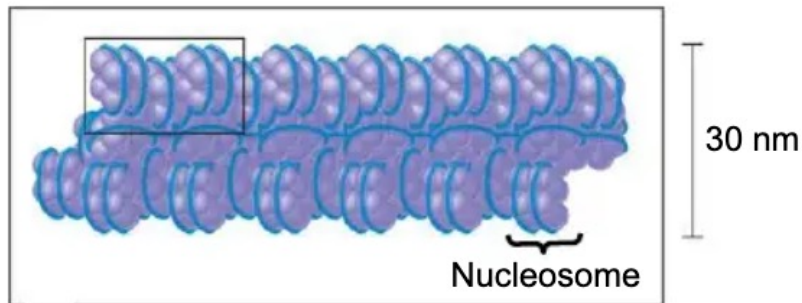
Video-modification-ac-methyl.mp4

- In electron micrographs
 - Unfolded chromatin has the appearance of beads on a string
- Each “bead” is a nucleosome
 - The basic unit of DNA packing

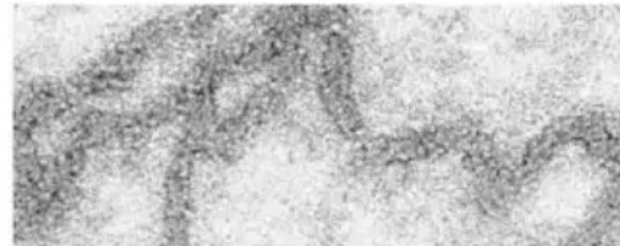


Higher Levels of DNA Packing

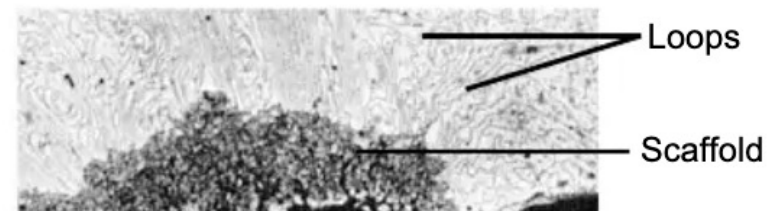
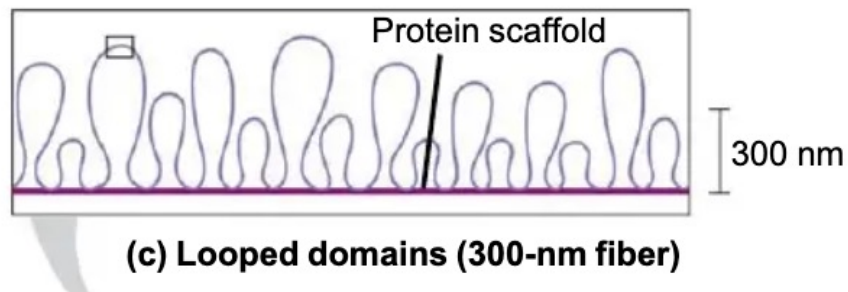
- The next level of packing
 - Forms the 30-nm chromatin fiber



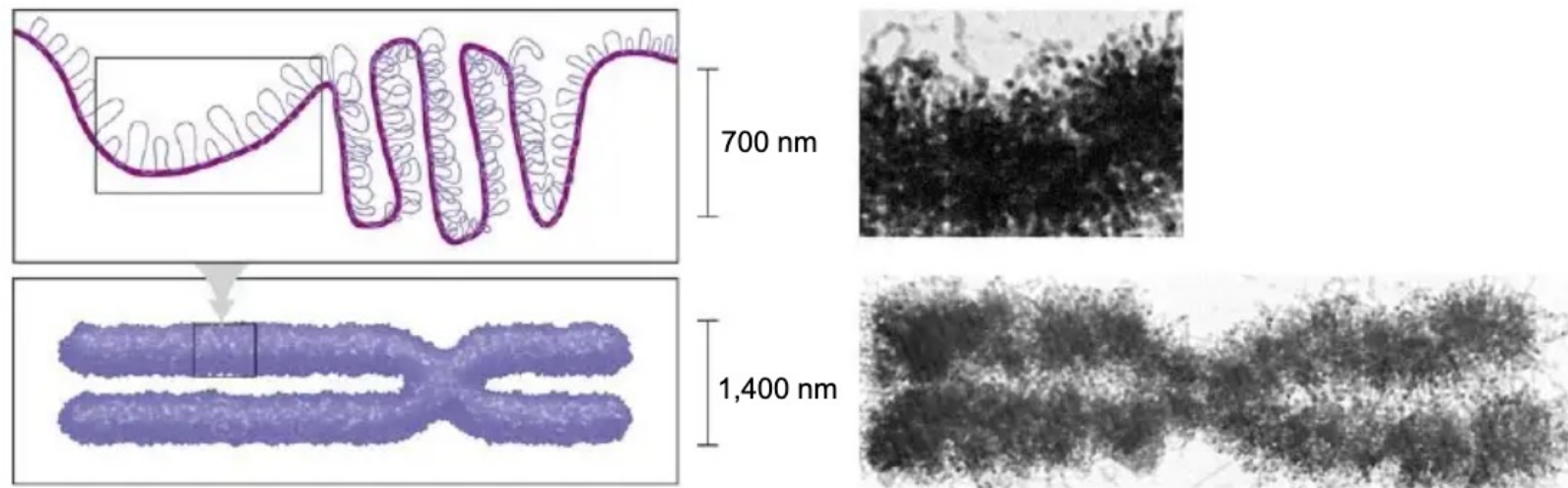
(b) 30-nm fiber



- The 30-nm fiber, in turn
 - Forms looped domains, making up a 300-nm fiber



- In a mitotic chromosome
 - The looped domains themselves coil and fold forming the characteristic metaphase chromosome



(d) Metaphase chromosome

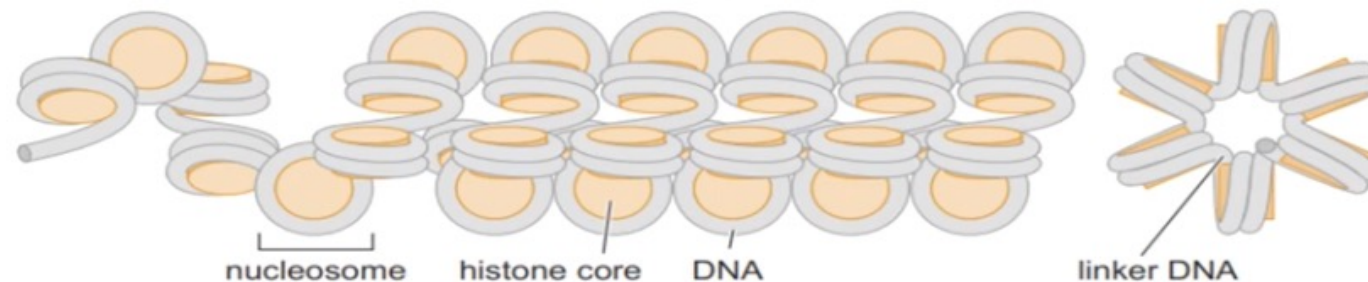
- In interphase cells
 - Most chromatin is in the highly extended form called euchromatin

30 nm CHROMATIN FIBER

- As salt concentrations are increased, the addition of histone H1 results in the nucleosomal DNA forming a 30-nm fiber (*in vitro*).
- There are two models for the structure of the 30-nm fiber:

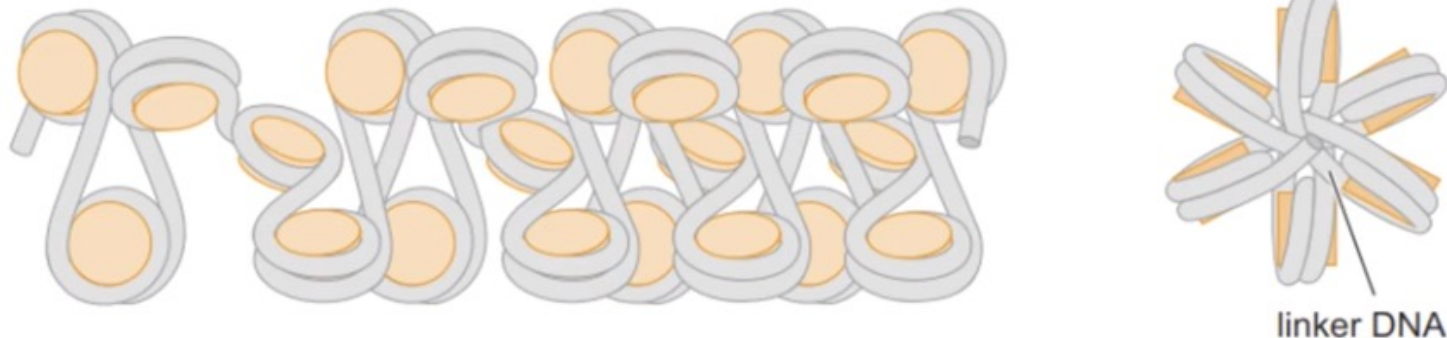
1. SOLENOID MODEL

- The 30-nm fiber is composed of nucleosome discs stacked on edge in the form of a helix, has a helical pitch of 11 nm.
- The linker DNA is buried in the center of the superhelix, but it never passes through the axis of the fiber.
- the linker DNA circles around the central axis as the DNA moves from one nucleosome to the next



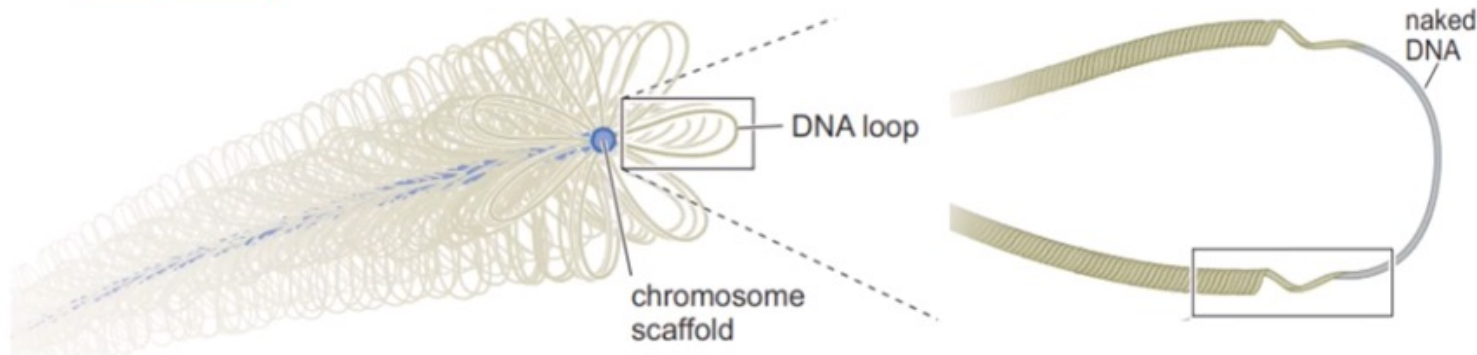
2. ZIGZAG MODEL

- The 30-nm fiber is a compacted form of these zigzag nucleosome arrays.
- The zigzag conformation requires the linker DNA to pass through the central axis of the fiber in a relatively straight form.
- Longer linker DNA favors this conformation.
- The form of the 30-nm fiber may not always be the same, and both forms of the 30-nm fiber may be found in cells depending on the local linker DNA length.

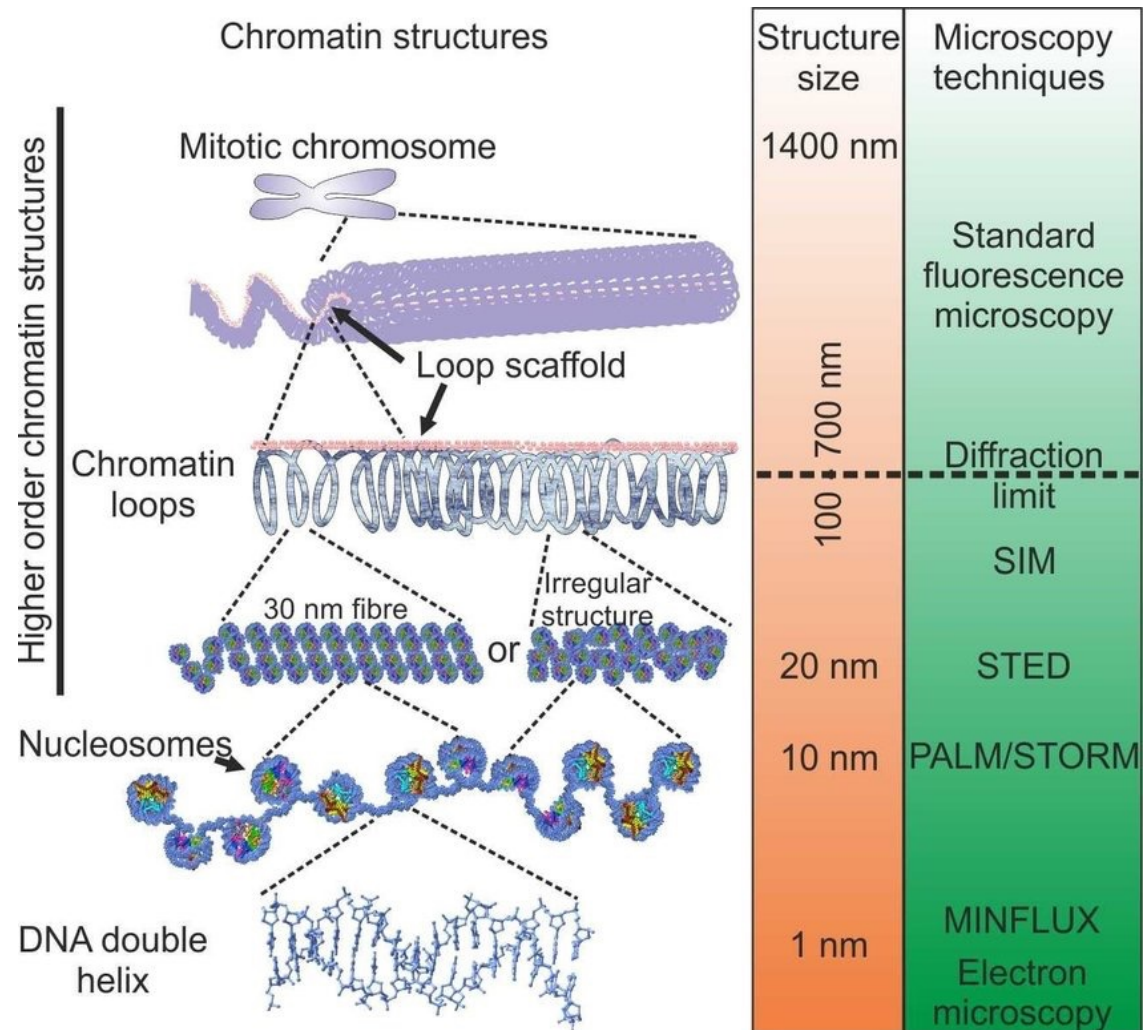


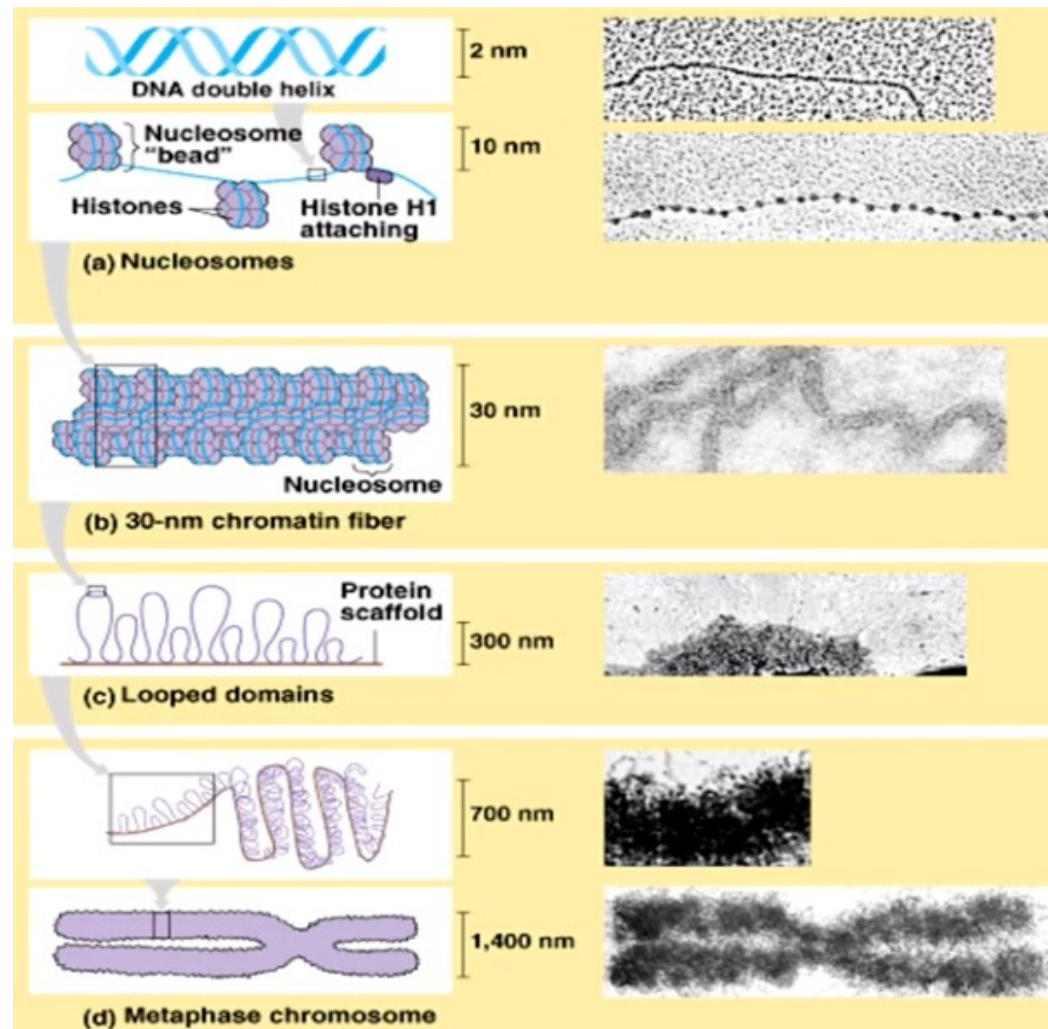
Further Compaction of DNA Involves Large Loops of Nucleosomal DNA

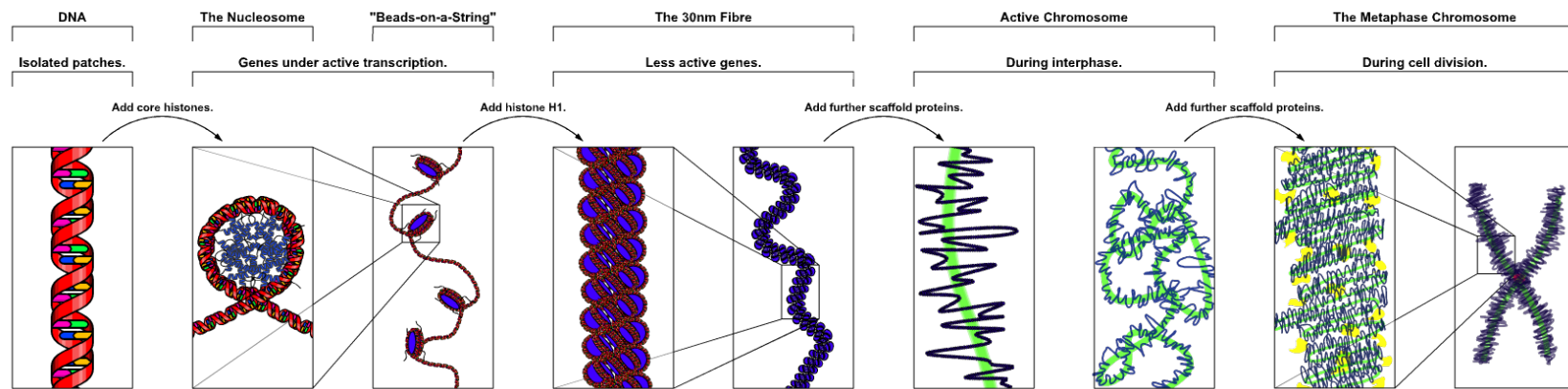
- the packaging of DNA into nucleosomes and the 30-nm fiber results in the compaction of the linear length of DNA by 40-fold *i.e.* insufficient to fit 1–2 m of DNA into a nucleus 1025 nm across.
- 30-nm fiber forms loops of 40–90 kb that are held together at their bases by a proteinaceous structure referred to as the **nuclear scaffold**.
- Two classes of proteins that contribute to the nuclear scaffold:
 1. Topoisomerase II (Topo II): The presence of Topo II at the bottom of each loop would ensure that the loops are topologically isolated from one another.
 2. SMC proteins: Condenses and holds sister chromatids together after chromosome duplication.



Video-Histone+ +Chromatin+ +Nucleosome+ +DNA+Packaging.mp4

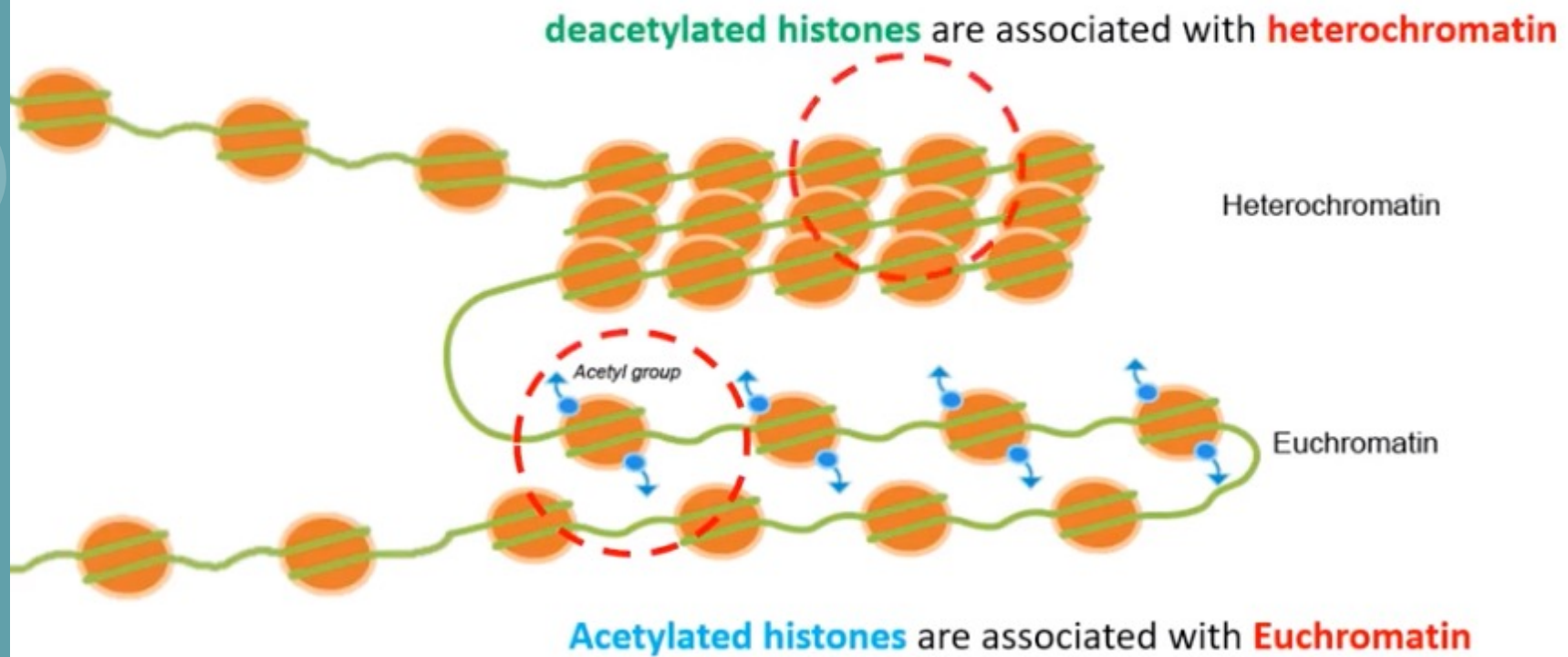


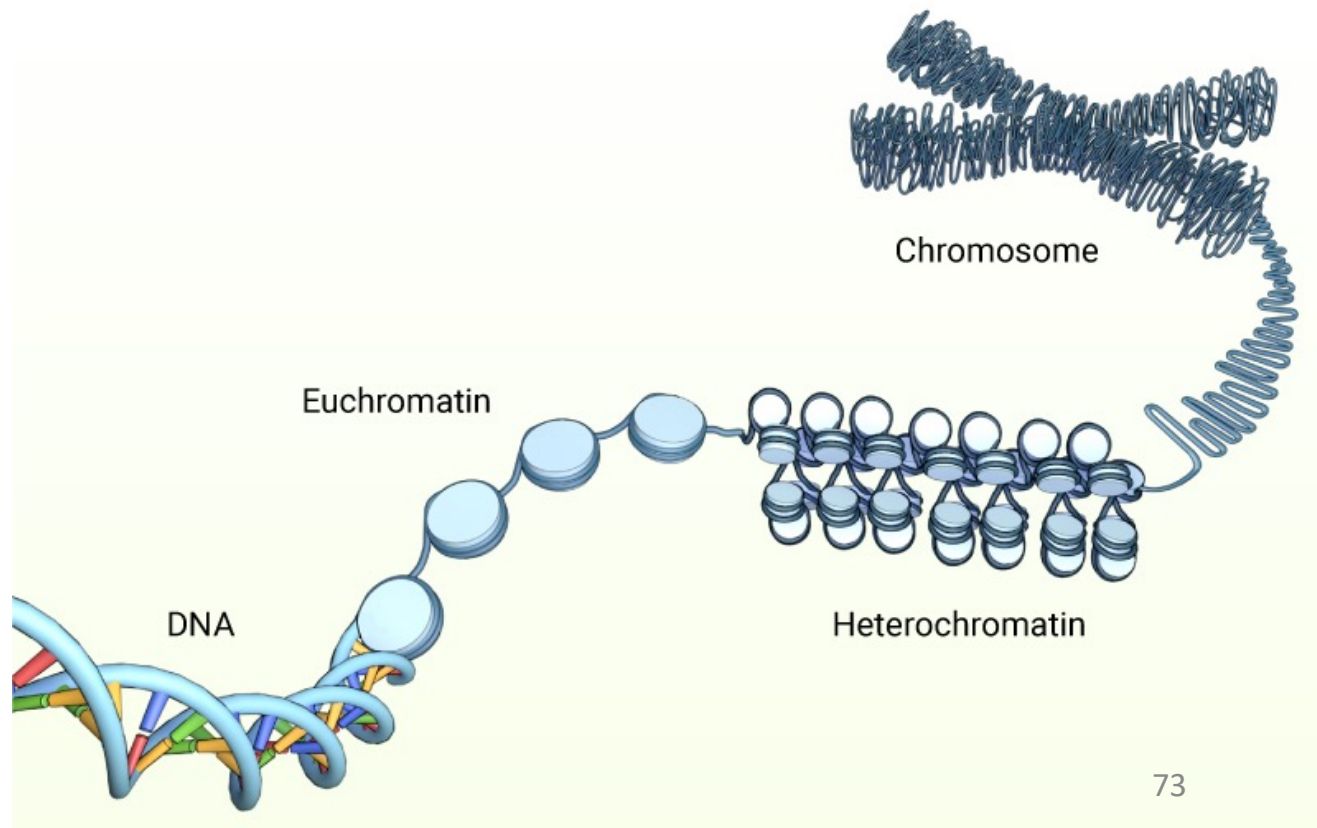
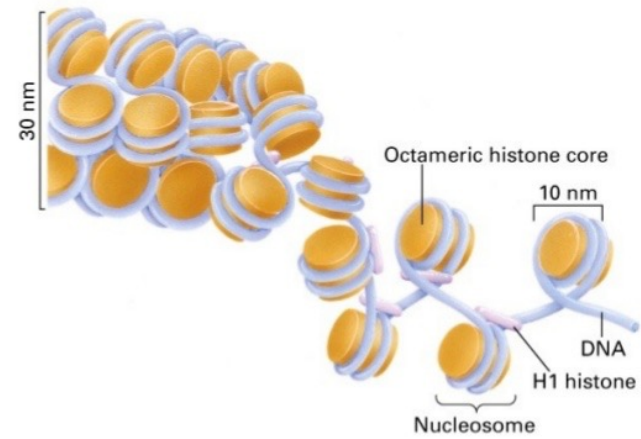
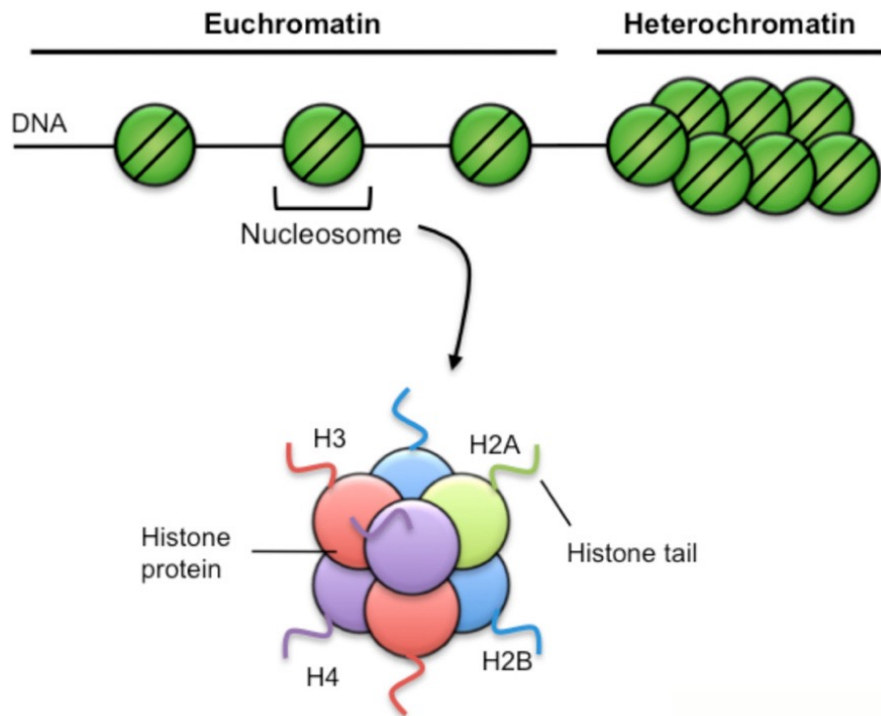


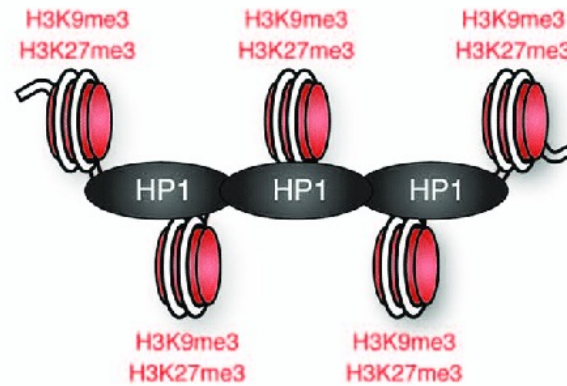


Heterochromatin VS Euchromatin

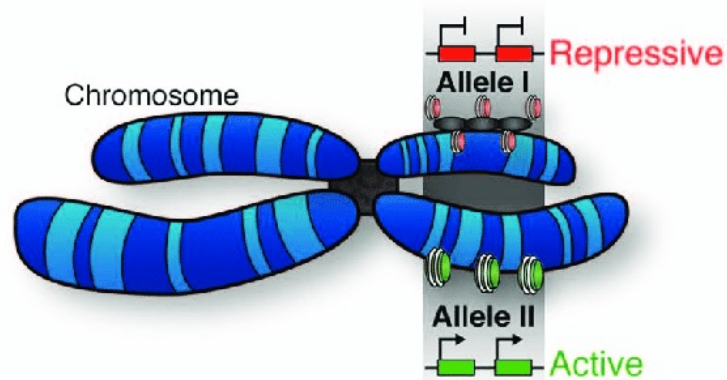




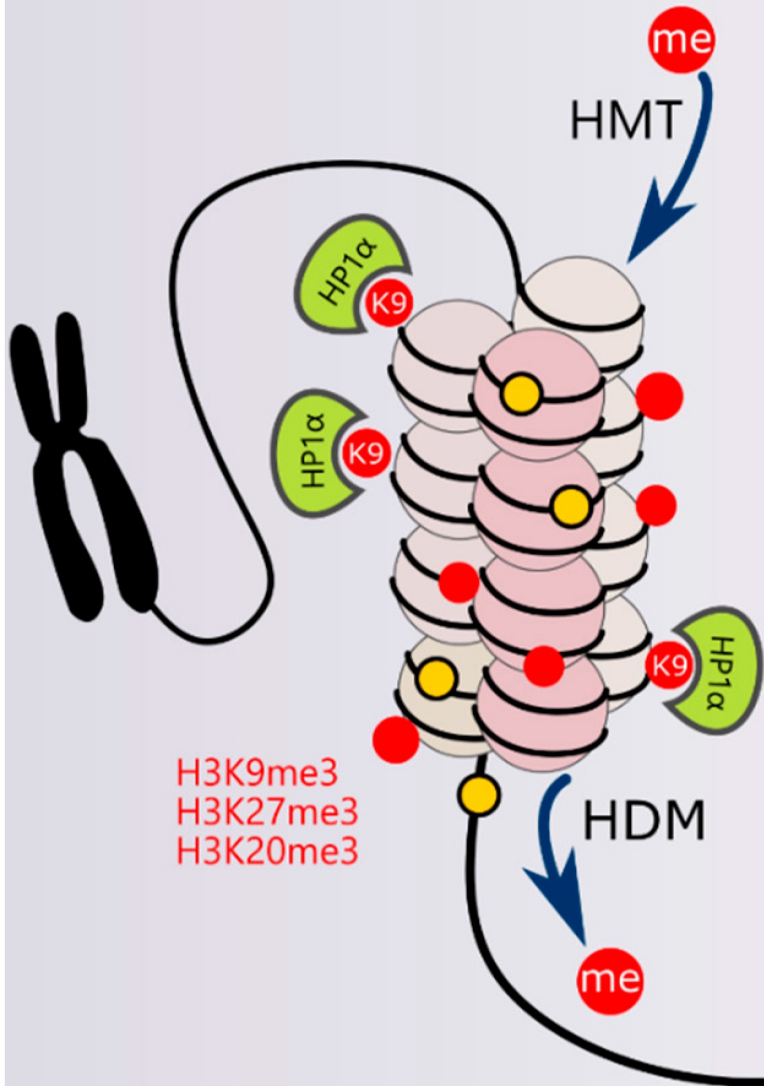


A**Open euchromatin****B****Silent heterochromatin****C****Bivalent chromatin**

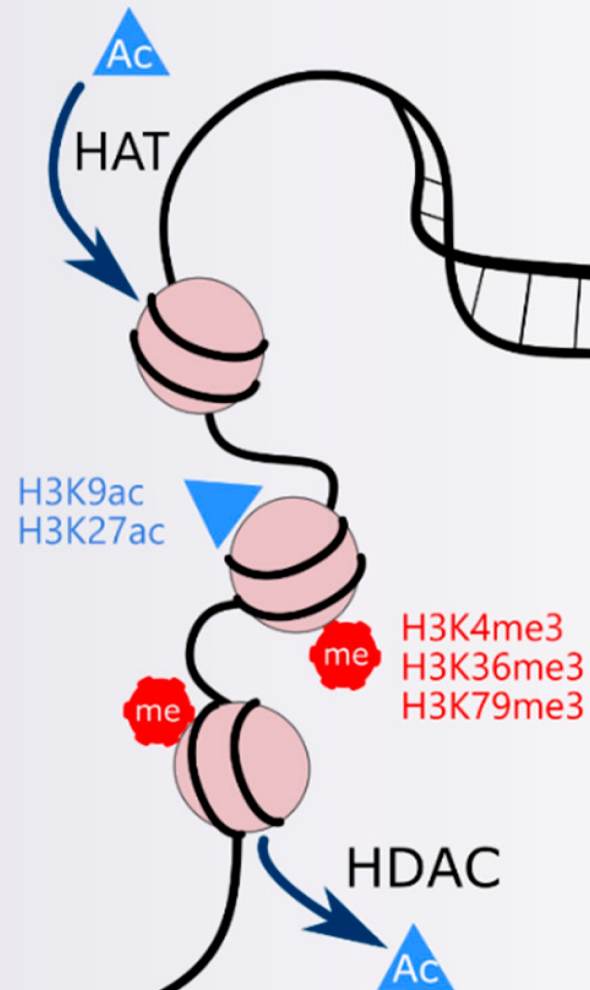
or



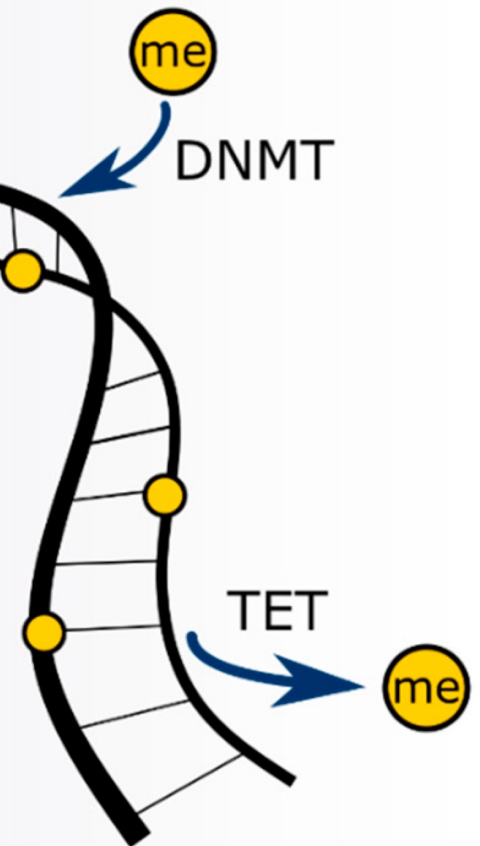
HETEROCHROMATIN



EUCHROMATIN



DNA METHYLATION

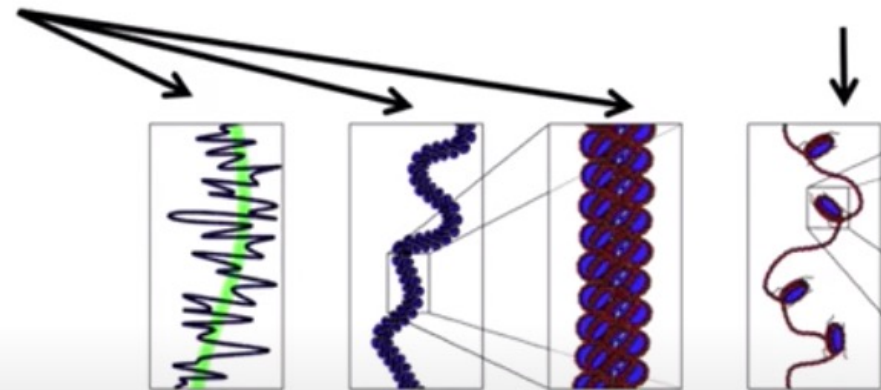


Heterochromatin versus euchromatin

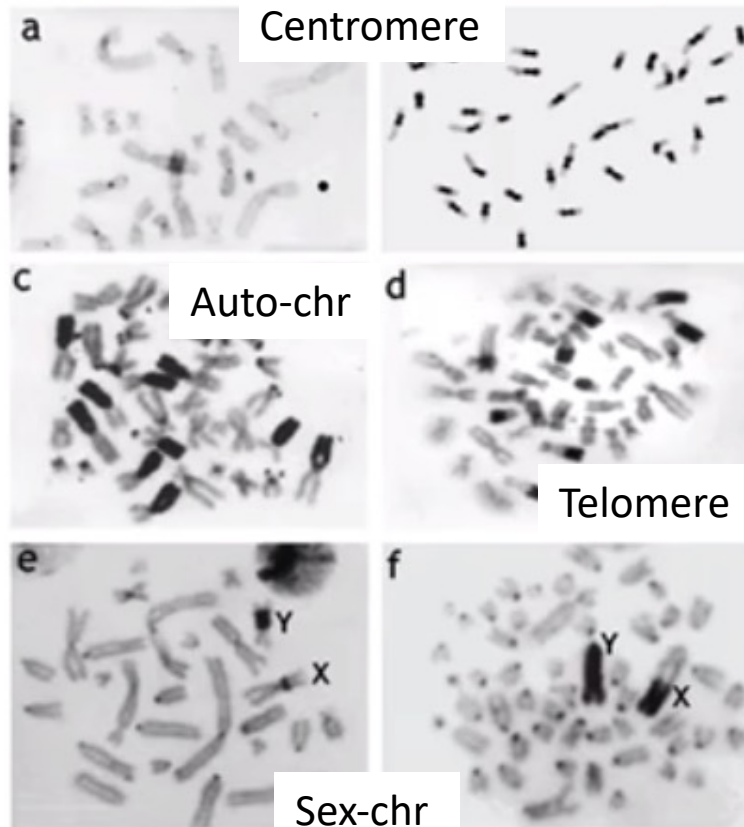
Heterochromatin – closed chromatin, densely stained

- **Constitutive** – same in all cell types
Performs a structural role
 - Centromeres
 - Telomeres
 - Portions of sex chromosomes, mainly Y

Euchromatin – open chromatin, less stained



Constitutive heterochromatin



Metaphase chromosomes from different mammals show different amounts of constitutive heterochromatin at:

1. Centromeres (a vs b)
2. Telomeres (d)
3. Autosomal chromosome arms (c)
4. Sex chromosomes (e and f)

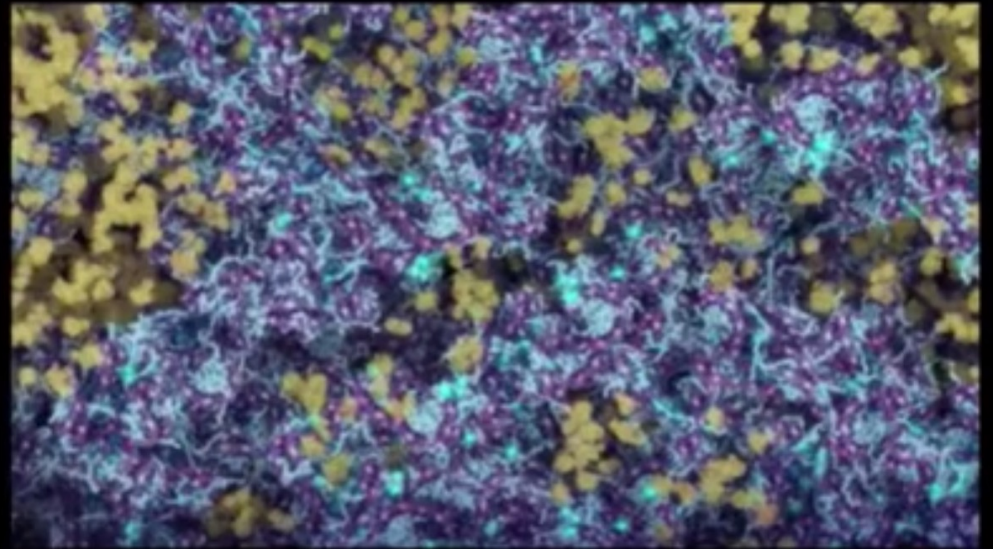
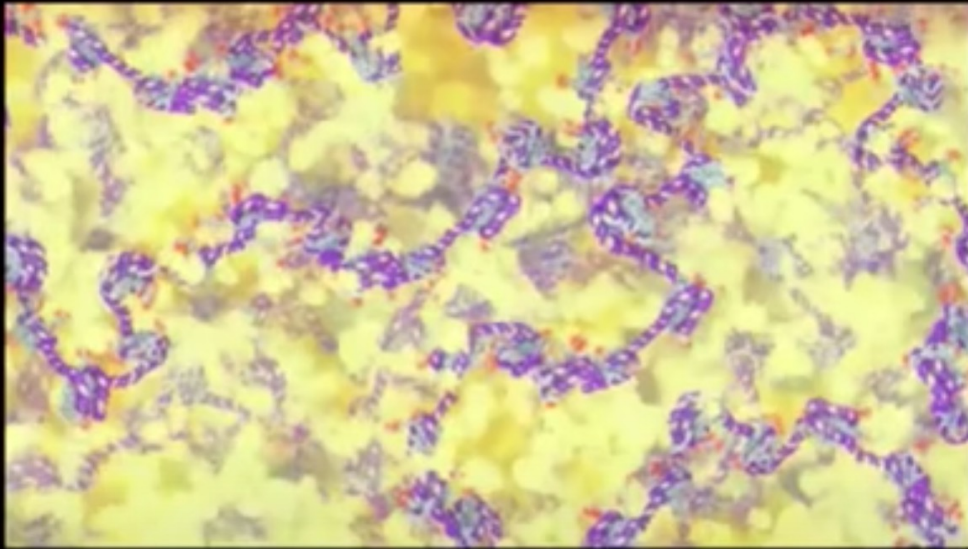
Different epigenetic marks are associated with euchromatin and heterochromatin

Heterochromatin has several functions

Gene silencing

Structural integrity of the genome

X chromosomes



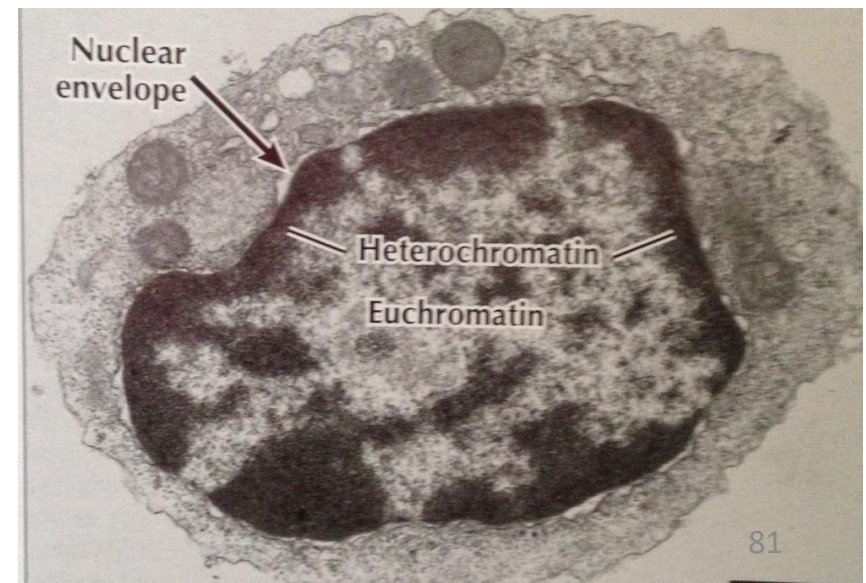
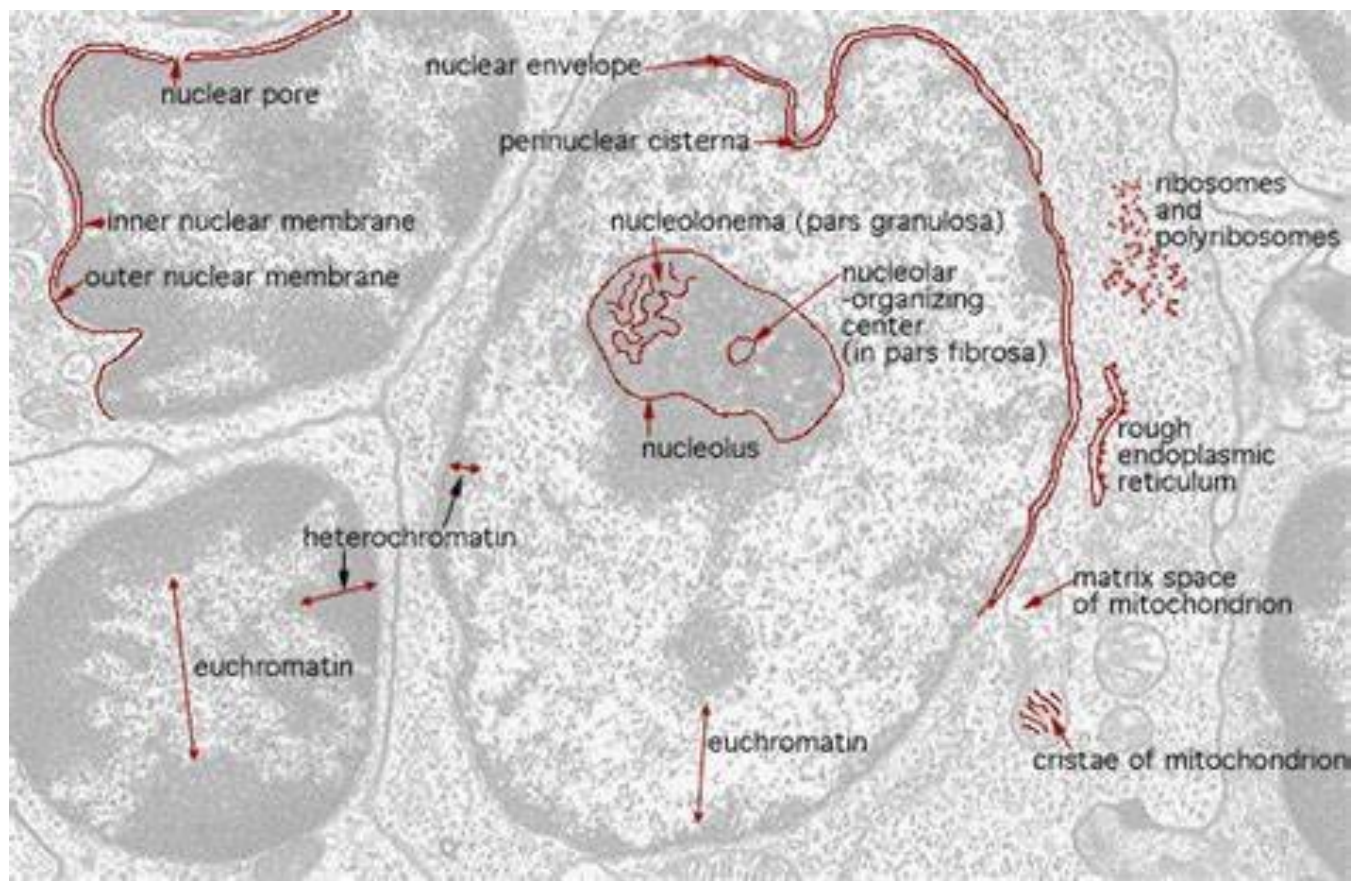
Euchromatin vs Heterochromatin

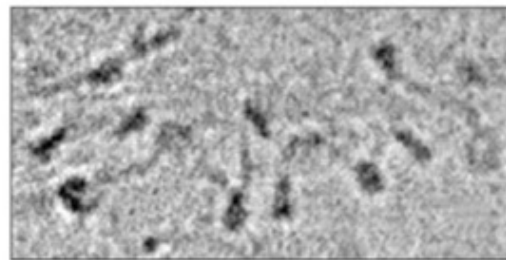
Euchromatin: appears decondensed or in open configuration
potential to be transcriptionally active

Heterochromatin: appears condensed or in closed configuration
replicates late in S phase
transcriptionally not active

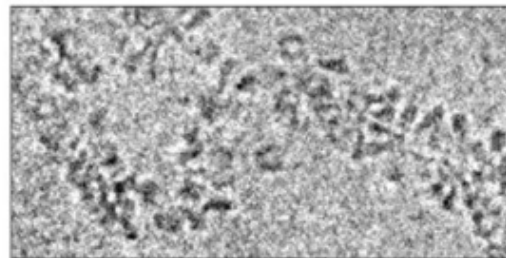
*?? Role of heterochromatin ?? A large percentage of human DNA is highly repetitive, does not code for anything (i.e., is not transcribed into mRNA). Function → related to chromatin structure and activity??
→ transcribed into microRNAs??*

- **Facultative heterochromatin:** sequences that are transcriptionally inactive in some cells but are active in other cells (one copy of an X chromosome)
- **Constitutive heterochromatin:** sequences that are transcriptionally inactive in all cells (centromere regions of all chromosomes); *highly repetitive* DNA



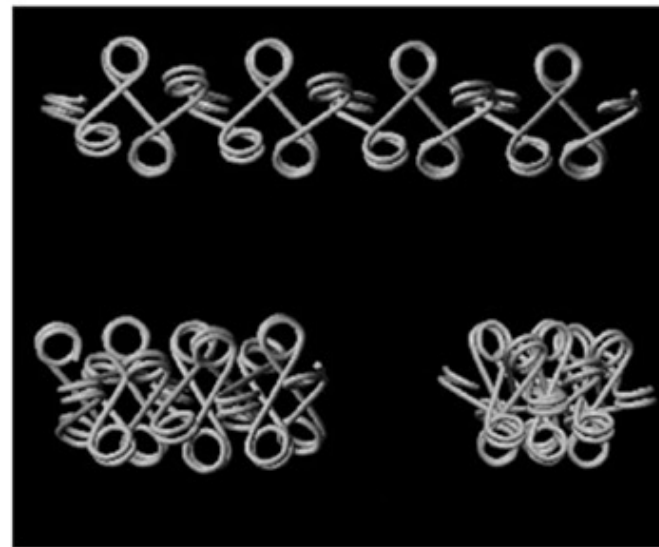


(A)



(B)

50 nm



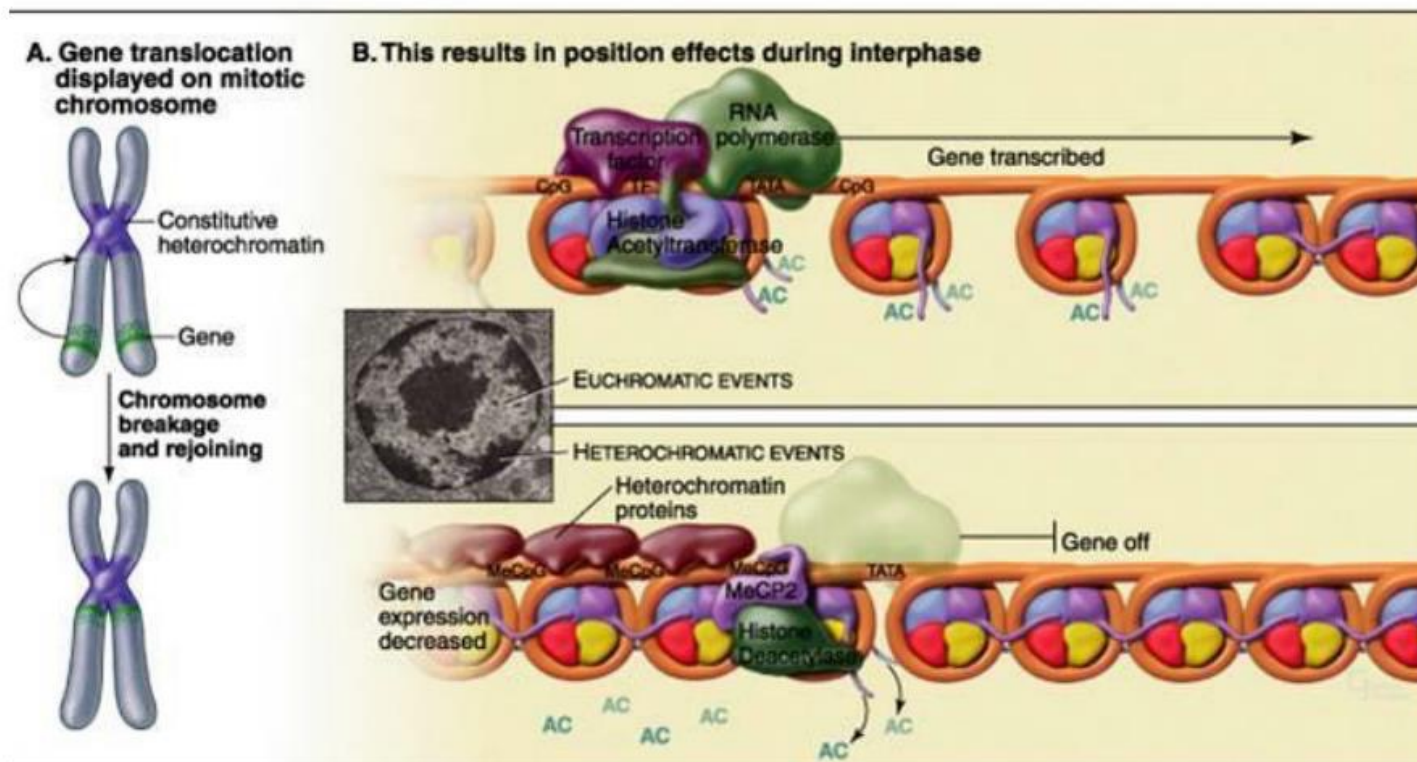
(C)

Figure 5-25 Essential Cell Biology, 2/e. (© 2004 Garland Science)

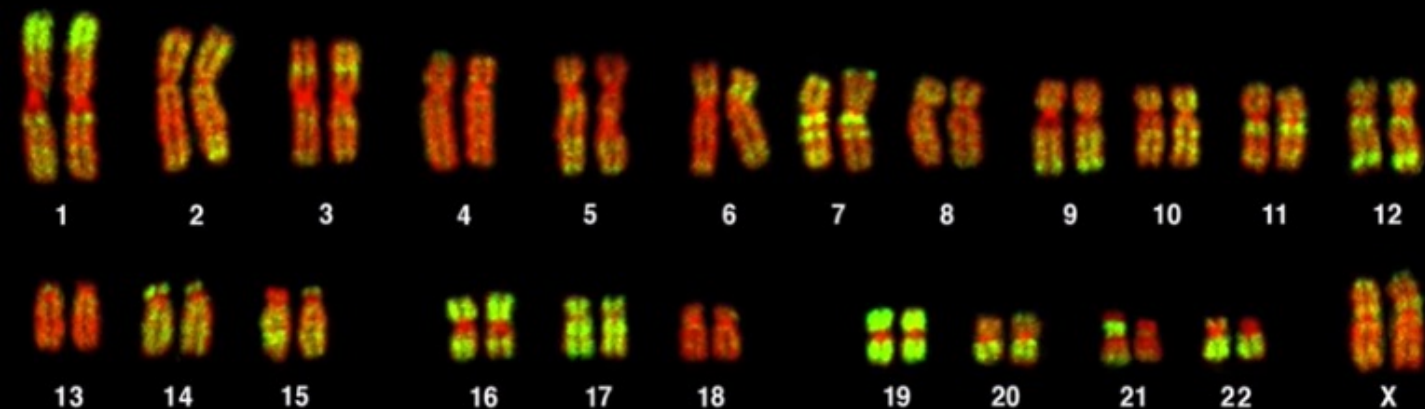
Heterochromatin vs Euchromatin

Modifications to histones and DNA direct formation of heterochromatin

- Deacetylation of H4
- Methylation of H3
- Methylation of CpG

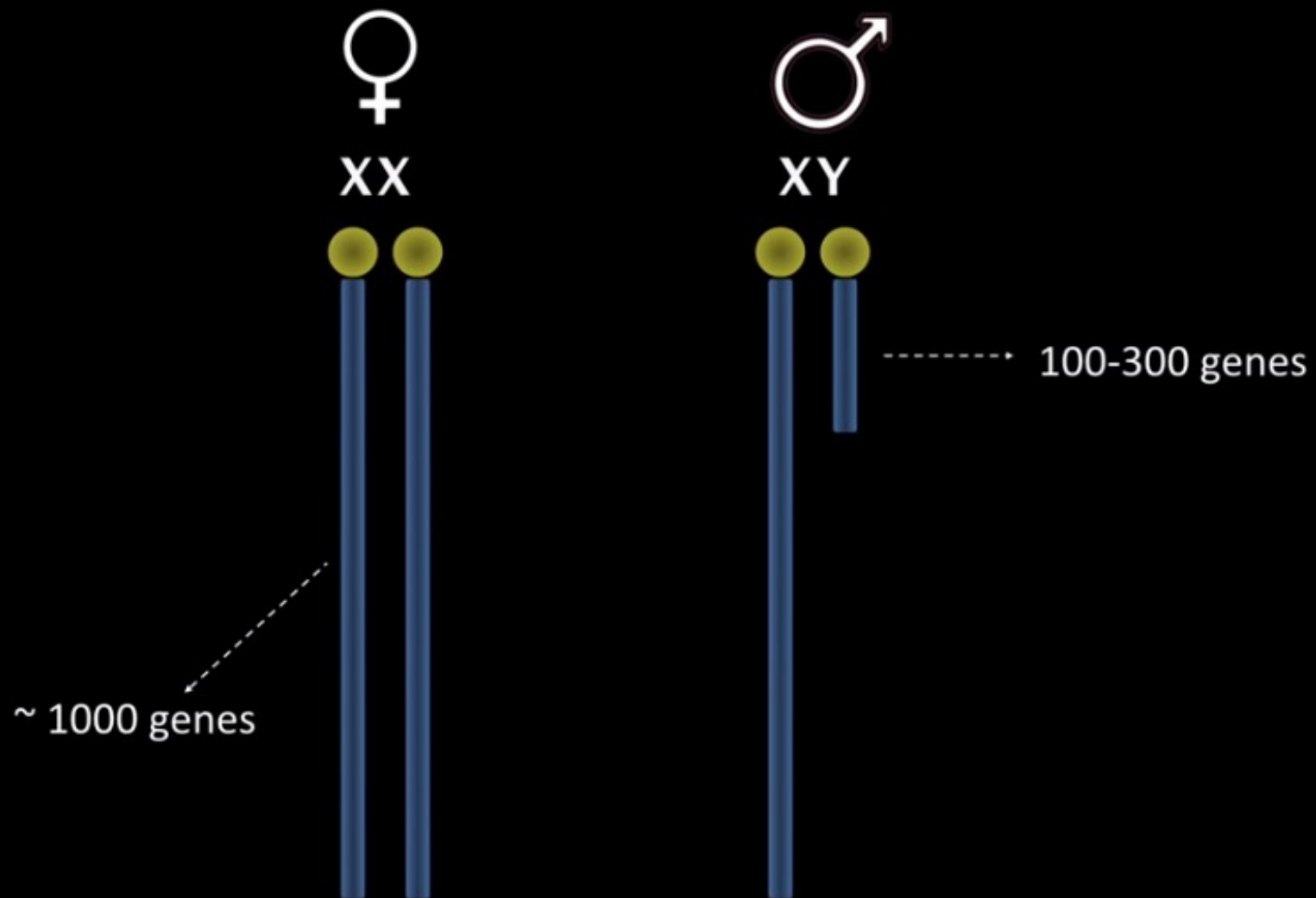


Why do we need to inactivate one X-chromosome?
In mammals, sex is determined by a pair of unequal “sex chromosomes”

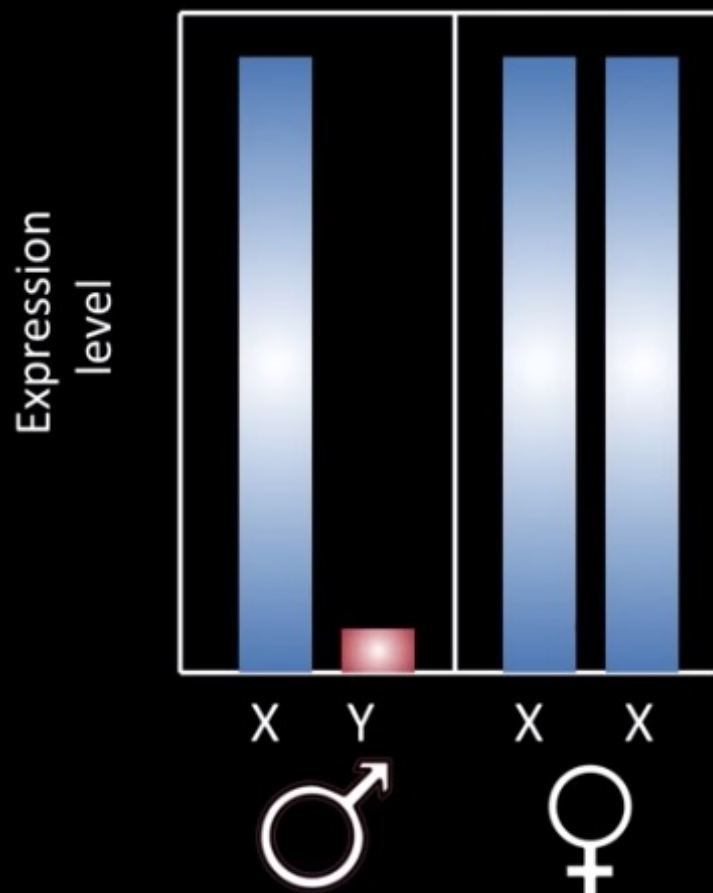


Sexually
Dimorphic
X, Y

A genetic inequality ...

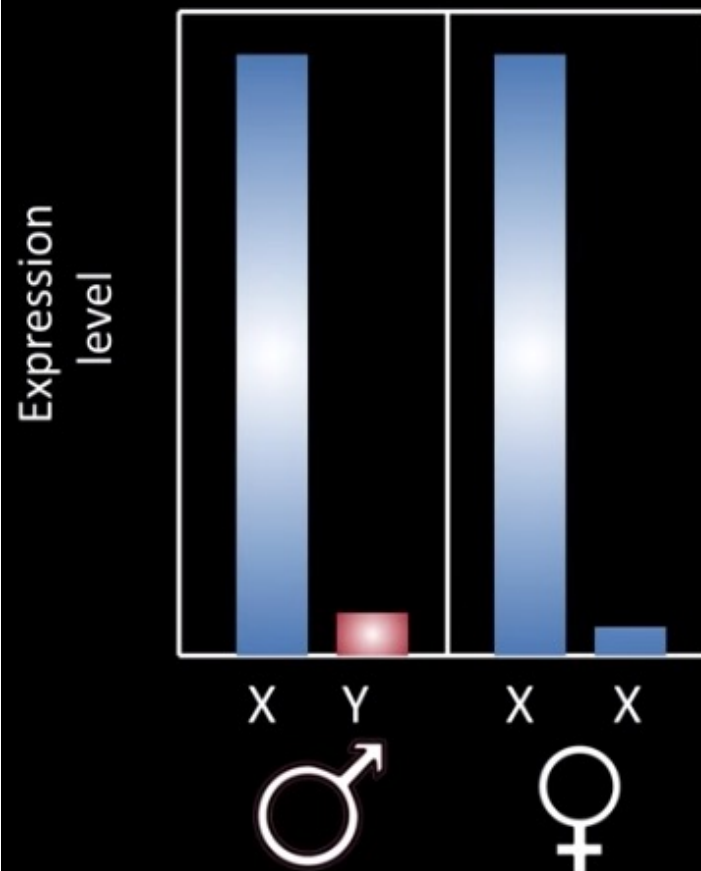


To compensate, mammals do something unique: X-chromosome inactivation (XCI)

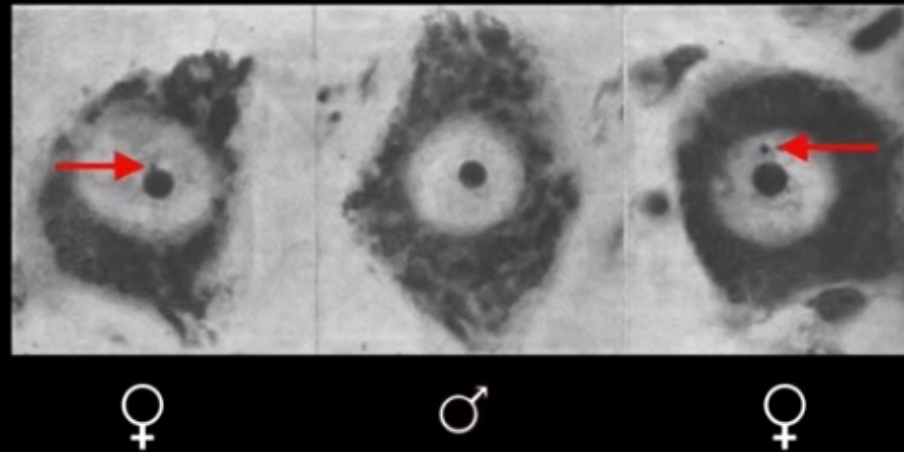


- The only chromosome capable of global silencing
- Mechanisms has been intensively studied for 50+ years
 - model for RNA & epigenetic silencing
 - numerous disease implications

To compensate, mammals do
something unique:
X-chromosome inactivation (XCI)

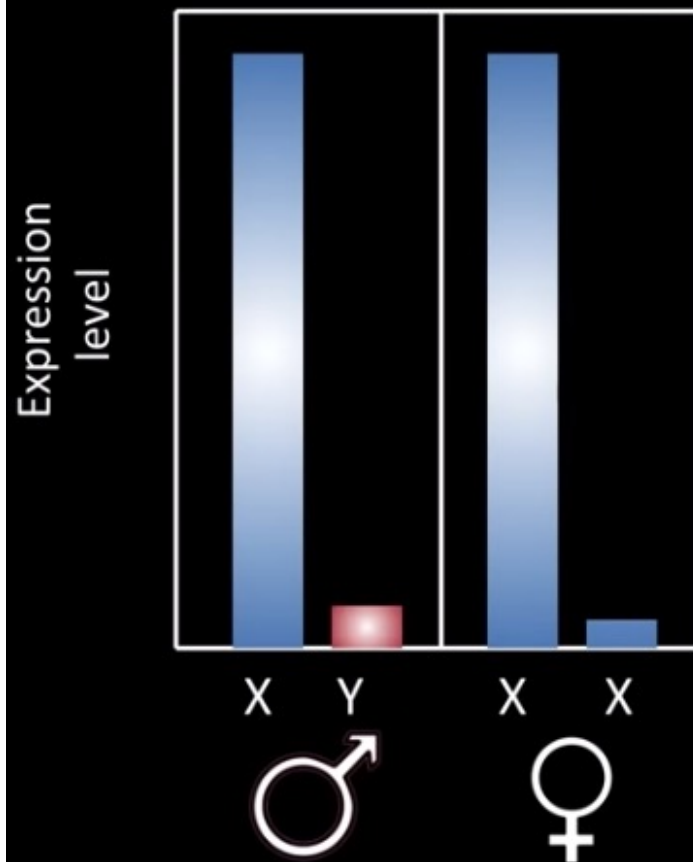


• Barr & Bertram *Nature* 1949

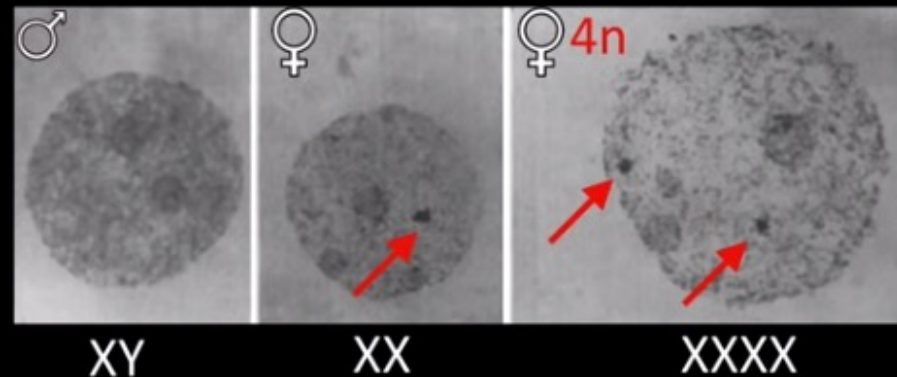
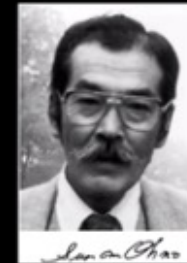


In cats: Could predict sex based on
presence of "Barr body"

To compensate, mammals do something unique: X-chromosome inactivation (XCI)

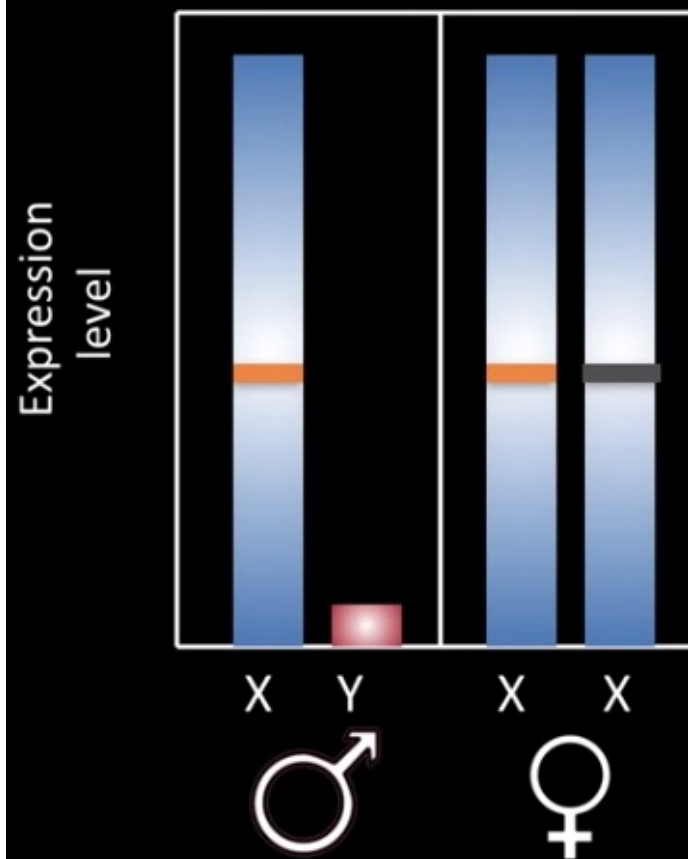


• S Ohno *Exp Cell Res* 1959



- In rats, # Barr bodies parallels # X-chromosomes
- Barr body = X. Condensed, bio-chemically and functionally distinct ??

To compensate, mammals do something unique: X-chromosome inactivation (XCI)

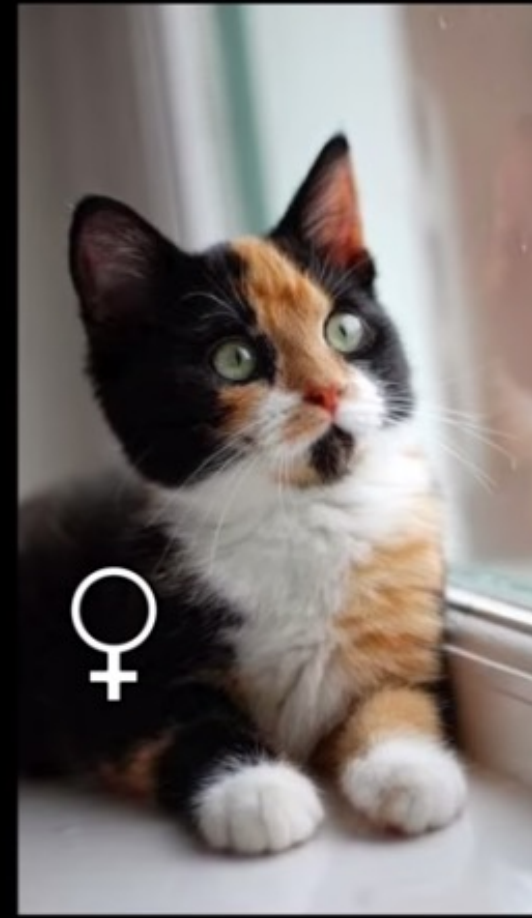
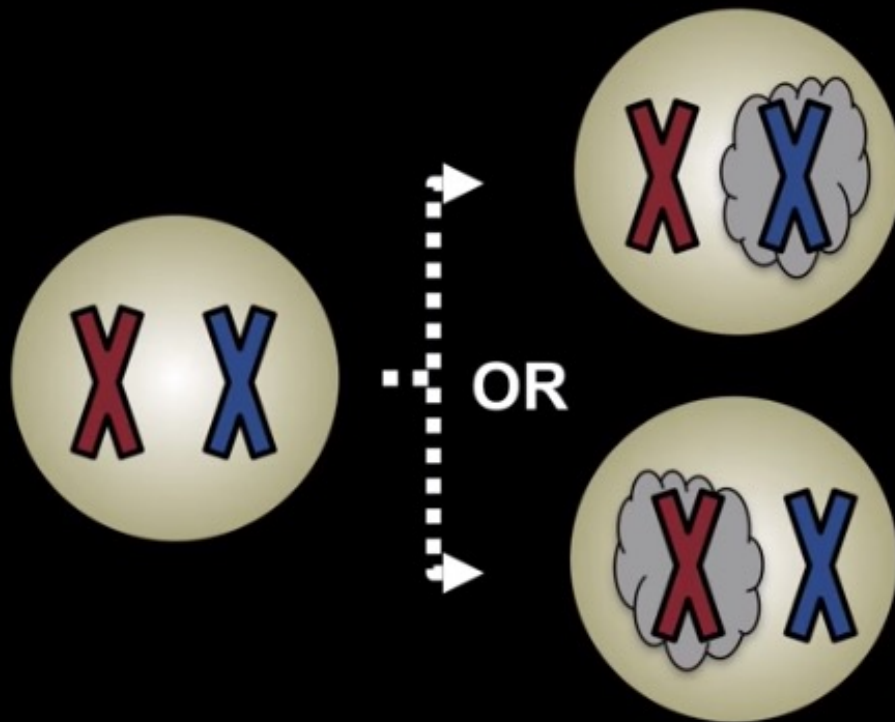


- MF Lyon *Nature* 1961

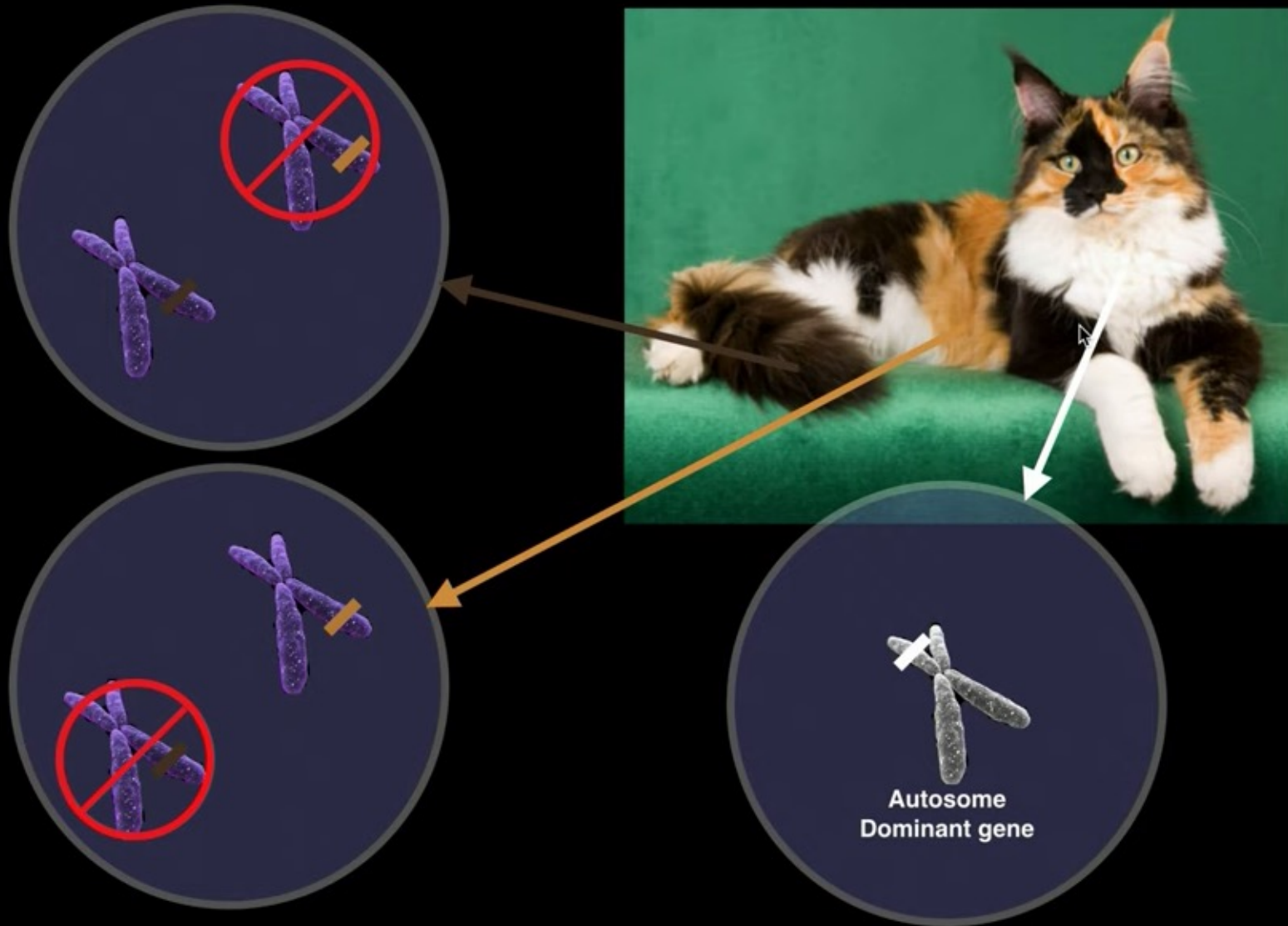


- Barr body = Inactivated X chromosome
- XCI = dosage compensation
- Random inactivation of 1X per cell

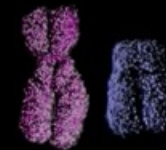
Choice of X = random
Every female is a mosaic



X-Chromosome Inactivation: Calico Cats



X-Chromosome Inactivation: The Barr Body



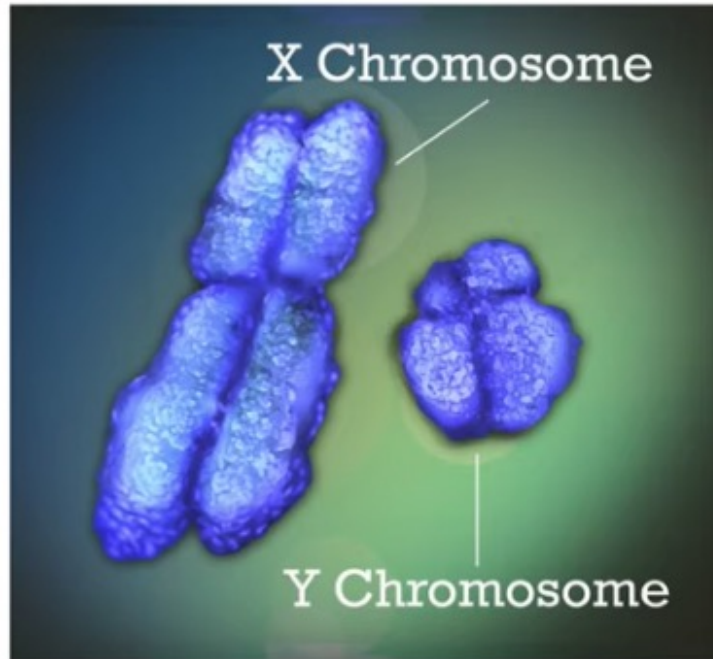
X-Chromosome Inactivation In Klinefelter's or Turner's Syndrome



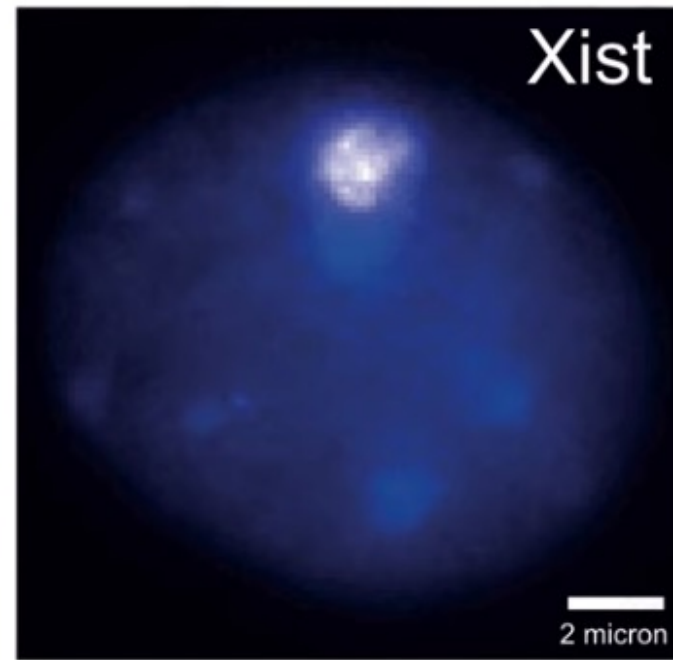
Lyon Hypothesis: All but one X-chromosome in any sex chromosome genotype is inactivated

Number of Barr Bodies = Number of X-chromosomes - 1

Xist: Master regulator of X inactivation



Flickr/ NIH Image Gallery

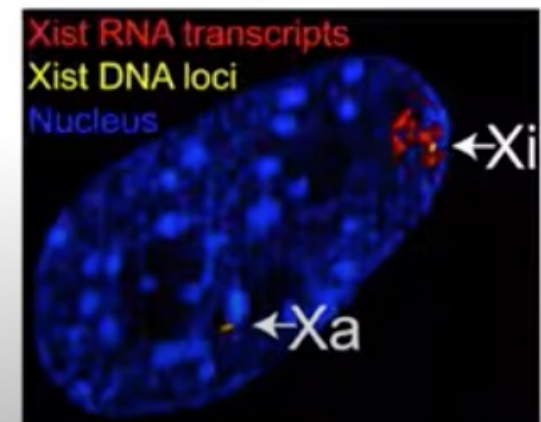


Giorgetti et al., *Nature*, 2016.

- ❖ 17 kb long noncoding RNA on X
- ❖ 200 molecules silences 160 Mb chromosome
- ❖ Escapees define sex differences in biology & disease

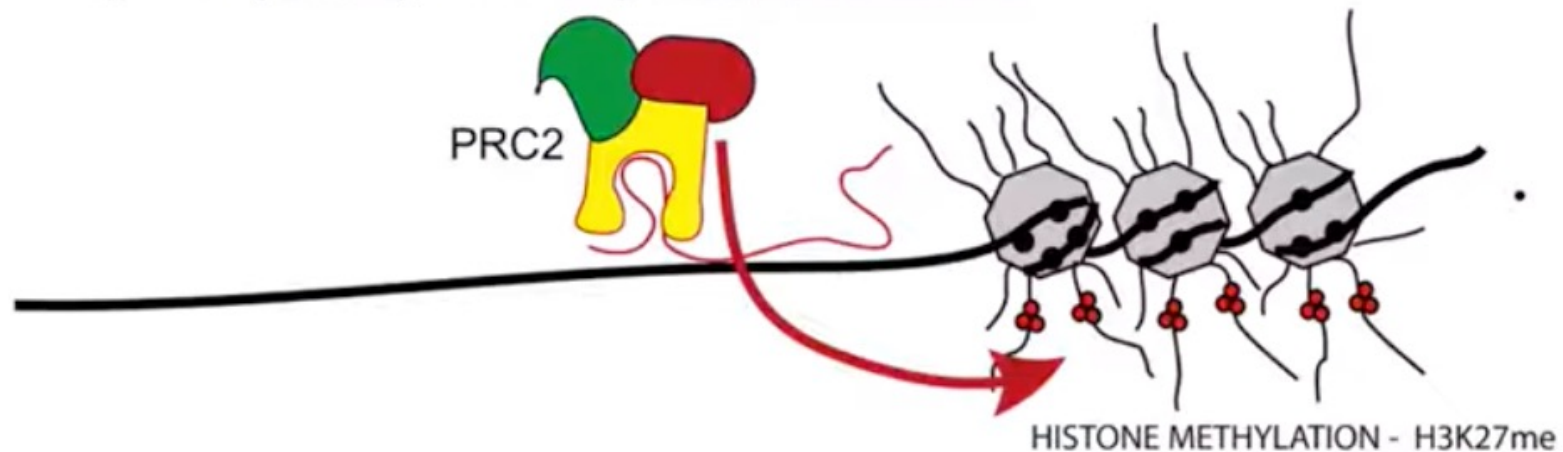
Xist – the critical determinant for X inactivation

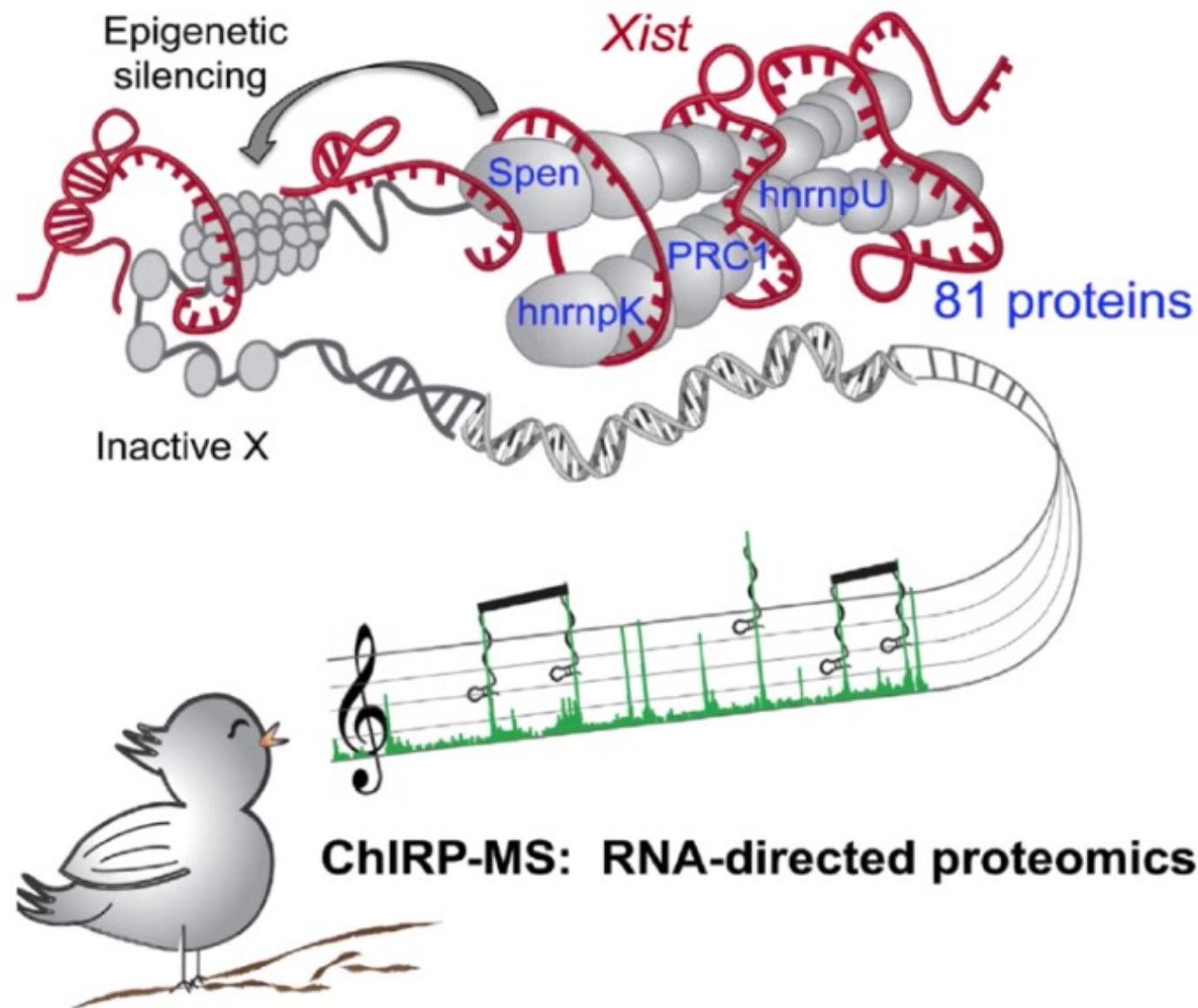
- 17 kb, spliced and polyadenylated, constrained to the nucleus and non-coding (no protein product)
- Expressed from just one of the two X chromosomes
- Expression is the first detectable event in X inactivation
- Determines the chromosome that will become the inactive X i.e. transcription of *Xist* results in silencing!
- *Xist* RNA coats the inactive X *in cis*



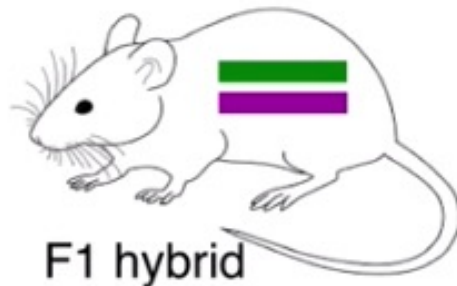
Xist – mechanism of action

- *Xist* contains repeats many repeats within the transcript
- Repeat A (RepA) is required for silencing function of *Xist*
- RepA binds the histone methyltransferase complex Polycomb Repressive Complex 2 (PRC2), which lays down H3K27me

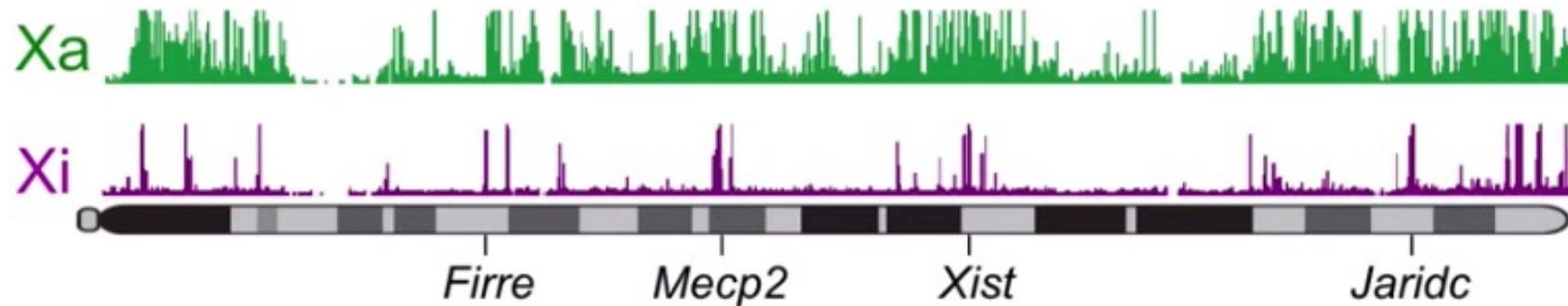




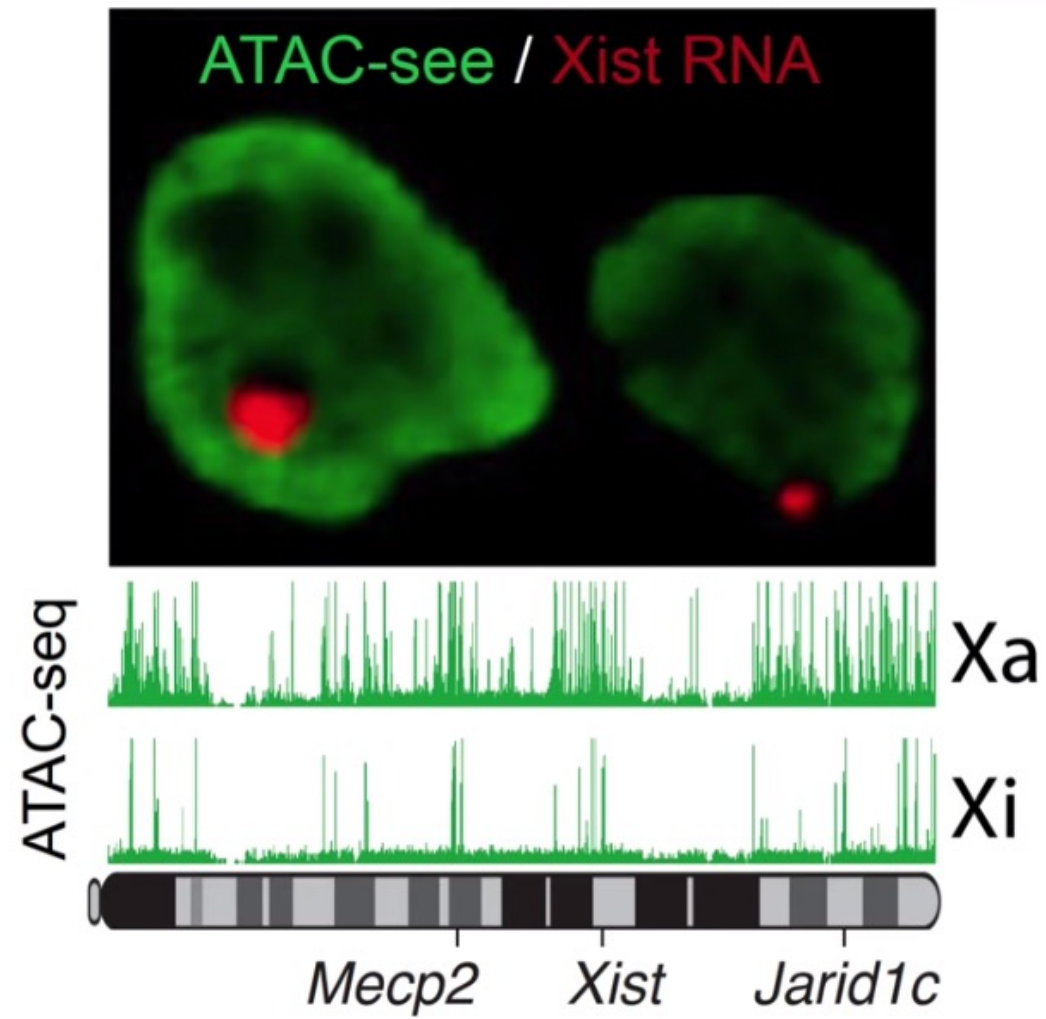
X inactivation: Allele-specific ATAC-seq



- 129S1 x castaneus F1 - SNP every 100 bp
- Allele-specific regulome in any cell types



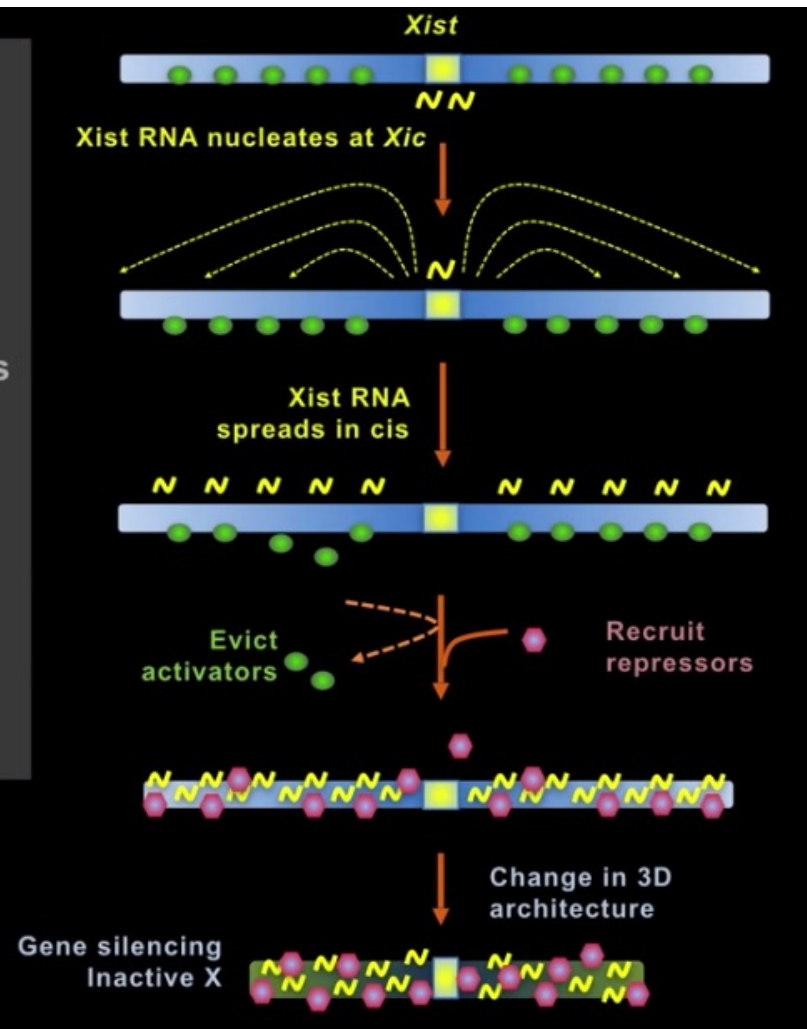
XCI escape in pieces: Promoters but not enhancers



Three functions of Xist:

- ◇ Repels activating factors
- ◇ Recruits repressive factors
- ◇ Alters 3D architecture

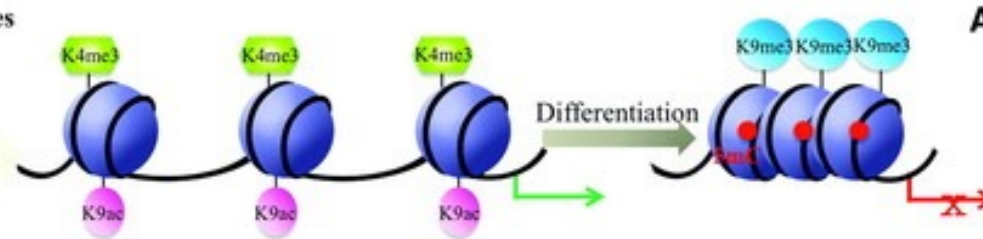
See Lecture 3



Chromatin state

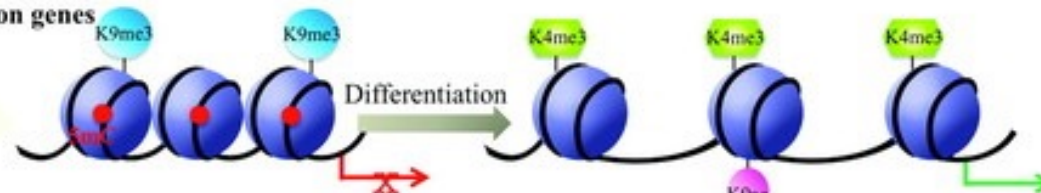
Pluripotency genes

MLL/Wdr5
CHD1
Brg1
GCN5
Tip60-p400
Jmjd1a
Jmjd2c
Kdm5b

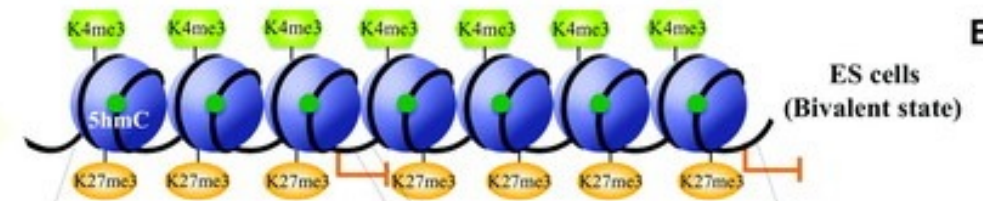


Differentiation genes

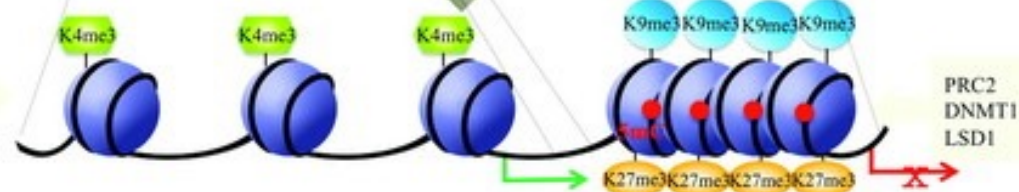
DNMTs
SETDB1

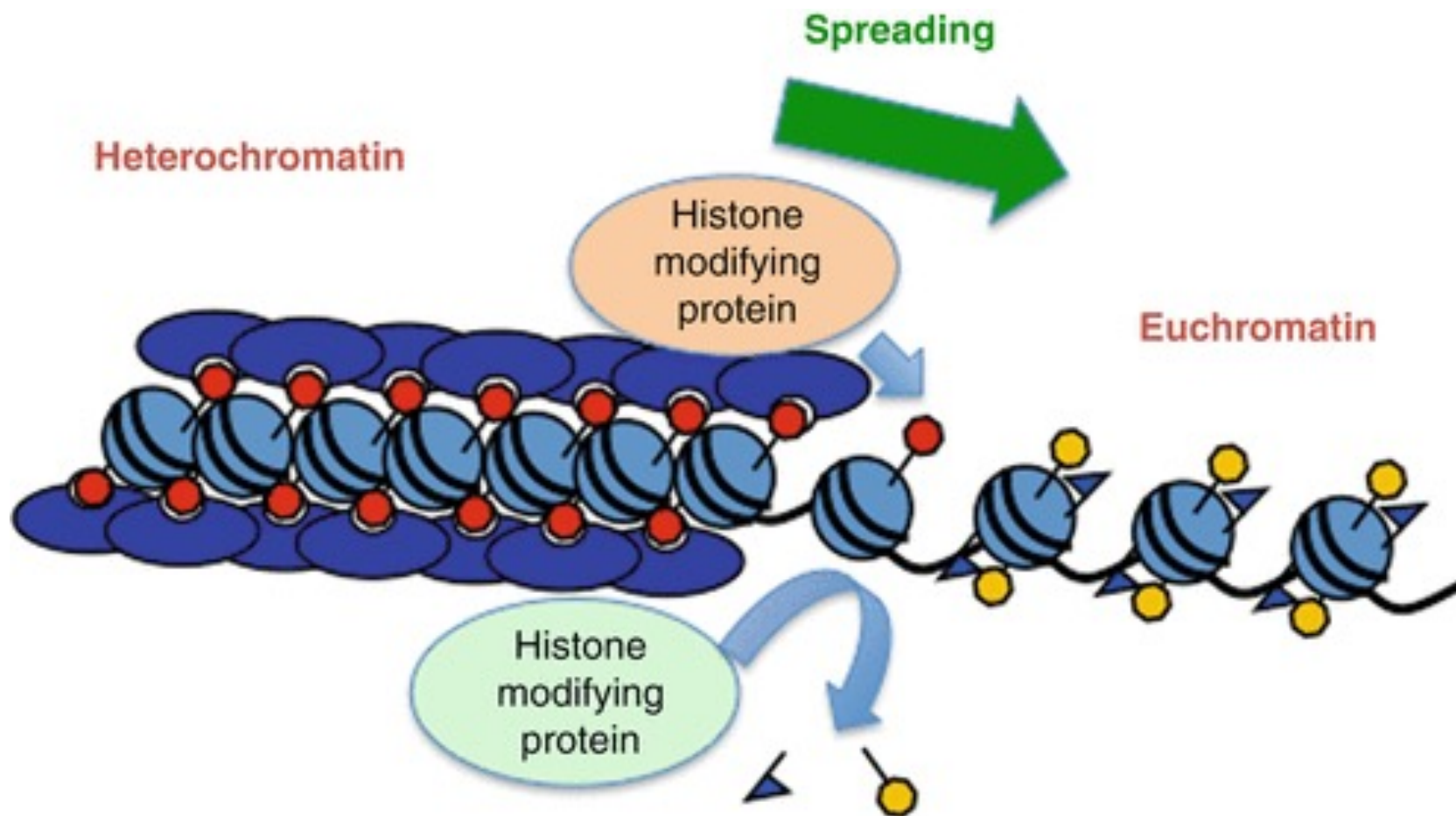


MLL1
MLL2
MLL/Dpy-30
PRC2
TET1



MLL2
Jmjd3/UTX

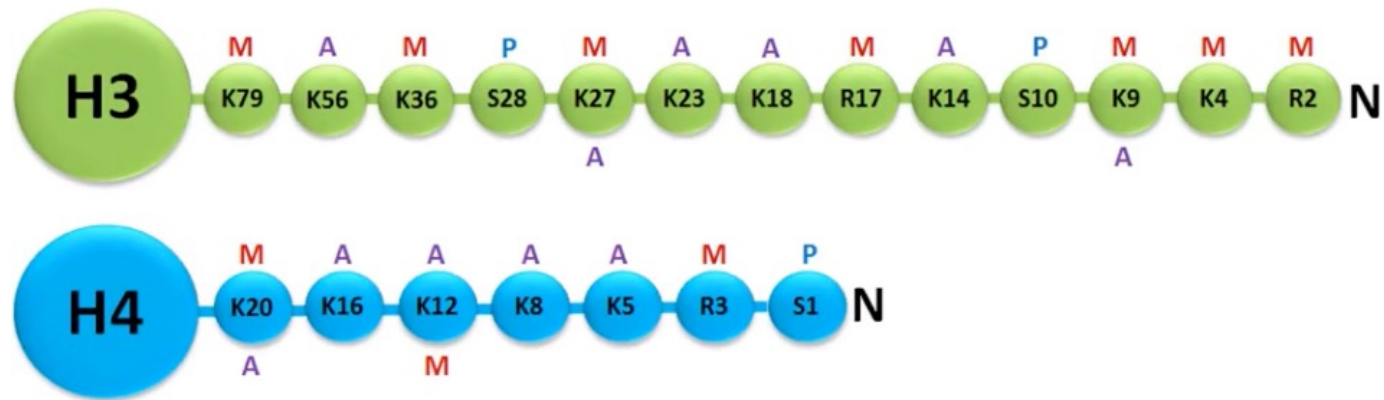




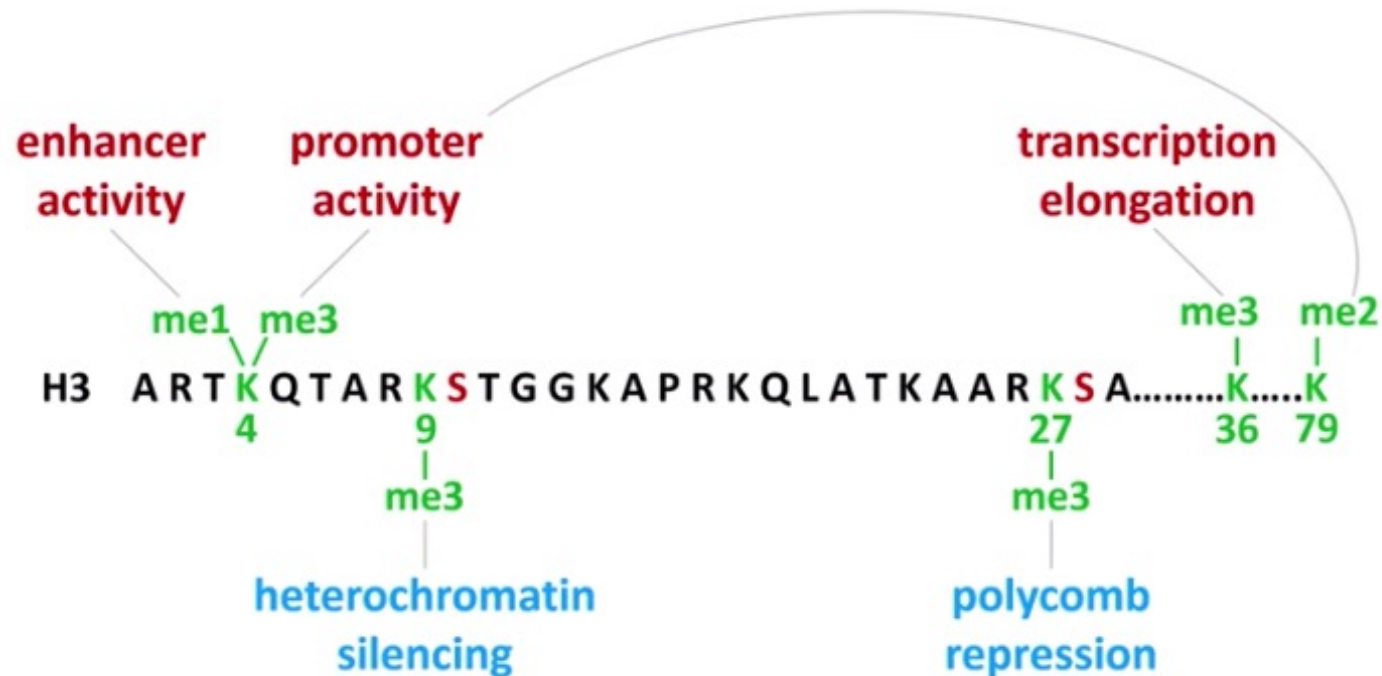
A core set of histone modifications help define chromatin states

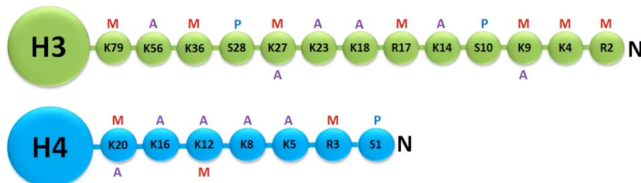
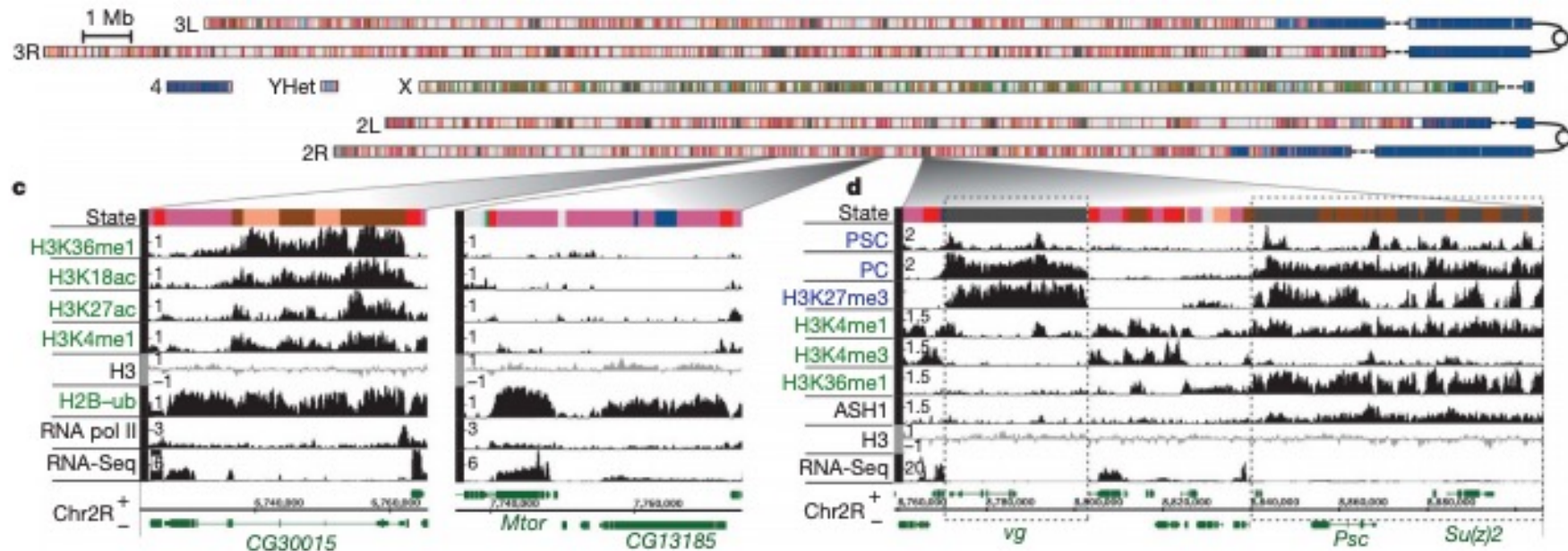
The following 7 core histone modifications are associated with specific transcription events across the genome

- H3K9ac active promoters
- H3K4me3 active promoters
- H3K4me1 enhancers and promoters
- H3K27ac enhancers and promoters
- H3K36me3 transcribed gene bodies
- H3K27me3 polycomb silenced promoters
- H3K9me3 heterochromatin



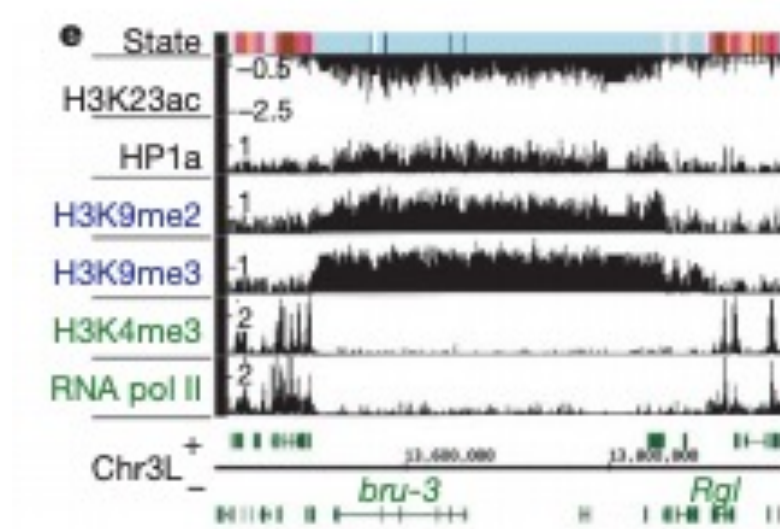
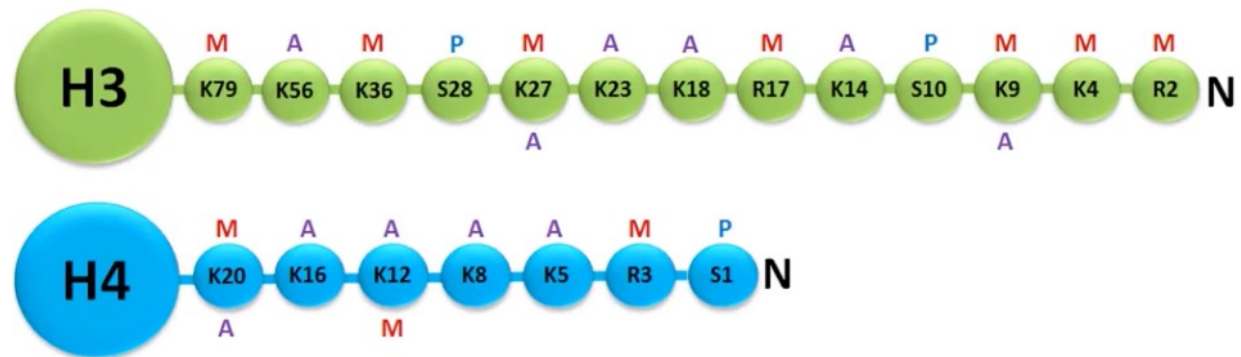
Lysine methylation is associated with transcription **activation** or **repression**

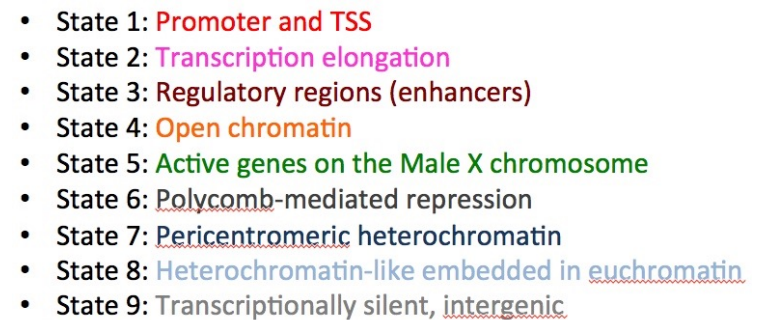




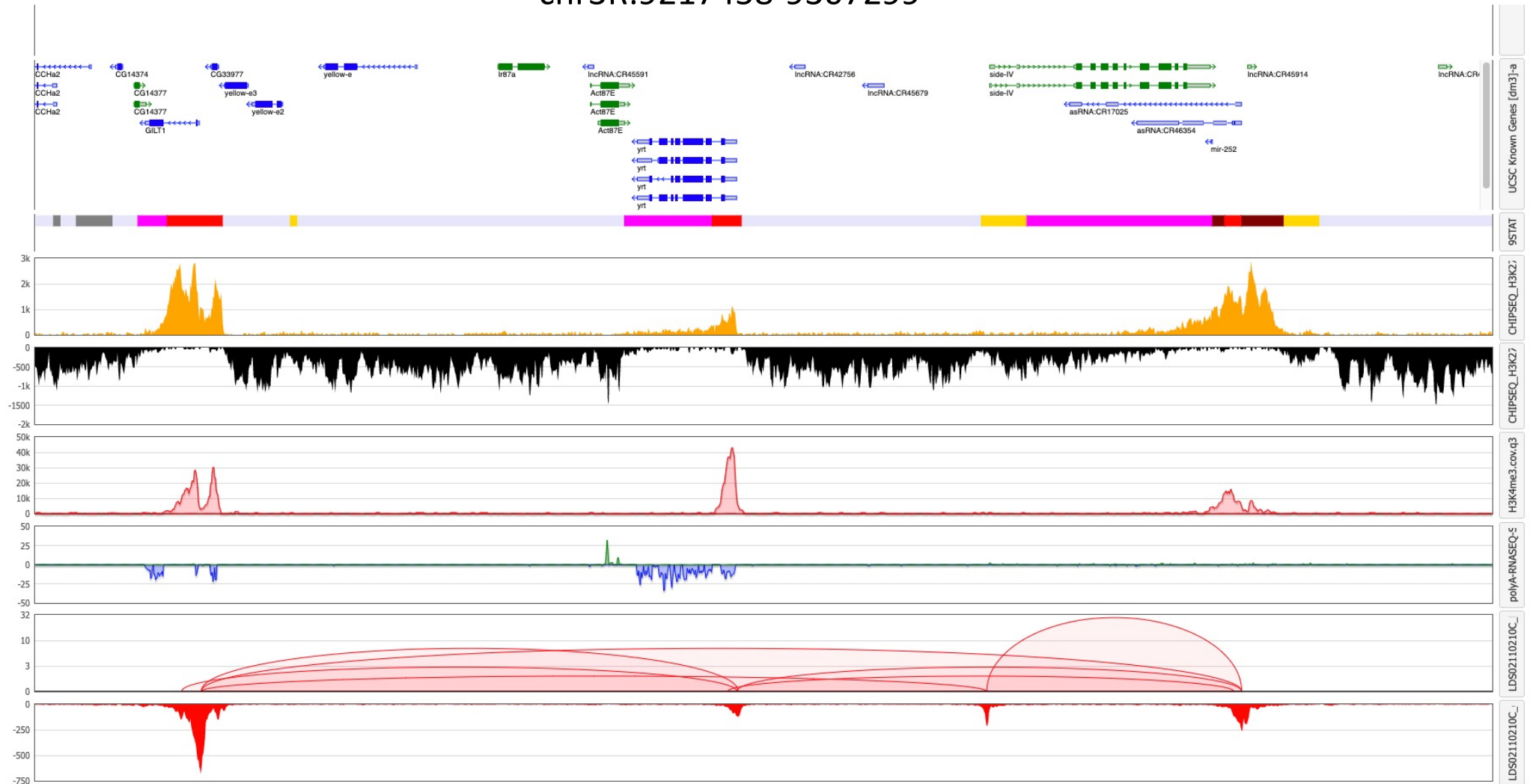
- State 1: **Promoter and TSS**
- State 2: **Transcription elongation**
- State 3: **Regulatory regions (enhancers)**
- State 4: **Open chromatin**
- State 5: **Active genes on the Male X chromosome**
- State 6: Polycomb-mediated repression
- State 7: Pericentromeric heterochromatin
- State 8: Heterochromatin-like embedded in euchromatin
- State 9: Transcriptionally silent, intergenic





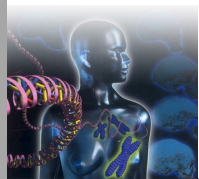
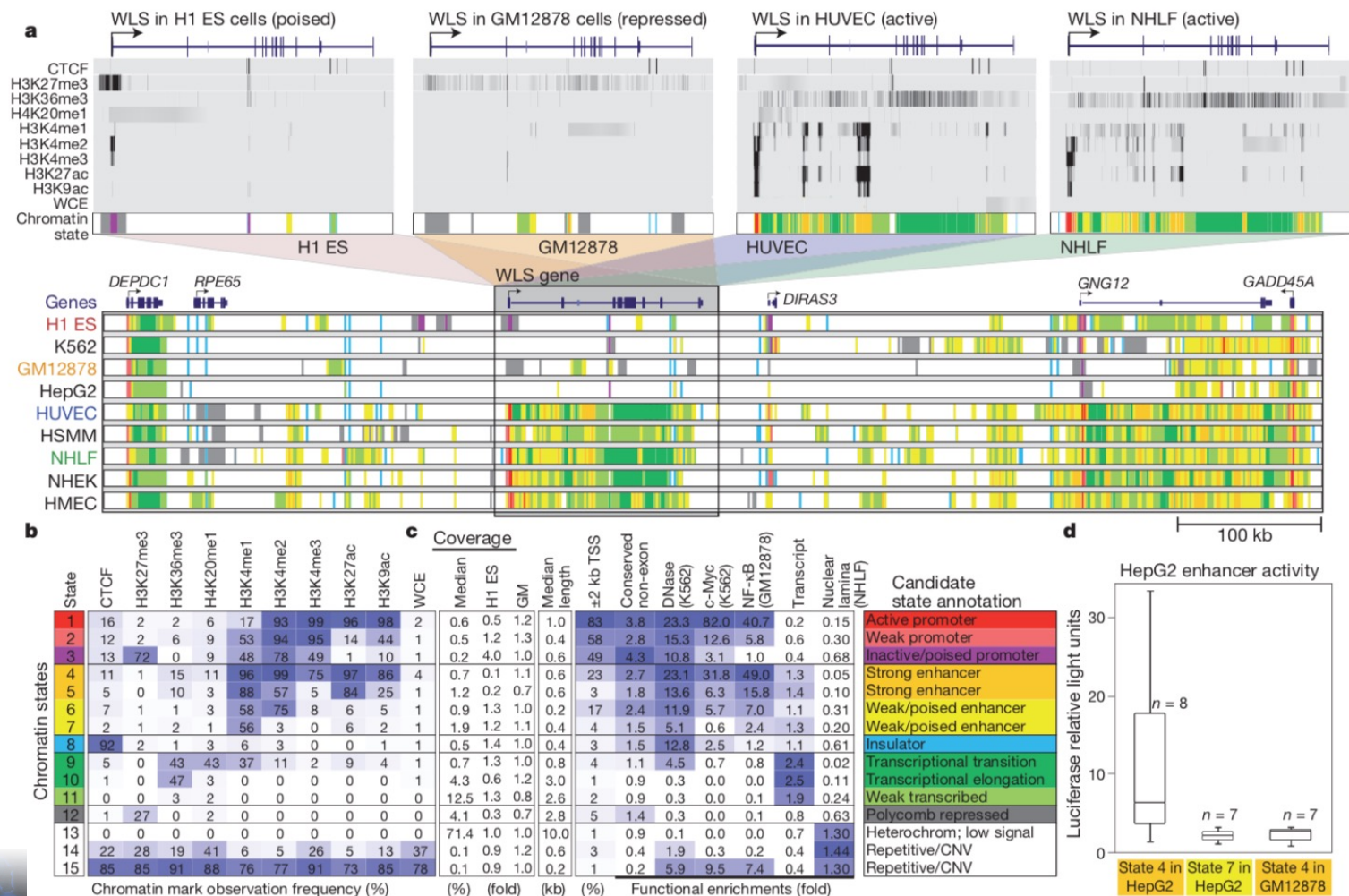


chr3R:9217458-9307299



- State 1: **Promoter and TSS**
- State 2: **Transcription elongation**
- State 3: **Regulatory regions (enhancers)**
- State 4: **Open chromatin**
- State 5: **Active genes on the Male X chromosome**
- State 6: **Polycomb-mediated repression**
- State 7: **Pericentromeric heterochromatin**
- State 8: **Heterochromatin-like embedded in euchromatin**
- State 9: **Transcriptionally silent, intergenic**





Chromatin State Maps

Roadmap Epigenomics Consortium

www.roadmapepigenomics.org

Chromatin state maps are available for 118 human cell types (mostly primary cells)

ChIP-seq profiles of the 7 core histone modifications are generated for each cell type

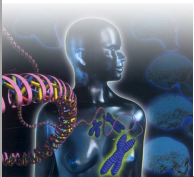
These profiles were computationally integrated using a Hidden Markov Model (HMM)

The genome can be segmented into as many as 25 chromatin states

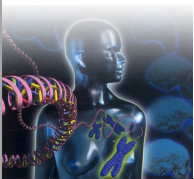
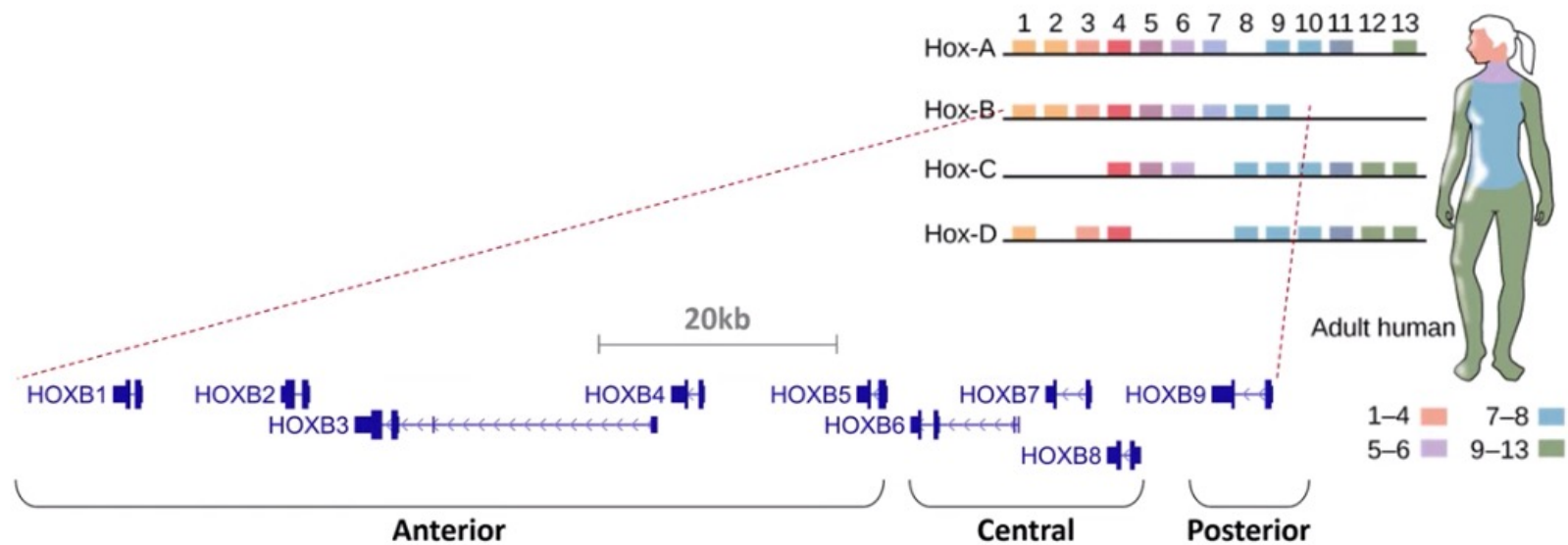
Genome browser tracks with different colours for each state can be used to highlight predicted transcription events of putative regulatory elements

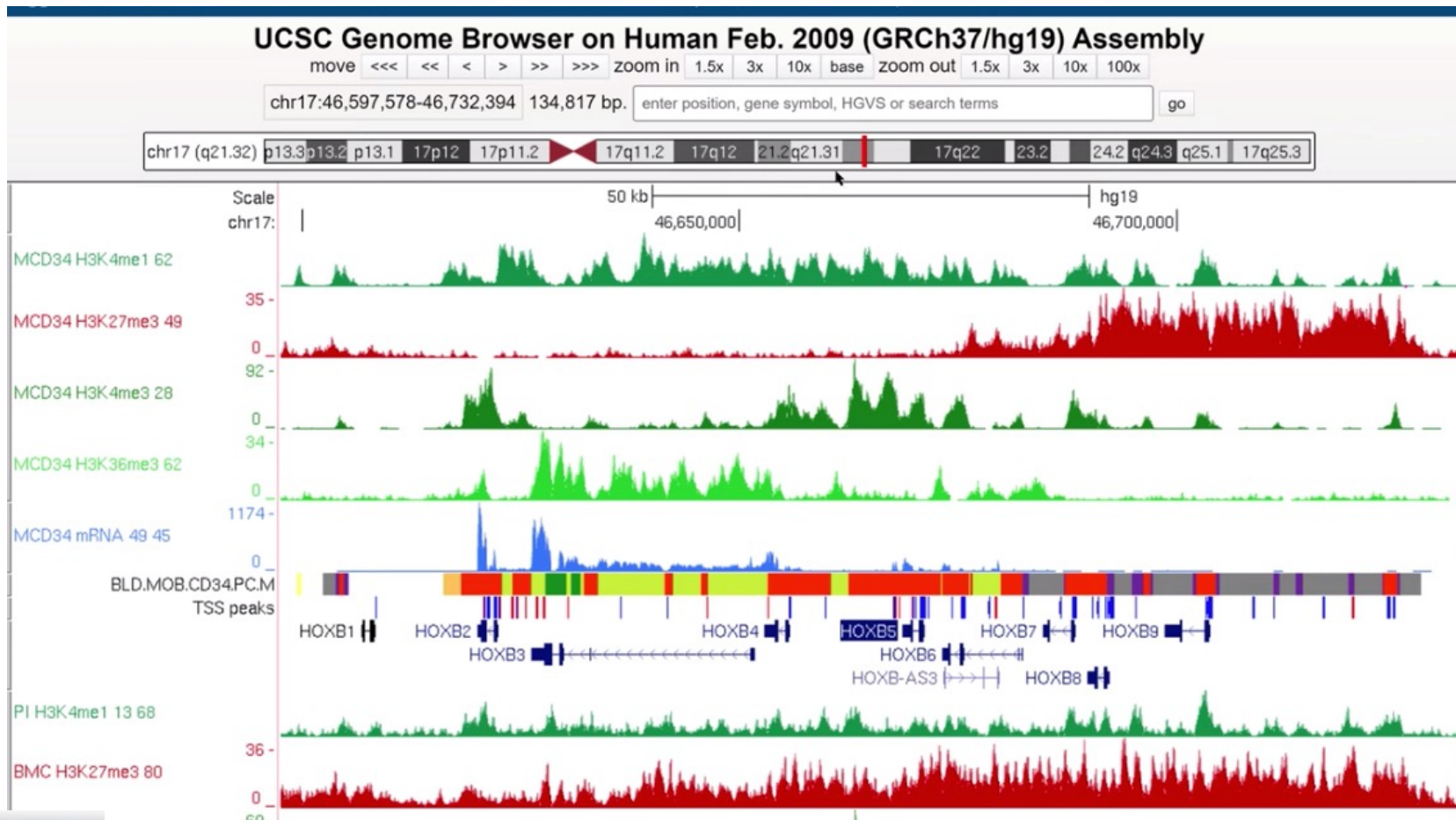
Imputed Chromatin State Maps

- State 1 - **Red** - TssA (Active TSS)
- State 2 - **Orange Red** - PromU (Promoter Upstream TSS)
- State 3 - **Orange Red** - PromD1 (Promoter Downstream TSS with DNase)
- State 4 - **Orange Red** - PromD2 (Promoter Downstream TSS)
- State 5 - **Green** - Tx5' (Transcription 5')
- State 6 - **Green** - Tx (Transcription)
- State 7 - **Green** - Tx3' (Transcription 3')
- State 8 - **Light Green** - TxWk (Weak transcription)
- State 9 - **GreenYellow** - TxReg (Transcription Regulatory)
- State 10 - **GreenYellow** - TxEnh5' (Transcription 5' Enhancer)
- State 11 - **GreenYellow** - TxEnh3' (Transcription 3' Enhancer)
- State 12 - **GreenYellow** - TxEnhW (Transcription Weak Enhancer)
- State 13 - **Orange** - EnhA1 (Active Enhancer 1)
- State 14 - **Orange** - EnhA2 (Active Enhancer 2)
- State 15 - **Orange** - EnhAF (Active Enhancer Flank)
- State 16 - **Yellow** - EnhW1 (Weak Enhancer 1)
- State 17 - **Yellow** - EnhW2 (Weak Enhancer 2)
- State 18 - **Yellow** - EnhAc (Enhancer Acetylation Only)
- State 19 - **Light Yellow** - DNase (DNase only)
- State 20 - **Medium Aquamarine** - ZNF/Rpts (ZNF genes & repeats)
- State 21 - **PaleTurquoise** - Het (Heterochromatin)
- State 22 - **Light Purple** - PromP (Poised Promoter)
- State 23 - **Purple** - PromBiv (Bivalent Promoter)
- State 24 - **Silver** - ReprPC (Repressed PolyComb)
- State 25 - **White** - Quies (Quiescent/Low)

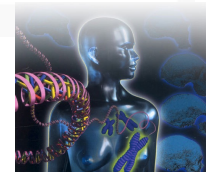
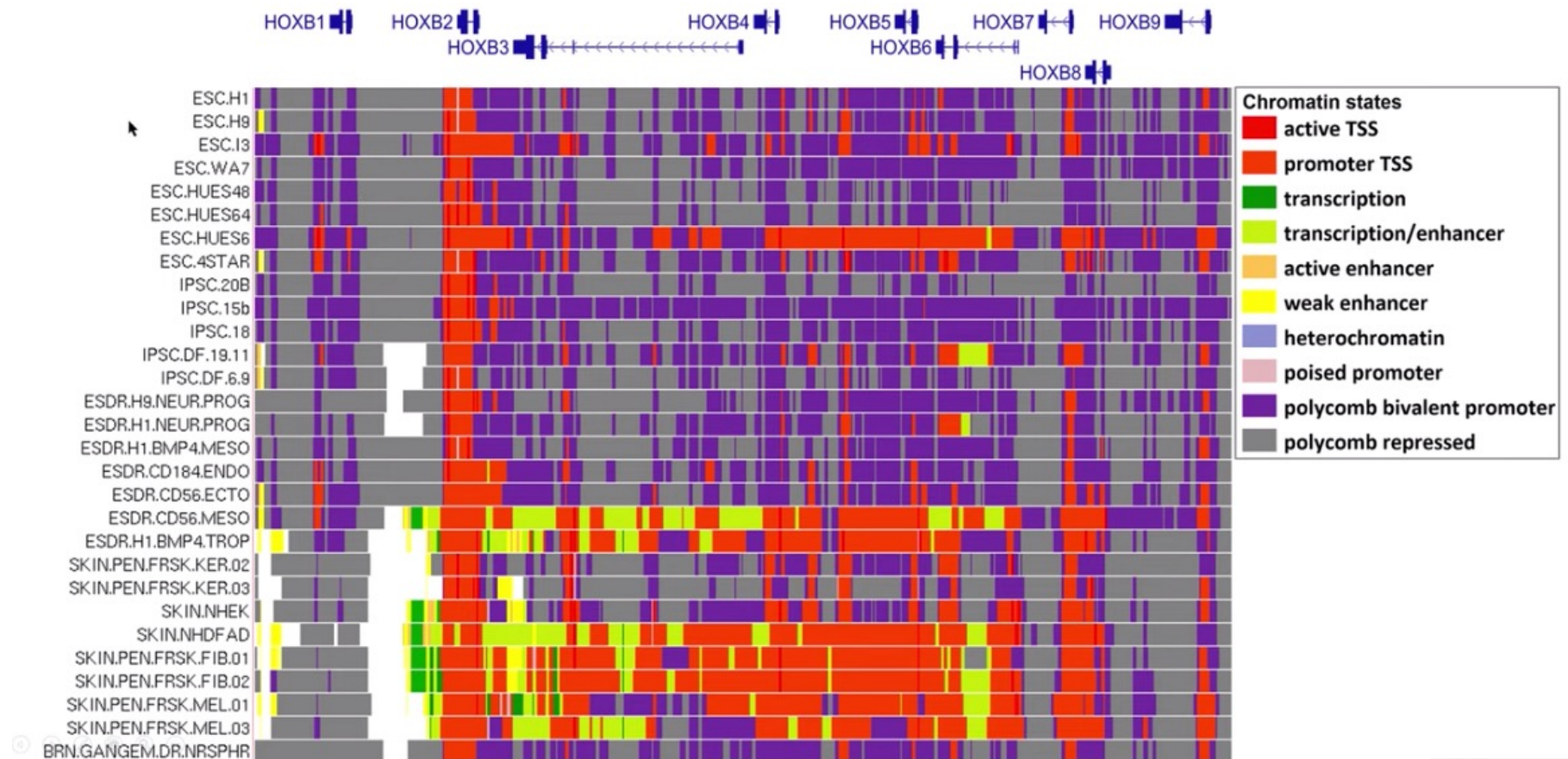


Chromatin states at a homeotic gene cluster in human tissues

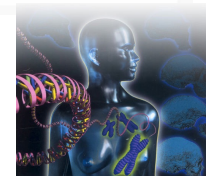
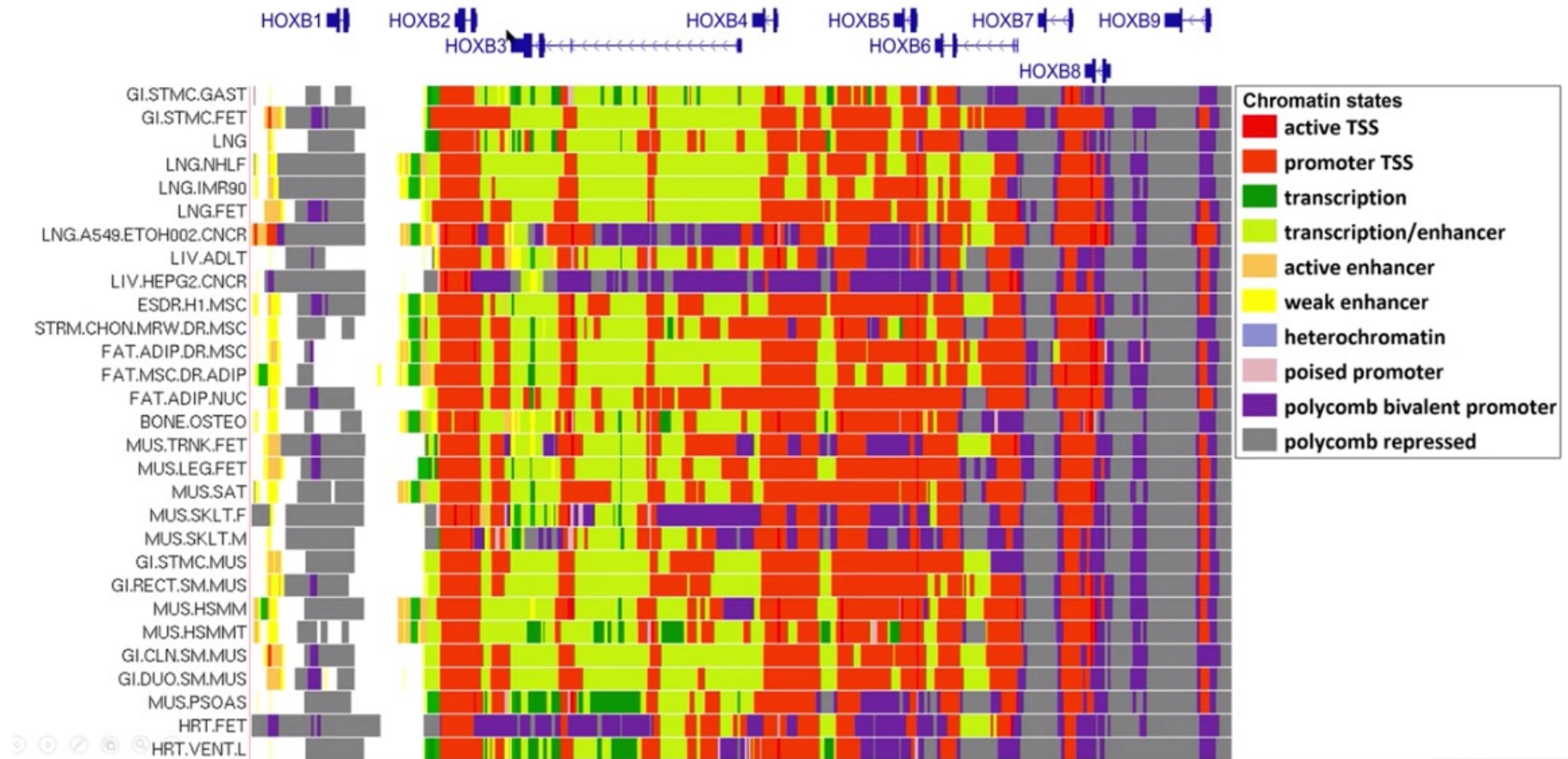




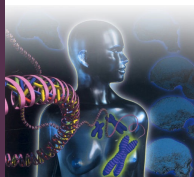
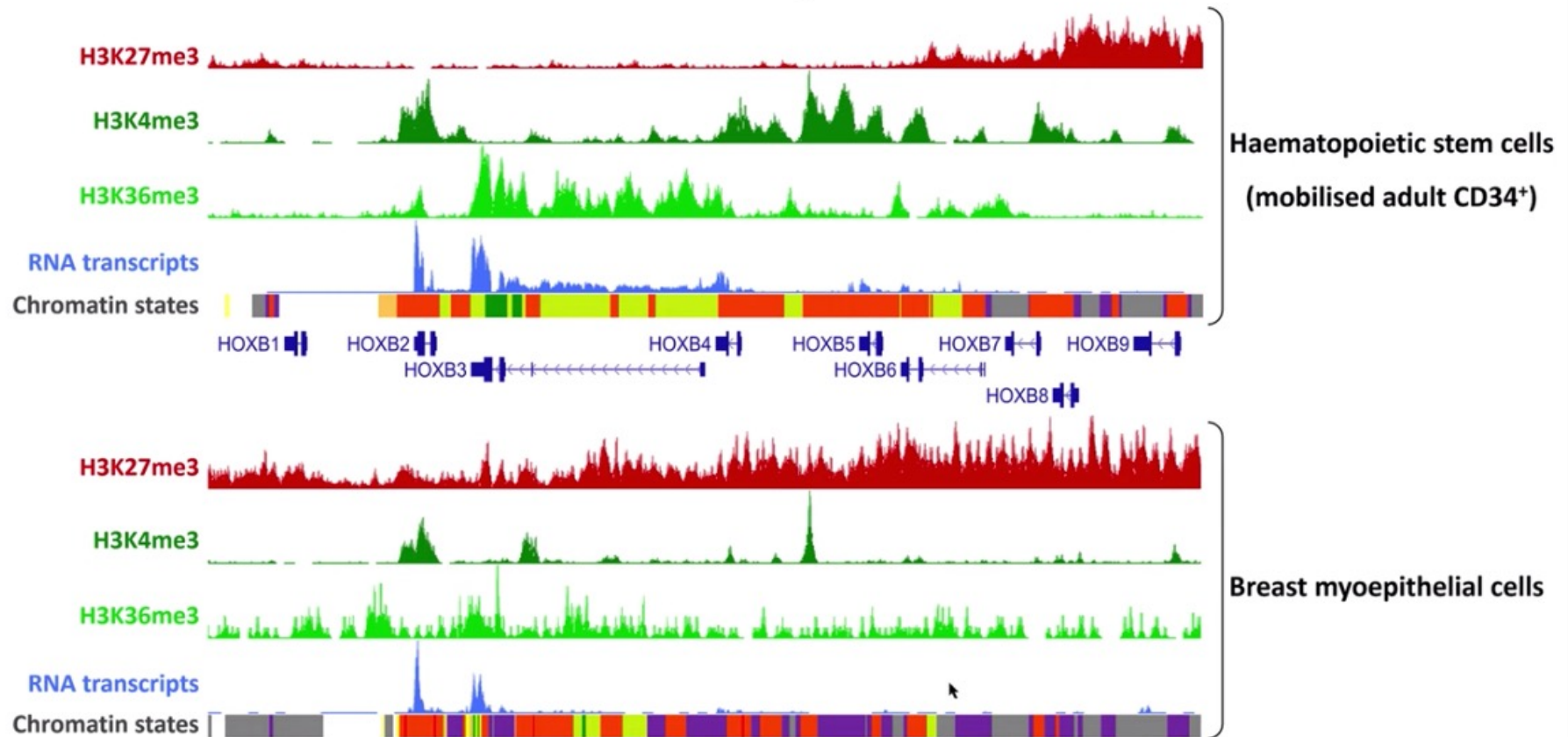
Chromatin states at a homeotic gene cluster in human tissues



Chromatin states at a homeotic gene cluster in human tissues



Chromatin states at a homeotic gene cluster in human tissues



Histone modifications mark transcription events

